510(k) SUBSTANTIAL EQUIVALENCE DETERMINATION
DECISION MEMORANDUM

A. 510(k) Number:

K182619

B. Purpose for Submission:

To establish performance claims and marketing clearance for the ePlex Blood Culture Identification Gram-Negative (BCID-GN) Panel

C. Measurand:

Nucleic acid targeted sequences from *Acinetobacter baumannii*, *Bacteroides fragilis*, *Citrobacter*, *Cronobacter sakazakii*, *Enterobacter cloacae* complex, *Enterobacter* (non-*cloacae* complex), *Escherichia coli*, *Fusobacterium necrophorum*, *Fusobacterium nucleatum*, *Haemophilus influenzae*, *Klebsiella oxytoca*, *Klebsiella pneumoniae* group, *Morganella morganii*, *Neisseria meningitidis*, *Proteus*, *Proteus mirabilis*, *Pseudomonas aeruginosa*, *Salmonella*, *Serratia*, *Serratia marcescens*, *Stenotrophomonas maltophilia*, CTX-M, IMP, KPC, NDM, OXA, and VIM, Pan Gram-Positive, Pan *Candida*

D. Type of Test:

A multiplexed nucleic acid test intended for use with the GenMark ePlex instrument for the simultaneous qualitative *in vitro* detection and identification of multiple bacterial nucleic acids from positive blood culture specimens.

E. Applicant:

GenMark Diagnostics, Incorporated

F. Proprietary and Established Names:

ePlex Blood Culture Identification Panel Gram Negative (BCID-GN)

Common Name: ePlex BCID-GN Panel

G. Regulatory Information:

1. Regulation section: 21 CFR 866.3365
2. Classification: Class II
3. Product code(s): PEN, PAM, PEO

4. Panel: Microbiology (83)

H. Intended Use:

1. Intended use(s):

The GenMark ePlex Blood Culture Identification Gram-Negative (BCID-GN) Panel is a qualitative nucleic acid multiplex *in vitro* diagnostic test intended for use on GenMark's ePlex Instrument for simultaneous qualitative detection and identification of multiple potentially pathogenic gram-negative bacterial organisms and select determinants associated with antimicrobial resistance in positive blood culture. In addition, the ePlex BCID-GN Panel is capable of detecting several gram-positive bacteria (Pan Gram-Positive assay) and several *Candida* species (Pan *Candida* assay). The ePlex BCID-GN Panel is performed directly on blood culture samples identified as positive by a continuous monitoring blood culture system and which contain gram-negative organism.

The following bacterial organisms and genes associated with antibiotic resistance are identified using the ePlex BCID-GN Panel: *Acinetobacter baumannii, Bacteroides fragilis, Citrobacter, Cronobacter sakazakii, Enterobacter cloacae* complex, *Enterobacter* (non-*cloacae* complex), *Escherichia coli, Fusobacterium necrophorum, Fusobacterium nucleatum, Haemophilus influenzae, Klebsiella oxytoca, Klebsiella pneumoniae* group, *Morganella morganii, Neisseria meningitidis, Proteus, Proteus mirabilis, Pseudomonas aeruginosa, Salmonella, Serratia, Serratia marcescens, Stenotrophomonas maltophilia,* CTX-M (*bla*$_{\text{CTX-M}}$), IMP (*bla*$_{\text{IMP}}$) , KPC (*bla*$_{\text{KPC}}$) , NDM (*bla*$_{\text{NDM}}$), OXA (*bla*$_{\text{OXA}}$) (OXA-23 and OXA-48 groups only), and VIM (*bla*$_{\text{VIM}}$).

The ePlex BCID-GN Panel contains assays for the detection of genetic determinants associated with resistance to antimicrobial agents including CTX-M(*bla*$_{\text{CTX-M}}$), which is associated with resistance to extended spectrum beta-lactamase (ESBL)-mediated resistance to penicillins, cephalosporins and monobactams, as well as OXA (*bla*$_{\text{OXA}}$) (OXA-23 and OXA-48 groups only), KPC (*bla*$_{\text{KPC}}$), and metallo-beta-lactamases IMP (*bla*$_{\text{IMP}}$), VIM (*bla*$_{\text{VIM}}$), and NDM (*bla*$_{\text{NDM}}$), which is associated with carbapenemase-mediated resistance. The antimicrobial resistance gene detected may or may not be associated with the agent responsible for disease. Negative results for these select antimicrobial resistance assays do not indicate susceptibility, as there are multiple mechanisms of resistance in gram-negative bacteria.

The ePlex BCID-GN Panel also contains targets designed to detect a broad range of organisms with a potentially misleading Gram stain result or organisms that may be missed by Gram staining altogether, for example in the case of co-infections. These include a broad Pan Gram-Positive assay (which is designed to detect *Bacillus cereus* group, *Bacillus subtilis* group, *Enterococcus, Staphylococcus*, and *Streptococcus*), as well as a Pan *Candida* assay, which is designed to detect four *Candida* species: *Candida albicans*, *Candida glabrata*, *Candida krusei*, and *Candida parapsilosis*.

The detection and identification of specific bacterial and fungal nucleic acids from

individuals exhibiting signs and/or symptoms of bloodstream infection aids in the diagnosis of bloodstream infection when used in conjunction with other clinical information. The results from the ePlex BCID-GN Panel are intended to be interpreted in conjunction with Gram stain results and should not be used as the sole basis for diagnosis, treatment, or other patient management decisions.

Negative results in the setting of a suspected bloodstream infection may be due to infection with pathogens that are not detected by this test. Positive results do not rule out co-infection with other organisms; the organism(s) detected by the ePlex BCID-GN Panel may not be the definite cause of disease. Additional laboratory testing (e.g. sub-culturing of positive blood cultures for identification of organisms not detected by ePlex BCID-GN Panel and for susceptibility testing, differentiation of mixed growth, and association of antimicrobial resistance marker genes to a specific organism) and clinical presentation must be taken into consideration in the final diagnosis of bloodstream infection.

2. Indication(s) for use:

   Same as intended use.

3. Special conditions for use statement(s):

   For prescription use only

   Limitations:

   - This product should not be used with blood culture media that contains charcoal.
   - Bacterial and fungal nucleic acids may be present in blood culture, independent of bacterial or fungal viability. Detection of an assay target does not guarantee that the corresponding bacteria or fungi are infectious or are the causative agents for clinical symptoms.
   - Decreased sensitivity has been observed for some targets in BD BACTEC Lytic Anaerobic blood culture bottles.
   - In mixed cultures, the ePlex BCID-GN Panel may not identify all organisms in the specimen, depending upon the concentration of each target present.
   - Antimicrobial resistance can occur via multiple mechanisms. A “Not Detected” result for the BCID-GP antimicrobial resistance gene assays does not indicate antimicrobial susceptibility. Subculturing and standard susceptibility testing of isolates is required to determine antimicrobial susceptibility.
   - The BCID-GN Pan *Candida* and Pan Gram-Positive assays are designed to detect *Candida* and gram-positive microorganisms in mixed blood cultures where these microorganisms might be missed by Gram stain. Lower than expected clinical sensitivity was observed for these analytes, which was likely due to the presence of *Candida* spp. or gram-positive microorganisms in mixed cultures at concentrations below the respective limits of detection for the Pan *Candida* and Pan GP targets.

- For *Enterococcus saccharolyticus, Serratia odorifera, and Streptococcus thoraltensis*, 100% detection was not observed at expected bottle positivity concentrations. Decreased sensitivity may be observed for these strains.
- False negative results may occur for specimens containing *Fusobacterium necrophorum* because the assay's limit of detection for *F. necrophorum* can be near the concentration observed at the time of bottle positivity.

4. Special instrument requirements:

   GenMark ePlex Instrument and Software

I. Device Description:

The ePlex Blood Culture Identification Gram-Negative (BCID-GN) Panel is based on the principles of competitive nucleic acid hybridization using a sandwich assay format, wherein a single-stranded target binds concurrently to a sequence-specific solution-phase signal probe and a solid-phase electrode-bound capture probe. The test employs nucleic acid extraction, target amplification via polymerase chain reaction (PCR) or reverse transcription (RT-PCR) and hybridization of target DNA. In the process, the double-stranded PCR amplicons are digested with exonuclease to generate single-stranded DNA suitable for hybridization.

Nucleic acid extraction from biological specimens occurs within the cartridge via cell lysis, nucleic acid capture onto magnetic beads, and release for amplification. The nucleic acid extraction is processed through microfluidic liquid handling. Once the nucleic acid targets are captured and inhibitors are washed away, the magnetic particles are delivered to the electrowetting environment on the printed circuit board (PCB) and the targets are eluted from the particles and amplified.

During hybridization, the single-stranded target DNA binds to a complementary, single-stranded capture probe immobilized on the working gold electrode surface. Single-stranded signal proves (labeled with electrochemically active ferrocenes) bind to specific target sequence/region adjacent to the capture probe. Simultaneous hybridization of target to signal probes and capture probe is detected by alternating current voltammetry (ACV). Each working electrode on the array contains specific capture probes, and sequential analysis of each electrode allows detection of multiple analyte targets.

A summary of microorganisms and resistance markers detected by the ePlex BCID-GN Panel are presented in Table 1:

Table 1: ePlex BCID-GN Panel, organism and resistance marker targets

Bacterial Targets	
Acinetobacter baumannii	*Klebsiella pneumoniae* group
Bacteroides fragilis	*Morganella morganii*
Citrobacter	*Neisseria meningitidis*
Cronobacter sakazakii	*Proteus*

Enterobacter cloacae complex	*Proteus mirabilis*
Enterobacter (non-*cloacae* complex)	*Pseudomonas aeruginosa*
Escherichia coli	*Salmonella*
Fusobacterium necrophorum	*Serratia*
Fusobacterium nucleatum	*Serratia marcescens*
Haemophilus influenzae	*Stenotrophomonas maltophilia*
Klebsiella oxytoca	
Antimicrobial Resistance Markers (See Table 5 For More Detail)	
CTX-M (*bla*$_{CTX-M}$)	NDM (*bla*$_{NDM}$)
IMP (*bla*$_{IMP}$)	OXA (*bla*$_{OXA}$)
KPC (*bla*$_{KPC}$)	VIM (*bla*$_{VIM}$)
Pan Targets	
Pan Gram-Positive	Pan *Candida*

Materials provided

ePlex Blood Culture Identification Gram-Negative Panel kit containing 12 individual test cartridges. Storage is at 2-8°C.

Materials needed but not provided

Equipment

- GenMark ePlex instrument and software
- Pipettes calibrated to deliver 50 µl
- Printer (optional)

Consumables

- Pipette tips, aerosol resistant, RNase/DNase-free
- Disposable, powder free gloves
- 10% bleach for decontamination of appropriate surfaces
- 70% ethanol or isopropyl alcohol
- 1.5 mL RNase/DNase-free microcentrifuge tube or equivalent

Interpretation of Results

Results interpretation of the ePlex BCID-GP Panel is performed by the ePlex instrument, and the interpretation of results on the ePlex BCID-GP Panel Detection Report for each targeted analyte is summarized in Table 2.

Table 2: Interpretation of results, ePlex BCID-GN Panel Detection Report

Target Result	Explanation	Action
Detected	The test was completed successfully and the target has generated signal above its defined threshold and the Internal Control was reported as PASS.	All results are displayed on the ePlex BCID-GN Panel Detection Report. Test is valid, report results.
Not Detected	The test was completed successfully and the target did not generate signal above its defined threshold and the Internal Control was reported as PASS.	All results are displayed on the ePlex BCID-GN Panel Detection Report. Test is valid, report results.
Invalid	The test has not successfully completed and results for this test are not valid. This may be due to an instrument or software error.	No results are displayed on the ePlex BCID-GN Panel Detection Report. Test is not valid, repeat test.

Genus and group assay result interpretation

While many results on the ePlex BCID-GN Panel are based on a single assay, the ePlex BCID-GN Panel uses two assays for each of the Proteus and Serratia results.

The ePlex BCID-GN Proteus result is based on two assays: the species-specific Proteus mirabilis and the broad Proteus assay. The broad Proteus assay will detect Proteus mirabilis, however, its primary purpose is to detect non-mirabilis Proteus species. See Table 3 for detailed Proteus target call logic.

Table 3: *Proteus* target results from ePlex BCID-GN Panel Detection Report

***Proteus* Result**	***Proteus* Assay**	***Proteus mirabilis* Assay**	**Description**
Not Detected	Negative	Negative	No *Proteus* species detected
Detected	Positive	Positive	*Proteus mirabilis* detected
Detected	Positive	Negative	Unspeciated *Proteus* detected

The ePlex BCID-GN Serratia result is based on two assays: the species-specific *Serratia marcescens* and the broad *Serratia* assay. The broad *Serratia* assay will detect *Serratia marcescens*, however, its primary purpose is to detect *Serratia* species other than *Serratia marcescens*. See Table 4 for detailed *Serratia* target call logic.

Table 4 *Serratia* target results from ePlex BCID-GN Panel Detection Report

***Serratia* Result**	***Serratia* Assay**	***Serratia marcescens* Assay**	**Description**
Not Detected	Negative	Negative	No *Serratia* species detected
Detected	Positive	Positive	*Serratia marcescens* detected
Detected	Positive	Negative	Unspeciated *Serratia* detected

Test results for antimicrobial resistance genes are only reported when an associated organism is detected in the same specimen. Table 5 specifies organism targets that are reported with each the six resistance markers on the ePlex BCID Panel (indicated by X in the table).

Table 5. Resistance marker organism associations

Organism	Resistance Gene					
	CTX-M	**IMP**	**KPC**	**NDM**	**OXA**	**VIM**
Acinetobacter baumannii	X	X	X	X	X	X
Bacteroides fragilis						
Citrobacter	X	X	X	X	X	X
Cronobacter sakazakii			X			
Enterobacter cloacae complex	X	X	X	X	X	X
Enterobacter (*non-cloacae* complex)	X	X	X	X	X	X
Escherichia coli	X	X	X	X	X	X
Fusobacterium necrophorum						
Fusobacterium nucleatum						
Haemophilus influenzae						
Klebsiella oxytoca	X	X	X	X	X	X
Klebsiella pneumoniae group	X	X	X	X	X	X
Morganella morganii	X	X	X	X	X	X
Neisseria meningitidis						
Proteus	X	X	X	X	X	X
Proteus mirabilis	X	X	X	X	X	X
Pseudomonas aeruginosa	X	X	X	X	X	X
Salmonella	X	X	X	X	X	X
Serratia	X	X	X	X	X	X
Serratia marcescens	X	X	X	X	X	X
Stenotrophomonas maltophilia	X					

Pan Assay Results

The ePlex BCID-GN Panel Pan Gram-Positive assay detects one or more of the following gram-positive organisms: *Enterococcus, Bacillus cereus* group, *Bacillus subtilis* group, *Staphylococcus,* and/or *Streptococcus* as shown in Table 6.

Table 6: Pan gram-positive target results from ePlex BCID-GN Panel Detection Report

Pan Gram-Positive Result	Description
Not Detected	No specified gram-positive organism detected.
Detected	One or more of the following gram-positive organisms has been detected: *Enterococcus, Bacillus cereus* group, *Bacillus subtilis* group, *Staphylococcus,* and/or *Streptococcus*. Additional testing for speciation is suggested.

The ePlex BCID-GN Panel Pan *Candida* detects one or more of the following *Candida* species: *Candida albicans, Candida glabrata, Candida krusei, and/or Candida parapsilosis* as shown in Table 7.

Table 7: Pan *Candida* Target Results from ePlex BCID-GN Panel Detection Report

Pan *Candida* Result	Description
Not Detected	No specified *Candida* species detected.
Detected	One or more of the following *Candida* organisms has been detected: *Candida albicans, Candida glabrata, Candida krusei,* and/or *Candida parapsilosis*. Additional testing for identification is recommended.

ePlex BCID-GN Panel reports:

Several different reports are available on the ePlex System. Results are provided in a printable format and may be viewed electronically or exported for additional analysis. Reports can be customized with account specific information such as the address, logo and institutional specific footers on each report. For more information on ePlex Reports, refer to the ePlex Operator Manual.

Detection Report

The ePlex BCID-GN Panel Detection Report includes the results for each individual sample run on the ePlex System. The Summary section indicates the overall test result and lists all detected targets in that sample. The Results section includes a list of all targets on the panel with an individual result for each target. Results are reported as Detected, Not Detected, or Invalid (displayed as a red **x**); results for the Internal Control are reported as PASS, FAIL, INVALID, or N/A.

External Control Report

The ePlex BCID-GN Panel External Control Report is generated for an external control that has been pre-defined in the ePlex BCID-GN Panel Software. For more information on defining external controls on the ePlex System, refer to the ePlex Operator Manual.

The Summary section indicates the overall result (PASS or FAIL status) and lists all detected targets for that external control. The Results section includes a list of all panel targets with the result, expected result, and PASS/FAIL status for each. Results are reported as Detected, Not Detected, or Invalid (displayed as a red **x**). A target is reported as PASS if the actual result matches the expected result (as defined for that control); a target is reported as FAIL if the actual result does not match the expected result. If the actual result for each target matches the expected result (all targets reported as PASS), the overall result for the external control is reported as PASS in the Summary section. If the actual result for any target does not match the expected result, the overall result for the external control is reported as FAIL in the Summary section.

Summary Report

The Summary Report allows the operator to use searchable criteria to create customized reports, using specified targets, dates, range of dates, sample, external control, test bay, or operator. For more information on creating Summary Reports, refer to the ePlex Operator Manual.

J. Substantial Equivalence Information:

1. Predicate device name(s):

   FilmArray Blood Culture Identification Panel

2. Predicate 510(k) number(s):

   K130914

3. Comparison with predicate:

Similarities		
Item	**New Device** ePlex Blood Culture Identification Gram Negative Panel (K182619)	**Predicate Device** FilmArray Blood Culture Identification Panel (K130914)
Classification	21 CFR 866.3365 Multiplex nucleic acid assay for identification of microorganisms and resistance markers from positive blood cultures	21 CFR 866.3365 Multiplex nucleic acid assay for identification of microorganisms and resistance markers from positive blood cultures
Device Class	Class II	Class II
Product Code	PEN, PAM, PEO	P PEN, PAM, PEO
Intended Use	The ePlex Blood Culture Identification Gram-Negative (BCID-GN) Panel is a qualitative nucleic acid multiplex in vitro diagnostic test intended for use on GenMark's ePlex instrument for simultaneous detection and identification of multiple potentially pathogenic gram negative bacterial organisms and select determinants of antimicrobial resistance in positive blood culture. In addition, the ePlex Panel is capable of detecting several gram-positive bacteria (Pan Gram-Positive assay), and several Candida species (Pan Candida assay). The ePlex BCID-GN Panel is performed directly on blood culture samples identified as positive by a continuous monitoring blood culture system that demonstrates the presence of organisms as confirmed by Gram stain.	The FilmArray Blood Culture Identification (BCID) Panel is a qualitative multiplexed nucleic acid-based in vitro diagnostic test intended for use with FilmArray systems. The FilmArray BCID Panel is capable of simultaneous detection and identification of multiple bacterial and yeast nucleic acids and select genetic determinants of antimicrobial resistance. The BCID assay is performed directly on blood culture samples identified as positive by a continuous monitoring blood culture system that demonstrate the presence of organisms as determined by Gram stain.
Organisms Detected	*Acinetobacter baumannii, Bacteroides fragilis, Citrobacter, Cronobacter sakazakii, Enterobacter cloacae* complex, *Enterobacter* (non-*cloacae*	*Enterococci, Listeria monocytogenes, Staphylococci* (including specific differentiation of *Staphylococcus aureus*), *Streptococci* (with specific

Similarities		
	complex), *Escherichia coli*, *Fusobacterium necrophorum*, *Fusobacterium nucleatum*, *Haemophilus influenzae*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Morganella morganii*, *Neisseria meningitidis*, *Proteus*, *Proteus mirabilis*, *Pseudomonas aeruginosa*, *Salmonella*, *Serratia*, *Serratia marcescens*, *Stenotrophomonas maltophilia*, Pan Gram-Positive (selected organisms), Pan Candida (selected species)	differentiation of *Streptococcus agalactiae*, *Streptococcus pneumoniae*, and *Streptococcus pyogenes*), *Acinetobacter baumannii*, *Enterobacteriaceae* (including specific differentiation of the *Enterobacter cloacae* complex, *Escherichia coli*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Proteus*, and *Serratia marcescens*), *Haemophilus influenzae*, *Neisseria meningitidis* (encapsulated), *Pseudomonas aeruginosa*, *Candida albicans*, *Candida glabrata*, *Candida krusei*, *Candida parapsilosis*, and *Candida tropicalis*
Analyte	DNA	DNA
Technological Principles	Multiplex nucleic acid amplification test (NAAT)	Multiplex nucleic acid amplification test (NAAT)
Specimen Type	Blood culture samples identified as positive by a continuous monitoring blood culture system that demonstrates the presence of organisms (Gram-positive, Gram-negative, yeasts) as confirmed by Gram stain	Blood culture samples identified as positive by a continuous monitoring blood culture system that demonstrates the presence of organisms (Gram-positive, Gram-negative, yeasts) as confirmed by Gram stain
Test Interpretation	Automated test interpretation and report generation. User cannot access raw data.	Automated test interpretation and report generation. User cannot access raw data.
Sample Preparation Method	Sample processing is automated in the ePlex BCID-GN cartridge	Sample processing is automated in the FilmArray BCID pouch
Controls	Each cartridge includes two types of internal controls that together monitor performance of each step of the testing process, including extraction, amplification and detection processes.	Two controls are included in each reagent pouch to control for each sample processing and both stages of PCR and melt analysis.

Differences		
Item	**New Device** ePlex Blood Culture Identification Gram Negative Panel (K182619)	**Predicate Device** FilmArray Blood Culture Identification Panel (K130914)
Resistance Genes	Assay detects *ctx*-M, *imp*, *kpc*, *ndm*, *oxa*, and *vim*	Assay detects *mec*A, *van*A, *van*B and *kpc*.
Chemistry	Reagents contained within cartridge to allow: sample lysis and nucleic acid extraction, PCR amplification, and hybridization-based electrochemical detection	Reagents contained with a pouch to allow: sample lysis, nucleic acid extraction, multiplex nested PCR followed by high resolution melting analysis to confirm identity of amplified product

Differences		
Item	**New Device** ePlex Blood Culture Identification Gram Negative Panel (K182619)	**Predicate Device** FilmArray Blood Culture Identification Panel (K130914)
Instrumentation	ePlex Instrument	FilmArray Instrument
Reagent Storage	Cartridge containing reagents is stored at refrigerated temperature	Reagents are stored at room temperature.
Gram Stain	Specimens tested have gram negative organisms based on Gram stain prior to testing.	Specimens tested have organisms present based on Gram stain.

K. Standard/Guidance Document Referenced:

- Class II Special Controls Guideline: Multiplex Nucleic Acid Assay for Identification of Microorganisms and Resistance Markers from Positive Blood Cultures (May 2015)
- CLSI MM17-A, Vol 28, No. 9, Verification and Validation of Multiplex Nucleic Acid Assays
- CLSI EP17-A2 Evaluation of Detection Capability for Clinical Laboratory Measurement Procedures; Approved Guideline – Second Edition (June 2012)
- CLSI EP07-A2: Interference Testing in Clinical Chemistry; Approved Guideline – Second Edition (November 2005)
- CLSI EP25-A Evaluation of Stability of In Vitro Diagnostic Reagents; Approved Guideline (May 2013)

L. Test Principle:

The ePlex instrument automates all aspects of sample processing and nucleic acid testing in a single cartridge associated with the ePlex BCID-GN Panel. Multiple bloodstream pathogens can be detected from a single blood culture specimen. Automation of DNA extraction, amplification, and target detection is accomplished within the cartridge using a combination of electrowetting and GenMark's eSensor technology. Electrowetting, or digital microfluidics, uses electrical fields to directly manipulate discrete droplets on the surface of a hydrophobically coated printed circuit board (PCB). Sample and reagents are moved in a programmable fashion in the ePlex cartridge to complete all portions of the sample processing and detection. eSensor technology is based on the principles of competitive DNA hybridization and electrochemical detection, which is highly specific and is not based on fluorescent or optical detection.

A sample is loaded onto the ePlex cartridge and nucleic acids are extracted and purified from the specimen via magnetic solid phase extraction. PCR is used to create double-stranded DNA, which is treated with exonuclease to create single-stranded DNA in preparation for eSensor detection.

The target DNA is mixed with ferrocene-labeled signal probes that are complementary to the specific targets on the panel. Target DNA hybridizes to its complementary signal probe and target-specific capture probes, which are bound to gold-plated electrodes, forming a hybridization complex as shown below in Figure 1. The presence of each target is determined by voltammetry which generates specific electrical signals from the ferrocene-labeled signal probe.

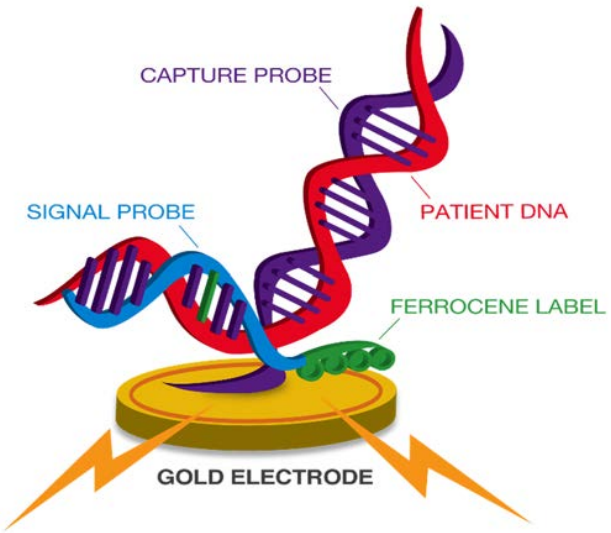


Figure 1

M. Performance Characteristics (if/when applicable):

1. Analytical performance:

a. Precision/Reproducibility:

A multisite reproducibility study was conducted at three testing sites (two external sites and one internal site). The study evaluated multiple sources of variability, such as site-to-site, lot-to-lot, day-to-day, and operator-to-operator variability. One ePlex instrument with four towers was used at each study site. Two operators each tested the reproducibility panel at each site on five nonconsecutive days with three unique lots of ePlex BCID-GN Panel cartridges.

The reproducibility panel consisted of a seven-member test panel comprised of six panel members representing 17 BCID-GN analytes (including five resistance markers) and one panel member containing an off-panel organism, *Cutibacterium granulosum*.

Panel members were prepared by spiking cultured isolates into negative blood culture matrix (NBM). The NBM was prepared by incubating 10 mL of negative whole blood in BD BACTEC Standard/10 Aerobic/F blood culture bottles per the manufacturer's instructions for a minimum of eight hours on a continuous monitoring blood culture system. After incubation and prior to use in analytical studies, the NBM was pre-screened to ensure the matrix is negative for all BCID-GN targets.

Each of the on-panel organisms was evaluated at two concentrations: (1) Bottle positivity

and (2) One log higher than bottle positivity (to mimic organism concentrations present after eight hours additional incubation on the blood culture instrument).

The organisms and concentrations evaluated in the study are summarized in Table 8.

Table 8: Bottle Positivity Concentrations

Organism	Bottle Positivity Concentration	Bottle Positivity +8 Hours Concentration
Acinetobacter baumannii (OXA+)	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Enterobacter cloacae (CTX-M+, KPC+)	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Escherichia coli (IMP+)	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Fusobacterium nucleatum	1 x 10^7 CFU/mL	1 x 10^8 CFU/mL
Haemophilus influenzae	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Klebsiella oxytoca	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Neisseria meningitidis	3 x 10^7 CFU/mL	3 x 10^8 CFU/mL
Pseudomonas aeruginosa (VIM+)	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Serratia marcescens	1 x 10^8 CFU/mL	1 x 10^9 CFU/mL
Candida albicans (Pan *Candida* target)	1 x 10^6 CFU/mL	1 x 10^7 CFU/mL
Staphylococcus aureus (Pan Gram-Positive target)	1 x 10^7 CFU/mL	1 x 10^8 CFU/mL
Cutibacterium granulosum	Incubated on instrument until bottle positivity plus eight hours	N/A

The seven panel members were tested 108 times each (excluding repeat testing): 3 replicates x 3 sites x 2 operators x 3 lots x 2 days of testing/operator/lot, for a minimum number of 756 tests.

A summary of results for the ePlex BCID-GN Panel reproducibility study is provided in Table 9. For the negative panel member, 100% of replicates generated the expected negative results for all ePlex BCID-GN Panel analytes.

Table 9. Reproducibility of the ePlex BCID-GN Panel

Analyte	Concentration Tested	Expected Results	Agreement with Expected Result			
			Site 1	Site 2	Site 3	All Sites (95% CI)
Organisms						
Acinetobacter baumannii	BP+8* 1x10^9 CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP** 1x10^8 CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	179/179	178/179	180/180	537/538 (99.0-100)
Enterobacter cloacae complex	BP+8 1x10^9 CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP 1x10^8 CFU/mL	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	Negative	Not Detected	180/180	179/179	180/180	539/539 (99.3-100)

Analyte	Concentration Tested	Expected Results	Agreement with Expected Result			
			Site 1	Site 2	Site 3	All Sites (95% CI)
Escherichia coli	BP+8 $1x10^9$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	35/35	36/36	107/107 (96.5-100)
	Negative	Not Detected	179/179	180/180	180/180	539/539 (99.3-100)
Fusobacterium nucleatum	BP+8 $1x10^8$ CFU/mL	Detected	36/36	35/35	36/36	107/107 (96.5-100)
	BP $1x10^7$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	179/179	180/180	180/180	539/539 (99.3-100)
Haemophilus influenzae	BP+8 $1x10^9$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	179/179	179/179	180/180	538/538 (96.6-100)
Klebsiella oxytoca	BP+8 $1x10^9$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	179/179	179/179	180/180	538/538 (96.6-100)
Neisseria meningitidis	BP+8 $3x10^8$ CFU/mL	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	BP $3x10^7$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	180/180	179/179	180/180	539/539 (99.3-100)
Pseudomonas aeruginosa	BP+8 $1x10^9$ CFU/mL	Detected	36/36	35/35	36/36	107/107 (96.5-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	179/179	180/180	180/180	539/539 (99.3-100)
Serratia	BP+8 $1x10^9$ CFU/mL	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	180/180	179/179	180/180	539/539 (99.3-100)
Serratia marcescens	BP+8 $1x10^9$ CFU/mL	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	180/180	179/179	180/180	539/539 (99.3-100)

Analyte	Concentration Tested	Expected Results	Agreement with Expected Result			
			Site 1	Site 2	Site 3	All Sites (95% CI)
Pan *Candida*	BP+8 $1x10^7$ CFU/m)	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	BP $1x10^6$ CFU/mL	Detected	35/36	36/36	36/36	107/108 (94.9-99.8)
	Negative	Not Detected	180/180	179/179	180/180	539/539 (99.3-100)
Pan Gram-Positive (*Staphylococcus aureus*)	BP+8 $1x10^8$ CFU/mL	Detected	34/36	35/35	36/36	105/107 (93.4-99.5)
	BP $1x10^7$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	179/179	179/180	180/180	538/539 (99.0-100)
Resistance Markers						
CTX-M (*Enterobacter cloacae*)	BP+8 $1x10^9$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP $1x10^8$ CFU/mL	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	Negative	Not Detected	144/144	143/143	144/144	431/431 (99.1-100)
IMP (*Escherichia coli*)	BP+8 $1x10^9$ CFU/mL	Detected	36/36	35/35	35/36	106/107 (94.9-99.8)
	BP $1x10^8$ CFU/mL	Detected	36/36	35/35	35/36	106/107 (94.9-99.8)
	Negative	Not Detected	143/143	144/144	144/144	431/431 (99.1-100)
KPC (*Enterobacter cloacae*)	BP+8 $1x10^9$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP $1x10^8$ CFU/mL	Detected	35/35	36/36	36/36	107/107 (96.5-100)
	Negative	Not Detected	144/144	143/1433	144/144	431/431 (99.1-100)
OXA (*Acinetobacter baumannii*)	BP+8 $1x10^9$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	143/143	143/143	144/144	430/430 (99.1-100)
VIM (*Pseudomonas aeruginosa*)	BP+8 $1x10^9$ CFU/mL	Detected	36/36	35/35	36/36	107/107 (96.5-100)
	BP $1x10^8$ CFU/mL	Detected	36/36	36/36	36/36	108/108 (96.6-100)
	Negative	Not Detected	143/143	144/144	144/144	431/431 (99.1-100)

*BP +8 – Bottle Positivity + 8 hours
**BP – Bottle Positivity

Five false positive results were observed in the BCID-GN Panel reproducibility study. *Bacteroides fragilis* (2), *Acinetobacter baumannii*, *Fusobacterium necrophorum*, and Pan Gram-Positive were detected in five different samples that had other organisms at positive concentrations. Four of the five samples with false positive results were observed at one testing site across two cartridge lots on four different days. On two of the days that

false positive results were observed (i.e., Pan Gram-Positive and *Acinetobacter baumannii* targets), external controls containing those targets were also tested. These results suggest a potential site-specific root cause (e.g., site-specific contamination) rather than a systematic performance issue with the BCID-GN Panel.

During the study, 812 runs were attempted with six pre-flight check failures (1.2% pre-flight errors) for which a new cartridge had to be tested. The pre-flight check is an ePlex system process that tests the integrity of the cartridge/instrument interface prior to initiating sample processing. An additional 10 runs were aborted by the system (1.2% system abort errors) resulting in 796 completed runs, including repeat runs due to invalid results.

Of the 756 samples initially tested, 715 (94.6%) had valid results and 41 (5.4%) had invalid results. Of the 40 samples re-tested, 39 had valid results and 1 had a second invalid result upon re-test. The final validity rate (95% CI) was 99.7% (99.0%-99.9%).

b. Linearity/assay reportable range:

Not applicable, qualitative assay.

c. Traceability, Stability, Expected values (controls, calibrators, or methods):

Assay Controls

Internal Controls

Each ePlex BCID-GN Panel cartridge includes internal controls that monitor performance of each step of the testing process, including extraction, amplification and detection of DNA targets.

Each of the eight amplification reactions has an internal control, and in each reaction, either the internal control or a target must generate signal above the defined threshold for a valid test result. Internal control results are interpreted by the ePlex Software and displayed on the ePlex BCID-GN Panel Reports as Internal Control with a result of PASS, FAIL, N/A, or INVALID. Table 10 includes details on the interpretation of Internal Control results.

Table 10: Internal control results

Internal Control Result	Explanation	Action
PASS	Signal above threshold has been detected from each amplification reaction. The test was completed and internal controls were successful, indicating valid results were generated.	All results are displayed on the ePlex BCID-GN Panel Detection Report. Test is valid, report results.
FAIL	Signal above threshold has not been detected from at least one amplification reaction. The test was completed but internal controls were not detected, indicating that results may not be valid.	No results are displayed on the ePlex BCID-GN Panel Detection Report. Test is not valid, repeat the test using a new cartridge.
N/A	The internal control in every amplification reaction does not generate signal above the threshold, but a target in every amplification reaction does generate signal above the threshold. The test was completed and internal controls were not successful, however detection of signal above the threshold for a target in every amplification reaction indicates valid results were generated.	All results are displayed on the ePlex BCID-GN Panel Detection Report. Test is valid, report results.
INVALID	An error has occurred during processing that prevents analysis of signal data. The test has not successfully completed and results for this test are not valid. This may be due to an instrument or software error.	No results are displayed on the ePlex BCID-GN Panel Detection Report. Test is not valid, repeat the test using a new cartridge.

External Controls

External controls are not provided with the ePlex BCID-GN Panel; however, the package insert includes the following recommendations for external controls:

- Positive and negative external controls should be tested with each new lot of reagents or monthly, whichever occurs first. Blood culture medium can be used as the negative control. Previously characterized positive samples or blood culture medium spiked with well characterized organisms can be used as the external positive control. External controls should be run in accordance with laboratory protocols and accrediting organizations, as applicable.

During the prospective clinical evaluation of the ePlex BCID-GN Panel, each testing site was provided with positive and negative external control samples. There were four organism mixes that served as positive controls and were prepared at concentrations near or below the concentration expected at bottle positivity. Negative blood culture matrix was used as the negative control. At least one positive and one negative external control were tested each day and cycled throughout the clinical study. The control mixes are described in Table 11.

Table 11. External controls used in the BCID-GN Panel Clinical study

External Controls	Expected Calls
Mix A	CTX-M-1, *Staphylococcus*, *Escherichia coli*, *Proteus mirabilis*, *Pseudomonas aeruginosa*, *Salmonella enterica*, *Serratia marcescens*, *E. cloacae* complex, *Enterobacter aerogenes*
Mix B	*Bacillus cereus*, IMP, OXA-23, *Candida albicans*, *Citrobacter freundii*, *Serratia proteamaculans*, *Neisseria meningitidis*, *Enterobacter amnigenus*, *Proteus vulgaris*
Mix C	*Bacteroides fragilis*, OXA-48, *Acinetobacter baumannii*, *Enterococcus* spp., *Fusobacterium necrophorum*, *Morganella morganii*, *Klebsiella oxytoca*, VIM
Mix D	*Cronobacter sakazakii*, *Stenotrophomonas maltophilia*, *Streptococcus* ssp., *Enterobacter asburiae*, *Haemophilus influenzae*, *Klebsiella pneumoniae*, NDM, *Fusobacterium nucleatum*, KPC
Negative control	Negative (Not Detected)

Over 90 days of investigational testing across 6 testing sites, there were 177 external controls tested with valid results: 25 Positive Control A, 22 Positive Control B, 21 Positive Control C, 21 Positive Control D, and 88 Negative Controls. Of these, 167/177 controls generated the expected results. The remaining 10 external control samples generated discordant results as follows:

3 external control samples had 1 or more false negative (FN) results:

- 1 Positive Control A sample had 1 FN *Pseudomonas aeruginosa* result.
- 1 Positive Control A sample had 1 FN *Enterobacter* (non-*cloacae* complex)
- result.
- 1 Positive Control A sample had 2 FN results: *Proteus mirabilis*, *Proteus*.

7 external control samples had 1 false positive (FP) result:

- 1 Negative Control sample had a 1 FP Pan Gram-Positive result.
- 2 Positive Control B samples each had 1 FP *Salmonella*.
- 2 Positive Control C samples each had 1 NDM FP result.
- 1 Positive Control C sample had 1 *Enterobacter cloacae* complex FP result.
- 1 Positive Control D sample had 1 VIM FP result.

Specimen stability

Clinical specimen stability claims were supported by an analytical study that investigated the effect of different specimen storage conditions/time periods on specimen integrity for specimens tested with the ePlex BCID-GN Panel. Three organism mixes containing representative targets from each of the eight multiplex primer pools on the ePlex BCID-GN Panel were used for this study. Organism mixes whose members were at concentrations consistent with those observed at bottle positivity were stored at ≤-80°C, ≤-20°C, 2-8°C, and at ambient temperature over several time points.

Baseline condition (T_0) was established with 20 replicates. Ten replicates were tested for each additional temperature and time point. All test conditions that were assessed in this study are summarized in Table 12.

Table 12. Specimen stability study design

Storage Condition	**Time Points**							
Ambient Conditions 15°C – 30°C	T_0	1day	3days	5days	7days	14days	30days	--
Refrigerated Conditions 2°C – 8°C	--	1day	3days	5days	7days	14days	30days	--
Frozen Conditions ≤-20°C and ≤-80°C	1 w	2 w	1mo	3mo	6mo	12mo	18mo	24mo
Bottle Incubator	Bottle positivity		12-hours post positivity					

The specimen stability criteria were as follows:

- Performance at each time point will be considered acceptable if the positivity rate for each analyte (either from the first 10 replicates or from the final 20 replicates in case of additional testing) is equivalent to the positivity rate for the corresponding analyte at time point T_0 (baseline time point).
- If a time point did not meet the positivity acceptance criteria, the next time point was evaluated. If the subsequent time point met the acceptance criteria, the study was continued. If two consecutive time points failed, the specimen stability was determined to be the last time point when the acceptance criteria were met.

Summaries of the specimen stability assessment data are in Tables 12-14:

Table 12. Ambient storage sample stability (Percent positivity and mean signal)

Target	**Day 0**		**Day 1**		**Day 3**		**Day 5**		**Day 7**		**Day 14**		**Day 30**	
	% Pos	**Mean nA**	**% Pos**	**Mean nA**	**% Pos**	**Mean nA**	**% Pos**	**Mean nA**	**% Pos**	**Mean nA**	**% Pos**	**Mean nA**	**% Pos**	**Mean nA**
Mix 1														
E. coli	100	391.5	100	597.8	100	673.5	100	72.0	100	725.8	100	754.9	100	758
H. influenzae	100	243.9	100	259.3	100	281.3	100	233.0	100	246.3	100	251.5	100	231.3
E. cloacae complex	100	230.1	100	516.3	100	723.4	100	818.4	100	886.6	100	937.3	100	919.2
Pan-*Candida*	100	233.1	100	247.0	100	217.8	100	247.0	100	184.7	100	233.7	100	245.3
OXA	100	735.7	100	746.1	100	726.0	100	816.5	100	795.2	100	823.0	100	824.0
CTX-M	95	292.9	100	333.5	100	340.2	100	310.4	100	350.4	100	337.9	100	387.8
Mix 2														
K. oxytoca	100	489.9	100	513.8	100	424.9	100	418.3	100#	492.5	100	532.7	100	512.2
P. aeruginosa	100	586.8	100	517.3	100	739	100	781.9	100#	783.4	100	524.7	100	800.4
A. baumannii	100	783.1	100	788.8	100	727.8	100	760.4	100#	759.5	100	799.9	100	803.1
E. non-cloacae complex	100	566.4	100	519.7	100	633.5	100	639	100#	669.6	100	472.3	100	587.8
IMP	100	492	100	491.5	100	479.2	100	457.5	100#	447.2	100	292.8	100	473.8
Mix 3														

Target	Day 0		Day 1		Day 3		Day 5		Day 7		Day 14		Day 30	
	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA
N. meningitidis	100	306	100	349.9	100	307.6	100	235.6	100^	277.8	100*	264.7	100	282.9
S. marcescens	100	479	100	530.7	100	498.9	100	545	100^	397.8	100*	368.1	100	375.6
B. fragilis	100	712.6	100	730.5	100	713.2	100	682.9	100^	739.8	100*	767.3	100	903.8
Pan Gram-Positive	100	461.3	100	511.4	100	625.5	100	633.4	100^	590.3	100*	544.7	100	489.6

Pan Gram-Positive false positive observed in 1 of 10 replicates
^ Mix 3; *M. morganii* and OXA-48 false positive observed in 1 of 20 replicates
* Mix 3; *P. aeruginosa* false positive observed in 1 of 20 replicates

Table 13. Refrigerated (2-8°C) storage sample stability (Percent positivity and mean signal)

Target	Day 0		Day 1		Day 3		Day 5		Day 7		Day 14		Day 30	
	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA
Mix 1														
E. coli	100	391.5	100	370.8	100	461.3	100	418.9	100	396.8	100	473	100	480.6
H. influenzae	100	243.9	100	226.6	100	308.2	100	279.2	100	268.7	100	285.1	100	218
E. cloacae complex	100	230.1	100	239.8	100	244.8	100	185.2	100	274.4	100	359.1	100	320.1
Pan-*Candida*	100	233.1	100	220	100	214.7	100	227.2	100	186.7	100	217.2	100	259.1
OXA	100	735.7	100	752.4	100	744.6	100	756.6	100	770.5	100	803.5	100	852.2
CTX-M	95	292.9	100	308.9	100	367.9	100	339.5	100	312.3	100	318.2	100	318.4
Mix 2														
K. oxytoca	100	489.9	100^	413	100	411.9	100	360.7	100	504.6	100	518	100*	461.2
P. aeruginosa	100	586.8	100^	457.1	100	755.7	100	627	100	734.9	100	388.6	100*	697.1
A. baumannii	100	783.1	100^	763.4	100	762.9	100	655.4	100	760.7	100	814.7	100*	691.3
E. non-cloacae complex	100	566.4	100^	519.3	100	661.8	100	553.7	100	660.2	100	388.2	100*	497.3
IMP	100	492	100^	404.8	100	490.7	100	431.8	100	467.3	100	244.1	**95***	443.4
Mix 3														
N. meningitidis	100	306	100	327.9	100	305.3	100	260.1	100	300.3	100	274.4	100	316.5
S. marcescens	100	479	100	526.9	100	442.1	100	486	100	337.1	100	303	100	309.5
B. fragilis	100	712.6	100	758.4	100	719	100	714.6	100	747.5	10	739.6	100	875
Pan Gram-Positive	100	461.3	100	579.4	100	551.5	100	613.4	100	536.1	100*	543.5	100	529.7

^ Mix 2, *Proteus* false positive detected in 1 of 20 replicates
* Mix 2 had 20 replicates

Table 14. Frozen (≤-20°C and ≤-70°C) Storage Sample Stability (Percent Positivity and Mean Signal)

Target	Day 0		Week 1		Week 2		Month 1		Month 3	
	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA
≤-20°C Condition										
Mix 1										
E. coli	100	391.5	100	534.7	100	478.7	100	547.2	100	377.1
H. influenzae	100	243.9	100	312.1	100	256.9	100	206.8	100	314.4

Target	Day 0		Week 1		Week 2		Month 1		Month 3	
	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA
E. cloacae complex	100	230.1	100	382.2	100	298.5	100	285.5	100	278.2
Pan-*Candida*	100	233.1	100	182.9	100	235.1	100	246.7	100	178.9
OXA	100	735.7	100	804.2	100	790	100	801	100	771.5
CTX-M	95	292.9	100	339.6	100	293.3	100	351.5	100	268.3
Mix 2										
K. oxytoca	100	489.9	100	422.3	100	525.2	100	445.9	100	505.4
P. aeruginosa	100	586.8	100	775.7	100	416.6	100	532.4	100	427.1
A. baumannii	100	783.1	100	738.2	100	826.2	100	767.6	100	529.5
E. non-cloacae complex	100	566.4	100	698.3	100	534.6	100	541.7	100	489.1
IMP	100	492	100	492.4	100	274.4	100	489.8	100	289.4
Mix 3										
N. meningitidis	100	306	100	336.4	100	244.4	100	279.5	100	436.4
S. marcescens	100	479	100	307.9	100	300.7	100	310	100	499.4
B. fragilis	100	712.6	100	798.5	100	725	100	919.6	100	599.6
Pan Gram-Positive	100	461.3	100	559.5	100	486.5	100	541	100	381.2
≤-70°C Condition										
Mix 1										
E. coli	100	391.5	100	538.5	100	502.6	100	410.2	100	290.9
H. influenzae	100	243.9	100	304.6	100	261.5	100	232.8	100	324
E. cloacae complex	100	230.1	100	367	100	296.8	100	210.6	100	275.6
Pan-*Candida*	100	233.1	100	190.7	100	222.6	100	271.4	100	146.5
OXA	100	735.7	100	798.2	100	784.4	100	770.7	100	757.4
CTX-M	100	292.9	100	298.4	100	286.9	100	296.8	100	256.7
Mix 2										
K. oxytoca	100	489.9	100	439.4	100	488.6	100	435.3	100	393.2
P. aeruginosa	100	586.8	100	790.9	100	403.4	100	604.8	100	315.2
A. baumannii	100	783.1	100	704.6	100	820.7	100	732.4	100	374.3
E. non-cloacae complex	100	566.4	100	664.6	100	501	100	528.2	100	368.8
IMP	100	492	100	476.1	100	268.8	100	482.1	100	304.2
Mix 3										
N. meningitidis	100	306	100	353.9	100	226.2	100	271.3	100	448.5
S. marcescens	100	479	100	324.7	100	277	100	359.3	100	428.8
B. fragilis	100	712.6	100	792.1	100	681.1	100	926.2	100	587.6
Pan Gram-Positive	100	461.3	100	575.5	100	480.7	100	562.8	100	412

The specimen stability study demonstrated that positive blood culture specimens can be stored up to 30 days at both ambient and refrigerated (2-8°C) temperatures as well as up to three months at both -20°C and -70°C without negatively impacting ePlex BCID-GN Panel performance. Study results also demonstrated that specimens can also be tested after incubation in a blood culture instrument for up to 12 hours post-bottle positivity.

Freeze-thaw Study

An analytical study was conducted to demonstrate the stability of clinical specimens after subjecting samples to one or two freeze-thaw cycles prior to testing on the BCID-GN Panel. One hundred and ten positive clinical samples representing 20 targeted analytes were tested fresh during the BCID-GN clinical study and selected to represent a variety of organisms present in positive blood cultures. All specimens contained gram-negative organisms on Gram stain. Samples were frozen at ≤-70°C and then subjected to one or two thawing cycles, followed by testing with the BCID-GN Panel. Study results are summarized in Table 15.

Table 15. Freeze/Thaw study (Positivity and mean signal)

Target	Total Replicates	Fresh (T0)		1X Freeze/Thaw		2X Freeze/Thaw	
		% Pos	Mean nA	% Pos	Mean nA	% Pos	Mean nA
Bacteroides fragilis	5	100	520.9	100	606.0	100	567.4
Citrobacter	3	100	364.6	100	643.3	100	537.8
Enterobacter (*non-cloacae* complex)	2	100	553.1	100	655.1	100	768.7
Enterobacter cloacae complex[A]	11	100	487.0	100	706.2	100	654.8
Escherichia coli	20	100	883.1	100	684.4	100	712.6
Haemophilus influenzae	2	100	378.2	100	468.0	100	427.3
Klebsiella oxytoca	3	100	362.8	100	514.1	100	562.0
Klebsiella pneumoniae[A]	17	100	439.7	100	467.7	100	534.0
Morganella morganii	2	100	556.6	100	522.5	100	567.9
Proteus	5	100	133.8	100	185.7	100	140.1
Proteus mirabilis	5	100	246.7	100	290.0	100	279.8
Pseudomonas aeruginosa[A]	9	100	609.9	100	650.5	100	645.3
Salmonella	2	100	93.1	100	94.3	100	101.0
Serratia	6	100	146.1	100	148.4	100	143.7
Serratia marcescens	5	100	206.3	100	378.4	100	389.1
Stenotrophomonas maltophilia	2	100	367.8	100	495.5	100	568.4
KPC*	1	100	713.9	100	797.7	100	837.6
CTX-M	4	100	273.6	100	348.8	100	351.8
Pane *Candida*[A]	7	100	162.4	100	153.4	100	188.9
Pan Gram-Positive	5	100	475.5	100	537.6	100	614.7

*There was only one sample tested for the KPC target. Therefore, the nA value is the actual observed value.

A. Four samples had false positive results but all expected targets were detected.
   i. Under the 1X freeze-thaw condition, one positive *Enterobacter cloacae* complex sample detected *Enterobacter cloacae* complex and OXA, making OXA a false positive result.
   ii. Under the 1X freeze-thaw condition, one positive Pan *Candida* sample detected Pan *Candida* and Pan Gram-Positive, making Pan Gram-Positive a false positive result.
   iii. Under the 2X freeze-thaw condition, one positive *Pseudomonas aeruginosa* sample detected *Pseudomonas aeruginosa* and *Stenotrophomonas maltophilia*, making *Stenotrophomonas maltophilia* a false positive result.
   iv. Under the 2X freeze-thaw condition, one positive *Klebsiella pneumoniae* sample detected *Klebsiella pneumoniae* and OXA, making OXA a false positive result.

Data from the freeze-thaw study support that clinical specimens are stable up to two freeze-thaw cycles and frozen specimens and previously frozen specimens are acceptable

for testing with the ePlex BCID-GN Panel.

Open Pouch Stability Study In-Consumable Sample Stability Study

An analytical study was conducted to demonstrate that the ePlex BC-GN cartridge (after the pouch has been opened) can be exposed to ambient conditions for up to two hours before loading the sample and testing on the ePlex instrument.

A second study was conducted to demonstrate that the cartridge can be stored at room temperature (15–30ºC) for two hours after the sample has been loaded but prior to running the cartridge in an ePlex bay.

For these two studies, the test panel was comprised of four organisms representing six of the eight multiplex primer pools and covering six targets on the ePlex BCID-GN Panel. Each organism was tested at a concentration approximating bottle positivity.

A total of 20 control replicates were also run for each organism. Control samples were loaded within 20 minutes after pouch opening and were run on the ePlex instrument within 20 minutes of sample loading.

The open pouch stability study demonstrated that ePlex BCID-GN cartridges are stable for up to 2 hours at room temperature after the pouch has been opened and prior to testing on the ePlex instrument. Results from the study are presented in Table 16.

Table 16: Open pouch stability study results

Target	% Positive	
	Sample tested within 20min after pouch opening	**Sample tested two hours after pouch opening**
Escherichia coli	100% (20/20)	100% (20/20)
Enterobacter cloacae complex	100% (20/20)	95% (19/20)
Haemophilus influenzae	100% (20/20)	100% (20/20)
Pan-*Candida*	100% (20/20)	95% (19/20)
CTX-M	95% (19/20)	100% (20/20)
OXA	100% (20/20)	100% (20/20)

The in-cartridge stability study demonstrated that the BCID-GN cartridge is stable for up to 2 hours at room temperature after the sample has been loaded prior to testing with the ePlex BCID-GN Panel. Results of the study are summarized in Table 17.

Table 17: In-Cartridge sample stability study results

Target	% Positive	
	Sample loaded in ≤20 minutes	**Sample loaded at 120 minutes**
Escherichia coli	100% (20/20)	100% (20/20)
Enterobacter cloacae complex	100% (20/20)	100% (20/20)
Haemophilus influenzae	100% (20/20)	100% (20/20)

Target	% Positive	
	Sample loaded in ≤20 minutes	Sample loaded at 120 minutes
Pan-*Candida*	100% (20/20)	100% (20/20)
CTX-M	95% (19/20)	95% (19/20)
OXA	100% (20/20)	100% (20/20)

d. Bottle Positivity: Organism Concentration Study

To approximate organism concentrations that may be present in positive blood culture bottles in a clinical setting, a study was conducted to establish the range of expected organism concentrations present in incubated blood cultures at bottle positivity (i.e., bottle ring) and eight hours after bottle positivity/bottle "ring." Representative organisms were evaluated, consisting of 14 genera that are detected by the ePlex BCID-GN Panel. The volume of negative human whole blood, determined by the manufacturer's instructions, was inoculated into each organism-appropriate bottle and incubated to bottle positivity and bottle positivity plus eight hours on a continuously monitoring blood culture system. At the time of positivity (and eight hours after positivity), the blood culture was removed from the instrument for determination of organism concentration (CFU/mL using a standard plate counting method) and ePlex BCID Panel testing. All estimated bottle positivity concentrations are equivalent or greater than the established Limit of Detection (LOD) for each of the assays of the ePlex BCID-GP Panel. The following bottle types were used for the study: BD BACTEC Plus Aerobic/F blood culture bottle (*E. faecium*, *S. aureus*, *S. anginosus*, *A. baumannii*, *E. cloacae*, *E. coli*, *H. influenzae*, *K. oxytoca*, *N. meningitidis*, *P. aeruginosa*, and *S. marcescens*), and BD BACTEC Lytic/10 Anaerobic/F (*B. fragilis* and *F. nucleatum*). Table 18 summarizes the concentration present at bottle positivity and bottle positivity plus eight hours for each organism evaluated.

Table 18: Bottle positivity study, organism concentrations

Organism	Average BP Concentration (CFU/mL)	Average BP+8hrs Concentration (CFU/mL)
Serratia marcescens	$1.2x10^9$	$2.2 x10^9$
Escherichia coli	$2.3 x10^8$	$1.5 x10^9$
Enterobacter cloacae	$2.8 x10^8$	$7.7 x10^8$
Neisseria meningitidis	$3.2 x10^7$	$2.1 x10^8$
Haemophilus influenzae	$6.9 x10^8$	$1.15 x10^9$
Pseudomonas aeruginosa	$1.6 x10^8$	$8.4 x10^8$
Fusobacterium nucleatum	$6.5 x10^7$	$4.9 x10^8$
Acinetobacter baumannii	$4.4 x10^8$	$3.8 x10^8$
Klebsiella oxytoca	$9.3 x10^8$	$1.5 x10^9$
Bacteroides fragilis	$4.7x10^8$	$6.7x10^9$
Staphylococcus aureus	$2.8 x10^7$	$2.1 x10^7$
Streptococcus anginosus	$4.1 x10^7$	$4.0 x10^8$

Organism	Average BP Concentration (CFU/mL)	Average BP+8hrs Concentration (CFU/mL)
Enterococcus faecium	4.9×10^7	3.6×10^7
Candida albicans	1.6×10^6	1.4×10^6

e. Detection limit:

The limit of detection (LoD) was determined for each analyte/group on the ePlex BCID-GN Panel using samples prepared in negative blood culture matrix consisting of EDTA human whole blood (amount consistent with the bottle manufacturer recommendations) spiked into a blood culture bottle and incubated for a minimum of eight hours in a continuously monitoring blood culture instrument. Quantitation of organism preparations was determined in CFU/mL using plating and colony count methods. Analytes were pooled together and serially diluted to create organism mixes containing no more than five organisms in negative blood/blood culture matrix. A minimum of 20 replicates per organism mix were tested to confirm the final LoD for each assay target. The limit of detection was defined as the lowest concentration at which each target is detected at least 95% of the time.

Results from the study demonstrated that all targets were detected in ≥95% of tested replicates at $\leq 1 \times 10^8$ CFU/mL for bacterial and resistance marker targets and $\leq 1 \times 10^6$ CFU/mL for fungal targets. It is noted that the LoD for *Fusobacterium necrophorum* approximates the concentration for this organism present at bottle positivity.

The confirmed LoD for ePlex BCID-GN Panel organism and resistance marker targets are shown in Table 19.

Table 19: LoD results summary

Target	Organism	Strain	LoD Concentration (CFU/mL)
Acinetobacter baumannii	*Acinetobacter baumannii*	NCTC 13421	1×10^6
	Acinetobacter baumannii	NCTC 13304	1×10^6
Bacteroides fragilis	*Bacteroides fragilis*	ATCC 25285	1×10^5
	Bacteroides fragilis	ATCC 43860	1×10^4
Citrobacter	*Citrobacter freundii*	NCTC 9750	1×10^6
	Citrobacter koseri	ATCC 27156	1×10^6
Cronobacter sakazakii	*Cronobacter sakazakii*	ATCC 29544	1×10^5
	Cronobacter sakazakii	ATCC 29004	1×10^6
Enterobacter (non-*cloacae* complex)	*Enterobacter aerogenes*	CDC#0074	1×10^6
	Enterobacter aerogenes	CDC#0161	1×10^5
	Enterobacter amnigenus	ATCC 33072	1×10^6
Enterobacter cloacae Complex	*Enterobacter cloacae*	CDC#0154	1×10^6
	Enterobacter asburiae	ATCC 35957	1×10^6

Target	Organism	Strain	LoD Concentration (CFU/mL)
	Enterobacter hormaechei	ATCC BAA-2082	1×10^6
Escherichia coli	*Escherichia coli*	CDC#0118	1×10^7
	Escherichia coli	NCTC 13441	1×10^6
Fusobacterium necrophorum	*Fusobacterium necrophorum*	ATCC 51357	1×10^8
	Fusobacterium necrophorum	ATCC 27852	1×10^7
Fusobacterium nucleatum	*Fusobacterium nucleatum*	ATCC 25586	1×10^7
	Fusobacterium nucleatum	ATCC 23726	1×10^5
Haemophilus influenzae	*Haemophilus influenzae*	ATCC 19418	1×10^5
	Haemophilus influenzae	ATCC 9006	1×10^7
Klebsiella oxytoca	*Klebsiella oxytoca*	ATCC 43165	1×10^7
	Klebsiella oxytoca	ATCC 8724	1×10^7
Klebsiella pneumoniae group	*Klebsiella pneumoniae*	CDC#0160	1×10^6
	Klebsiella pneumoniae	CDC#0107	1×10^6
Morganella morganii	*Morganella morganii*	ATCC 25829	1×10^7
	Morganella morganii	CDC#0133	1×10^7
Neisseria meningitidis	*Neisseria meningitidis*	ATCC 13090	1×10^5
	Neisseria meningitidis	ATCC 13102	1×10^4
Proteus	*Proteus vulgaris*	ATCC 6896	1×10^7
	Proteus vulgaris	ATCC 6380	1×10^7
Proteus mirabilis	*Proteus mirabilis*	CDC#0159	1×10^6
	Proteus mirabilis	ATCC 43071	1×10^6
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	CDC#0103	1×10^6
	Pseudomonas aeruginosa	NCTC 13437	1×10^6
Salmonella	*Salmonella bongori*	ATCC 43975	1×10^5
	Salmonella enterica	ATCC 6962	1×10^5
Serratia	*Serratia liquefaciens*	ATCC 27592	1×10^6
	Serratia plymuthica	ATCC 53858	1×10^7
Serratia marcescens	*Serratia marcescens*	ATCC 14041	1×10^7
	Serratia marcescens	ATCC 14756	1×10^5
Stenotrophomonas maltophilia	*Stenotrophomonas maltophilia*	ATCC 13637	1×10^6
	Stenotrophomonas maltophilia	ATCC 17666	1×10^7
Pan Candida	*Candia albicans*	ATCC 10231	1×10^6
	Candia glabrata	ATCC 15126	1×10^5
Pan Gram-Positive	*Enterococcus faecalis*	ATCC 51575	1×10^5
	Enterococcus faecium	ATCC 31282	1×10^7
	Bacillus subtilis	ATCC 21008	1×10^6
	Staphylococcus aureus	ATCC BAA-2313	1×10^5
	Streptococcus agalactiae	ATCC 13813	1×10^6
	Streptococcus anginosus	ATCC 33397	1×10^6

Target	Organism	Strain	LoD Concentration (CFU/mL)
CTX-M	*Escherichia coli*	NCTC 13441	1×10^4
	Klebsiella pneumoniae	CDC#0107	1×10^5
IMP	*Enterobacter aerogenes*	CDC#0161	1×10^6
	Pseudomonas aeruginosa	CDC#0103	1×10^5
KPC	*Enterobacter hormaechei*	ATCC BAA-2082	1×10^6
	Morganella morganii	CDC#0133	1×10^6
NDM	*Escherichia coli*	CDC#0118	1×10^5
	Proteus mirabilis	CDC#0159	1×10^5
OXA	*Acinetobacter baumannii* (OXA-23)	NCTC 13421	1×10^5
	Acinetobacter baumannii (OXA-27)	NCTC 13304	1×10^5
	Enterobacter aerogenes (OXA-48)	CDC#0074	1×10^6
	Klebsiella pneumoniae (OXA-48)	CDC#0160	1×10^6
VIM	*Enterobacter cloacae*	CDC#0154	1×10^6
	Pseudomonas aeruginosa	NCTC 13437	1×10^5

f. Analytical Reactivity (Inclusivity)

The analytical reactivity of the BCID-GN Panel was evaluated using a collection of 336 bacterial and fungal strains/isolates. Organisms were tested at concentrations expected at bottle positivity with bacteria tested at 1×10^9 CFU/mL or lower and fungal strains tested at 1×10^6 CFU/mL. For organisms that were not evaluated in the bottle positivity study, test samples were prepared at 4.5×10^8 CFU/mL (i.e., average concentrations for all gram-negative organisms evaluated in the bottle positivity study). Each strain evaluated for inclusivity was tested in triplicate and if not detected in initial testing, the strain was retested at a higher concentration. If detected at the high concentration(s), the species/isolate is indicated as detected with reduced sensitivity and the concentration is indicated in footnotes under Table 20 below. All strains evaluated generated 'detected' results by the ePlex BCID-GN Panel with the following exceptions:

- Strains of *Citrobacter amalonaticus*, *Citrobacter farmer*, *Citrobacter gillenii*, *Citrobacter murliniae* and *Citrobacter sedlakii* were not detected by the BCID-GN Panel at any concentration evaluated (up to 1×10^9 CFU/mL).

- *Enterobacter kobei* and *Enterobacter cancerogenous* were not detected by the ePlex BCID-GN Panel.

- *Enterobacter cowanii* was not detected by the ePlex BCID-GN *Enterobacter* non-*cloacae* complex assay. Instead, this species generated detected results with the *Enterobacter cloacae* complex. It is noted that this species is not included in the *Enterobacter cloacae* complex.

- The Pan Gram-positive assay target may have reduced sensitivity for

Streptococcus thoraltensis and *Enterococcus saccharolyticus* (1/6 replicates detected for both organisms at 1 x 10^7 CFU/mL). Additional testing at 4 x 10^8 (*Streptococcus thoraltensis*) and 1x 10^8 (*Enterococcus saccharolyticus*) generated 3/3 replicates detected for the Pan Gram-positive target.

- For one strain each of *Enterobacter amnigenus* and *Morganella morganii*, two of three initially tested replicates were detected. Additional testing for each strain generated positive results for three additional replicates for a total of five of six replicates tested.

- One strain of *Serratia odorifera* was not detected at 1 x 10^8 CFU/mL and was detected in only one of three replicates at 1 x 10^9 CFU/mL. *In silico* analysis suggest that this species will not be detected by the BCID-GN Panel.

- In initial testing of samples prepared at 1 x 10^6 CFU/mL, the Pan *Candida* target was detected for one of six replicate samples containing *Candida krusei* (ATCC 34135) and two of six replicates of *Candida parapsilosis* (ATCC 90018). Additional testing was performed with blood culture samples from culture bottles grown on a blood culture instrument until bottle positivity. For this additional testing, the Pan *Candida* target was detected for all six replicates of each Candida strain.

Organisms and associated resistance markers evaluated in the analytical reactivity study and detected by the ePlex BCID-GN Panel are shown in Table 20. Additional strains were evaluated as part of the LoD study (See Table 19).

Table 20: Analytical Reactivity (Inclusivity)

Organism	Strain
Acinetobacter baumannii	
Acinetobacter baumannii	CDC#0052
	NCTC 13302
	NCTC 13303
	NCTC 13305
	NCTC 13420
	NCTC 13422
	NCTC 13423
Acinetobacter baumannii (NDM-1)	CDC#0033
Acinetobacter baumannii (OXA-23)	ATCC BAA-1605
	CDC#0045
	CDC#0056
	NCTC 13301
	NCTC 13424
Acinetobacter spp. (IMP only)	JMI4084[A]
Bacteroides fragilis	
Bacteroides fragilis	ATCC 23745
	ATCC 700786
	NCTC 9343

Organism	Strain
Citrobacter	
Citrobacter braakii	ATCC 43162
	ATCC 51113
Citrobacter freundii	ATCC 6879
	ATCC 8090
Citrobacter freundii (CTX)	JMI2047
Citrobacter freundii (KPC-2)	CDC#0116
Citrobacter koseri	ATCC 25409
	ATCC 27028
	ATCC 29225
	ATCC 29936
Citrobacter species (CTX-15, NDM-1)	CDC #0157
Citrobacter werkmanii	ATCC 51114
Citrobacter youngae	ATCC 29935
Cronobacter sakazakii	
Cronobacter sakazakii	ATCC 12868
	ATCC BAA-894
	FSL F6-0023

Organism	Strain
***Enterobacter* (non-cloacae complex)**	
Enterobacter aerogenes	ATCC 13048
	ATCC 29010
	ATCC 51697
Enterobacter amnigenus	ATCC 33731
	ATCC 51816[B]
Enterobacter gergoviae	ATCC 33028
	ATCC 33426
***Enterobacter cloacae* complex**	
Enterobacter asburiae	ATCC 35954
	ATCC 35955
	ATCC 35956
Enterobacter cloacae (CTX-15)	CDC#0038
Enterobacter cloacae (CTX-9)	NCTC 13464
Enterobacter cloacae (CTX-15, KPC-2)	CDC#0163
Enterobacter cloacae (CTX, NDM)	JMI53571
Enterobacter cloacae subsp. *cloacae*	ATCC 23355
	ATCC 35030
Enterobacter cloacae subsp. *dissolvens*	ATCC 23373
Enterobacter hormaechei	ATCC 700323
Enterobacter hormaechei subsp. *hormaechei*	ATCC 49162
Enterobacter hormaechei subsp. *oharae*	ATCC 49163
Enterobacter hormaechei subsp. *steigerwaltii*	CIP108489T
Enterobacter ludwigii	DSM-16688
Escherichia coli	
Escherichia coli	ATCC 14948
	ATCC 25922
	ATCC 33605
	ATCC 33876
	ATCC 35150
	ATCC 4157
	ATCC 43888
	ATCC 51446
	ATCC 51755
	ATCC 53498
	ATCC 700728
	NCIMB 8545
	NCTC 8620
	ATCC 9637
	ATCC BAA-196
	ATCC BAA-197
	ATCC BAA-198
	ATCC BAA-199
	ATCC BAA-200
	ATCC BAA-201
	ATCC BAA-202

Organism	Strain
	ATCC BAA-203
	ATCC BAA-204
	LMC_243094647
	LMC_243098776
	LMC_243098947
	LMC_243108047
	LMC_243109799
	LMC_243112411
	LMC_244006281
	LMC_244006433
	LMC_244008038
	LMC_244012579
	NCTC 13351
	NCTC 10279
	ATCC 10536
	ATCC 10538
	ATCC 10799
	ATCC 11229
	ATCC 13762
	ATCC 14169
Escherichia coli (CTX-14)	CDC#0086
Escherichia coli (CTX-15)	ATCC BAA-2326
	NCTC 13353
	NCTC 13400
	NCTC 13450
	NCTC 13451
Escherichia coli (CTX-3)	NCTC 13452
Escherichia coli (CTX-1)	NCTC 13461
Escherichia coli (CTX-2)	NCTC 13462
Escherichia coli (CTX-8)	NCTC 13463
Escherichia coli (CTX-15, NDM-6)	CDC#0137
Escherichia coli (CTX-15, NDM-7)	CDC#0162
Escherichia coli (IMP)	NCTC 13476
Escherichia coli (KPC)	ATCC BAA-2340
Escherichia coli (NDM-5)	CDC#0150
Escherichia coli (OXA)	LMC_DR00012
Escherichia coli (VIM)	JMI32465
Fusobacterium necrophorum	
Fusobacterium necrophorum subsp. *necrophorum*	ATCC 25286
	NCTC 10575
	NCTC 10577
Fusobacterium nucleatum	
Fusobacterium nucleatum subsp. *nucleatum*	ATCC 31647
Fusobacterium nucleatum subsp. *fusiforme*	ATCC 51190
Fusobacterium nucleatum subsp. *vincentii*	ATCC 49256
Haemophilus influenzae	

Organism	Strain
Haemophilus influenzae	ATCC 33930
	ATCC 43065
	ATCC 43163
	NCTC 11931
	NCTC 12699
	NCTC 8143
Haemophilus influenzae Type b	ATCC 10211
Haemophilus influenzae Type c	ATCC 9007
Haemophilus influenzae Type d	ATCC 9332
Haemophilus influenzae Type e	NCTC 8472
Haemophilus influenzae Type f	ATCC 9833
Klebsiella oxytoca	
Klebsiella oxytoca	ATCC 43086
	ATCC 43863
	ATCC 49131
	ATCC 700324
	ATCC 51817
Klebsiella oxytoca (KPC-3)	CDC#0147
Klebsiella pneumoniae group	
Klebsiella pneumoniae (CTX-15)	CDC#0109
Klebsiella pneumoniae (CTX-25)	NCTC 13465
Klebsiella pneumoniae (CTX, KPC)	IMH-C2261309
Klebsiella pneumoniae (CTX, NDM-1)	NCTC 13443
Klebsiella pneumoniae (CTX-15; NDM-1; OXA-232)	CDC#0153
Klebsiella pneumoniae (CTX-15, OXA-232)	CDC#0075
	CDC#0066
Klebsiella pneumoniae (CTX-15, OXA-181)	CDC#0039
	CDC#0140
	CDC#0141
	CDC#0142
Klebsiella pneumoniae (IMP-4)	CDC#0034
	CDC#0080
Klebsiella pneumoniae (KPC-3)	CDC#0125
	CDC#0112
	CDC#0113
Klebsiella pneumoniae (KPC)	ATCC BAA-1705
	IMH-C2260742
	IMH-C3151729
	IMH-C4151728
	IMH-C4171868
Klebsiella pneumoniae (OXA-48)	NCTC 13442

Organism	Strain
Klebsiella pneumoniae (CTX-15; VIM-27)	CDC#0040
Klebsiella pneumoniae (VIM-1)	CDC#0135
	NCTC 13439
	NCTC 13440
Klebsiella pneumoniae subsp. *Ozaenae*	ATCC 11296
Klebsiella pneumoniae subsp. *pneumoniae*	ATCC 13883
	ATCC 27736
	ATCC 51503
	ATCC 51504
Klebsiella quasipneumoniae	ATCC 700603
Klebsiella pneumoniae subsp. *rhinoscleromatis*	ATCC 9436
Klebsiella variicola	ATCC BAA-830
Morganella morganii	
Morganella morganii	ATCC 25830
	GM148-209
Morganella morganii (CTX-15; NDM-1)	CDC#0057[C]
Neisseria meningitidis	
Neisseria meningitidis Serotype A	ATCC 13077
Neisseria meningitidis Serotype B	NCTC 10026
Neisseria meningitidis Serotype W135	NCTC 11203
Neisseria meningitidis Serotype Y	ATCC 35561
Proteus	
Proteus hauseri	ATCC 13315
Proteus mirabilis	ATCC 33583
	ATCC BAA-663
Proteus mirabilis (IMP)	JMI955389
Proteus mirabilis (KPC-6)	CDC#0155
Proteus penneri	ATCC 35197
Proteus vulgaris	ATCC 33420
	ATCC 49132
	ATCC 8427
	NCTC 4636
Pseudomonas aeruginosa	
Pseudomonas aeruginosa (IMP-14)	CDC#0092
Pseudomonas aeruginosa (IMP-1)	CDC#0241
Pseudomonas aeruginosa (IMP)	CDC#0439
Pseudomonas aeruginosa (KPC-5)	CDC#0090
Pseudomonas aeruginosa (VIM-2)	CDC#0100
Pseudomonas aeruginosa (VIM-4)	CDC#0054

Organism	Strain
Salmonella	
Salmonella enterica serovar *4,[5],12:i*	FSL S5-0580
Salmonella enterica serovar *Agona*	ATCC 51957
Salmonella enterica serovar *Bareilly*	ATCC 9115
Salmonella enterica serovar *Braenderup*	ATCC 700136
Salmonella enterica serovar *Enteritidis*	ATCC BAA-708
Salmonella enterica serovar *Hadar*	ATCC 51956
Salmonella enterica serovar *Heidelberg*	ATCC 8326
Salmonella enterica serovar *Infantis*	ATCC BAA-1675
Salmonella enterica serovar *Javiana*	ATCC 10721
Salmonella enterica serovar *Montevideo*	ATCC 8387
Salmonella enterica serovar *Muenchen*	ATCC 8388
Salmonella enterica serovar *Oranienburg*	ATCC 9239
Salmonella enterica serovar *Paratyphi B*	FSL S5-0447
Salmonella enterica serovar *Saintpaul*	ATCC 9712
Salmonella enterica serovar *Thompson*	ATCC 8391
Salmonella enterica serovar *Typhi*	ATCC 19430
Salmonella enterica subsp. *arizonae*	ATCC 13314
Salmonella enterica subsp. *diarizonae*	ATCC 12325
Salmonella enterica subsp. *enterica* serovar *Typhimurium*	ATCC 14028
Salmonella enterica subsp. *houtenae*	ATCC 29834
Salmonella enterica subsp. *indica*	ATCC BAA-1578
Salmonella enterica subsp. *salamae*	ATCC 6959
Salmonella enterica subsp. *enterica* serovar *Mississippi*	FSL A4-0633
Salmonella enterica subsp. *enterica* serovar *Schwarzengrund*	FSL S5-0458
Serratia	
Serratia ficaria	ATCC 33105
Serratia fonticola	ATCC 29844
Serratia grimesii	ATCC 14460
Serratia marcescens	ATCC 13880
	ATCC 43861
	ATCC 43862
Serratia marcescens (CTX)	JMI10244
Serratia rubidaea	ATCC 27593
	ATCC 29025
Stenotrophomonas maltophilia	
Stenotrophomonas maltophilia	ATCC 13636
	GM148-207
	GM148-208
Pan-Gram Positive	
Bacillus amyloliquefaciens	ATCC 23845
Bacillus atrophaeus	ATCC 49337
Bacillus cereus	ATCC 10876
Bacillus licheniformis	ATCC 21039
Bacillus thuringiensis	ATCC 35646
Enterococcus avium	ATCC 14025
Enterococcus casseliflavus	ATCC 700327
Enterococcus faecalis	JMI876745
Enterococcus gallinarum	ATCC 49573
Enterococcus hirae	ATCC 49479
Enterococcus raffinosus	ATCC 49464
Enterococcus saccharolyticus	ATCC 43076[D]
Staphylococcus capitis	NRS866
Staphylococcus chromogenes	ATCC 43764
Staphylococcus cohnii	ATCC 29974
Staphylococcus epidermidis	ATCC 35984
Staphylococcus gallinarum	ATCC 700401
Staphylococcus haemolyticus	ATCC 29970
Staphylococcus hominis	ATCC 27844
Staphylococcus hyicus	ATCC 11249
Staphylococcus lentus	ATCC 700403
Staphylococcus lugdunensis	ATCC 49576
Staphylococcus pasteuri	ATCC 51128
Staphylococcus vitulinus	ATCC 51699
Streptococcus constellatus	ATCC 27513
Streptococcus dysgalactiae	ATCC 35666
Streptococcus equi	ATCC 9528
Streptococcus gallolyticus	ATCC 9809
Streptococcus gordonii	ATCC 35557

Organism	Strain
Streptococcus infantis	ATCC 700779
Streptococcus intermedius	ATCC 27335
Streptococcus mitis	ATCC 49456
Streptococcus oralis	ATCC 35037
Streptococcus parasanguinis	ATCC 15909
Streptococcus pneumoniae	ATCC 8335
Streptococcus pyogenes	ATCC 12344
Streptococcus salivarius	ATCC 7073
Streptococcus thoraltensis	ATCC 700865[E]

Organism	Strain
Pan Candida	
Candida albicans	ATCC 24433
	ATCC 90028
Candida glabrata	ATCC 2001
	ATCC 66032
Candida krusei	ATCC 14243
	ATCC 32196
	ATCC 34135[F]
Candida parapsilosis	ATCC 22019
	ATCC 58895
	ATCC 90018[G]

A. Non-target species used to evaluate resistance marker.
B. 5/6 replicates detected at 2.0 x 10^8 CFU/mL.
C. 5/6 replicates detected at 4.5 x 10^8 CFU/mL.
D. Strain may have reduced sensitivity and was not 100% detected at concentrations <1 x 10^8 CFU/mL.
E. Strain may have reduced sensitivity and was not 100% detected at concentrations <4 x 10^8 CFU/mL.
F. In initial testing, 1/6 replicates was detected at 1x 10^6 CFU/mL; during additional testing, 3/3 replicates detected at bottle positivity.
G. In initial testing, 2/6 replicates were detected at 1 x 10^6 CFU/mL; during additional testing, 6/6 replicates detected at bottle positivity.

Predicted reactivity (*In silico* analysis), genus and group assays

In silico analysis was performed for the broader genus or group-level assays of the ePlex BCID-GN Panel including: *Citrobacter*, *Enterobacter cloacae* complex, *Enterobacter* non-*cloacae* complex, *Proteus*, *Serratia*, Pan *Candida*, and Pan Gram-Positive assays. Tables 21-28 below include results from this analysis including species for which detection is predicted, species for which detection is predicted for a percentage of strain sequences, and species for which detection is not predicted.

Note: The performance of the ePlex BCID-GN Panel has not been established for the organisms listed in the tables below with the exception of species that were evaluated in the analytical reactivity study which are indicated with an asterisk.

Table 21: Predicted (*in silico*) reactivity for *Citrobacter*

Detection Predicted for ≥95% of target sequences		
*Citrobacter koseri**	*Citrobacter intermedius*	*Citrobacter amalonaticus*[A]*
Detection Predicted for 85%-94% of target sequences		
*Citrobacter freundii**	*Citrobacter braakii**	
Detection Predicted for <85.0% of target sequences		
*Citrobacter werkmanii** (66.7%)	*Citrobacter youngae** (50.0%)	
Detection Not Predicted		
Citrobacter europaeus	*Citrobacter gillenii*	

A. Detection predicted *in silico*, however ATCCBAA-2563 was not detected in wet testing. See **Analytical Reactivity (Inclusivity)** Study.

Table 22: Predicted (*in silico*) reactivity for *Enterobacter cloacae* complex

Detection Predicted for ≥95% of target sequences		
*Enterobacter cloacae**	*Enterobacter asburiae**	*Enterobacter hormaechei**
Enterobacter xiangfangensis		
Detection Predicted for 85%-94% of target sequences		
None Identified		
Detection Predicted for <85.0% of target sequences		
*Enterobacter ludwigii** (68.4%)	*Enterobacter nimipressuralis* (25.0%)	
Detection Not Predicted		
Enterobacter kobei	*Enterobacter cancerogenus*	

Table 23: Predicted (*in silico*) reactivity for *Enterobacter* (non-*cloacae* complex)

Detection Predicted for ≥95% of target sequences		
*Enterobacter aerogenes**	*Enterobacter gergoviae**	
Detection Predicted for 85%-94% of target sequences		
None Identified		
Detection Predicted for <85.0% of target sequences		

*Enterobacter amnigenus** (62.5%)		
Detection Not Predicted		
None Identified		

Table 24: Predicted (*in silico*) reactivity for *Klebsiella pneumoniae* group

Detection Predicted for ≥95% of target sequences		
*Klebsiella pneumoniae**	*Klebsiella quasipneumoniae**	*Klebsiella variicola**
Detection Predicted for 85%-94% of target sequences		
None Identified		
Detection Predicted for <85.0% of target sequences		
None Identified		
Detection Not Predicted		
None Identified		

Table 25: Predicted (*in silico*) reactivity for *Proteus*

Detection Predicted for ≥95% of target sequences		
*Proteus mirabilis**	*Proteus penneri**	*Proteus vulgaris**
*Proteus hauseri**	*Proteus cibarius*	
Detection Predicted for 85%-94% of target sequences		
None Identified		
Detection Predicted for <85.0% of target sequences		
None Identified		
Detection Not Predicted		
Proteus myxofaciens		

Table 26: Predicted (*in silico*) reactivity for S*erratia*

Detection Predicted for ≥95% of target sequences		
*Serratia marcescens**	*Serratia grimesii**	*Serratia rubidaea**
*Serratia ficaria**	*Serratia liquefaciens**	*Serratia proteamaculans*
*Serratia fonticola**	*Serratia plymuthica**	
Detection Predicted for 85%-94% of target sequences		
None Identified		
Detection Predicted for <85.0% of target sequences		
Serratia quinivorans (33.3%)		
Detection Not Predicted		
Serratia nematodiphila	*Serratia odorifera*[A*]	*Serratia ureilytica*

A. Not predicted *in silico*, however ATCC 33077 was intermittently detected in wet testing (1/3 replicates detected). See Analytical Reactivity (Inclusivity) Study.

Table 27: Predicted (*in silico*) reactivity for Pan *Candida*

Detection Predicted for ≥95% of target sequences		
*Candida albicans**	*Candida glabrata**	*Candida krusei**
*Candida parapsilosis**		
Detection Predicted for 85%-94% of target sequences		
None Identified		
Detection Predicted for <85% of target sequences		
None Identified		
Detection Not Predicted		
Detection was not predicted by bioinformatic analysis for other *Candida* species for which sequence data was available.		

Table 28: Predicted (*in silico*) reactivity for Pan Gram-Positive

Detection Predicted for ≥95% of target sequences		
Bacillus		
*Bacillus amyloliquefaciens**	*Bacillus paralicheniformis*	*Bacillus toyonensis*
*Bacillus atrophaeus**	*Bacillus siamensis*	*Bacillus vallismortis*
Bacillus bombysepticus	*Bacillus subtilis**	*Bacillus velezensis*
*Bacillus licheniformis**	*Bacillus tequilensis*	*Bacillus weihenstephanensis*
Bacillus methylotrophicus	*Bacillus thuringiensis**	
Enterococcus		
*Enterococcus avium**	*Enterococcus faecium**	*Enterococcus raffinosus**
Enterococcus dispar	*Enterococcus flavescens*	*Enterococcus saccharolyticus**
Enterococcus durans	*Enterococcus malodoratus*	*Enterococcus thailandicus*
*Enterococcus faecalis**	*Enterococcus pseudoavium*	
Staphylococcus		
*Staphylococcus aureus**	*Staphylococcus haemolyticus**	*Staphylococcus pseudintermedius*
Staphylococcus agnetis	*Staphylococcus hominis**	*Staphylococcus pseudolugdunensis*
Staphylococcus argensis	*Staphylococcus hominis* subsp. *novobiosepticus*	*Staphylococcus pulvereri*
Staphylococcus argenteus	*Staphylococcus hyicus**	*Staphylococcus rostri*
Staphylococcus auricularis	*Staphylococcus jettensis*	*Staphylococcus saprophyticus*
*Staphylococcus capitis**	*Staphylococcus kloosii*	*Staphylococcus schleiferi*
Staphylococcus caprae	*Staphylococcus lentus**	*Staphylococcus schweitzeri*
Staphylococcus carnosus	*Staphylococcus lugdunensis**	*Staphylococcus sciuri*
*Staphylococcus chromogenes**	*Staphylococcus lutrae*	*Staphylococcus simiae*

*Staphylococcus cohnii**	*Staphylococcus massiliensis*	*Staphylococcus simulans*
Staphylococcus delphini	*Staphylococcus microti*	*Staphylococcus stepanovicii*
Staphylococcus devriesei	*Staphylococcus muscae*	*Staphylococcus succinus*
*Staphylococcus epidermidis**	*Staphylococcus nepalensis*	*Staphylococcus vitulinus**
Staphylococcus equorum	*Staphylococcus pasteuri**	*Staphylococcus warneri*
Staphylococcus felis	*Staphylococcus petrasii*	*Staphylococcus xylosus*
Staphylococcus fleurettii	*Staphylococcus pettenkoferi*	
*Staphylococcus gallinarum**	*Staphylococcus piscifermentans*	
Streptococcus		
Streptococcus agalactiae	*Streptococcus infantarius*	*Streptococcus phocae*
Streptococcus alactolyticus	*Streptococcus infantis**	*Streptococcus pneumoniae**
Streptococcus anginosus	*Streptococcus intermedius**	*Streptococcus porcinus*
Streptococcus australis	*Streptococcus intestinalis*	*Streptococcus porcorum*
Streptococcus caballi	*Streptococcus lactarius*	*Streptococcus pseudopneumoniae*
Streptococcus constellatus	*Streptococcus loxodontisalivarius*	*Streptococcus pseudoporcinus*
Streptococcus criceti	*Streptococcus luteciae*	*Strepotococcus pyogenes*
Streptococcus cristatus	*Streptococcus lutetiensis*	*Streptococcus rifensis*
Streptococcus danieliae	*Streptococcus macedonicus*	*Streptococcus rubneri*
Streptococcus dentasini	*Streptococcus marimammalium*	*Streptococcus salivarius**
Streptococcus dentisani	*Streptococcus massiliensis*	*Streptococcus saliviloxodontae*
Streptococcus didelphis	*Streptococcus mitis**	*Streptococcus sanguinis*
Streptococcus difficilis	*Streptococcus moroccensis*	*Streptococcus seminale*
Streptococcus dysgalactiae subsp. *dysgalactiae*	*Streptococcus oligofermentans*	*Streptococcus sinensis*
Streptococcus dysgalactiae subsp. *equisimilis*	*Streptococcus oralis**	*Streptococcus suis*
*Streptococcus dysgalactiae**	*Streptococcus oricebi*	*Streptococcus thermophilus*
*Streptococcus equi**	*Streptococcus orisratti*	*Streptococcus thoraltensis**
Streptococcus equinus	*Streptococcus panodentis*	*Streptococcus tigurinus*
Streptococcus fryi	*Streptococcus parasanguinis**	*Streptococcus troglodytae*
*Streptococcus gallolyticus**	*Streptococcus parasuis*	*Streptococcus troglodytidis*
*Streptococcus gordonii**	*Streptococcus parauberis*	*Streptococcus urinalis*
Streptococcus himalayensis	*Streptococcus pasteuri*	*Streptococcus ursoris*
Streptococcus hongkongensis	*Streptococcus pasteurianus*	*Streptococcus vestibularis*
Streptococcus hyointestinalis	*Streptococcus peroris*	*Streptococcus waiu*
Detection Predicted for 85%-94% of target sequences		
*Bacillus cereus**	*Enterococcus hirae**	*Staphylococcus saccharolyticus*

*Enterococcus casseliflavus**	*Staphylococcus arlettae*	*Streptococcus bovis*
Enterococcus cecorum	*Staphylococcus condimenti*	*Streptococcus uberis*
Enterococcus gallinarum	*Staphylococcus intermedius*	
Detection Predicted for <85.0% of target sequences		
Bacillus mojavensis (77.8%)	*Bacillus sonorensis (83.3%)*	*Streptococcus halichoeri (66.7%)*
Streptococcus ratti (75.0%)		
Detection Not Predicted		
Bacillus pseudomycoides	*Enterococcus sulfureus*	*Streptococcus hyovaginalis*
Enterococcus aquimarinus	*Enterococcus termitis*	*Streptococcus ictaluri*
Enterococcus asini	*Enterococcus ureasiticus*	*Streptococcus iniae*
Enterococcus caccae	*Enterococcus ureilyticus*	*Streptococcus lactis*
Enterococcus camelliae	*Enterococcus villorum*	*Streptococcus macacae*
Enterococcus canintestini	*Staphylococcus caseolyticus*	*Streptococcus marmotae*
Enterococcus canis	*Streptococcus acidominimus*	*Streptococcus merionis*
Enterococcus columbae	*Streptococcus azizii*	*Streptococcus milleri*
Enterococcus devriesei	*Streptococcus cameli*	*Streptococcus minor*
Enterococcus haemoperoxidus	*Streptococcus canis*	*Streptococcus oriloxodontae*
Enterococcus hawaiiensis	*Streptococcus castoreus*	*Streptococcus orisasini*
Enterococcus hermanniensis	*Streptococcus cremoris*	*Streptococcus orisuis*
Enterococcus italicus	*Streptococcus criae*	*Streptococcus ovis*
Enterococcus mundtii	*Streptococcus cuniculi*	*Streptococcus pharyngis*
Enterococcus pallens	*Streptococcus dentapri*	*Streptococcus pluranimalium*
Enterococcus pernyi	*Streptococcus dentiloxodontae*	*Streptococcus plurextorum*
Enterococcus phoeniculicola	*Streptococcus dentirousetti*	*Streptococcus plutanimalium*
Enterococcus plantarum	*Streptococcus devriesei*	*Streptococcus porci*
Enterococcus quebecensis	*Streptococcus downei*	*Streptococcus rupicaprae*
Enterococcus ratti	*Streptococcus entericus*	*Streptococcus sobrinus*
Enterococcus rivorum	*Streptococcus ferus*	*Streptococcus tangierensis*
Enterococcus rotai	*Streptococcus gallinaceus*	
Enterococcus silesiacus	*Streptococcus henryi*	

Predicted Reactivity (*In silico* analysis) for resistance markers

Known variants of the six resistance markers reported by the ePlex BCID-GN Panel were each assessed for predicted reactivity using *in silico* analysis. BLAST searches to support *in silico* analyses were conducted in March 2018; therefore, any new resistance gene variants deposited into GenBank after March 2018 have not been evaluated.

Tables 29-39 below describe the predicted reactivity for CTX-M, IMP, KPC, NDM, OXA, and VIM resistance marker assays of the ePlex BCID GN Panel. Strains that were tested as part of the Inclusivity study are marked with an asterisk. Table 40 includes all variants that are not predicted to be detected by the ePlex BCID-GN Panel.

The CTX-M assay on the ePlex BCID-GN Panel is designed to detect the following CTX-M groups: CTX-M-1, CTX-M-2, CTX-M-8, CTX-M-9, and CTX-M-25. Results from *in silico* results for variants included in each of these CTX-M groups are presented in Table 29-33.

Table 29: Predicted (*in silico*) reactivity for CTX-M-1 group

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	CTX-M-15
Citrobacter	*Citrobacter freundii*	CTX-M-3
		CTX-M-15*
		CTX-M-30
		CTX-M-55
	Citrobacter koseri	CTX-M-3
		CTX-M-15
Enterobacter (non-*cloacae* Complex)	*Enterobacter aerogenes*	CTX-M-3
		CTX-M-15
	Enterobacter gergoviae	CTX-M-15
Enterobacter cloacae Complex	*Enterobacter asburiae*	CTX-M-15
	Enterobacter cloacae	CTX-M-3
		CTX-M-15*
		CTX-M-22
		CTX-M-37
		CTX-M-89
		CTX-M-177
	Enterobacter hormaechei	CTX-M-15
Escherichia coli	*Escherichia coli*	CTX-M-1*
		CTX-M-3*
		CTX-M-10
		CTX-M-12
		CTX-M-14*
		CTX-M-15*
		CTX-M-22
		CTX-M-23
		CTX-M-28
		CTX-M-29
		CTX-M-32
		CTX-M-33
		CTX-M-34
		CTX-M-36
		CTX-M-38
		CTX-M-42
		CTX-M-55
		CTX-M-58
		CTX-M-61
		CTX-M-65
		CTX-M-69
		CTX-M-79
		CTX-M-82
		CTX-M-101
		CTX-M-103
		CTX-M-117
		CTX-M-123
		CTX-M-127
		CTX-M-132
		CTX-M-138
		CTX-M-139
		CTX-M-142
		CTX-M-144
		CTX-M-150
		CTX-M-158
		CTX-M-163
		CTX-M-166
		CTX-M-169
		CTX-M-170
		CTX-M-171
		CTX-M-172
		CTX-M-174
		CTX-M-175
		CTX-M-179
		CTX-M-180
		CTX-M-181
		CTX-M-182
		CTX-M-184
		CTX-M-186
Klebsiella oxytoca	*Klebsiella oxytoca*	CTX-M-3
		CTX-M-15
		CTX-M-28
		CTX-M-162

Target	Associated Organism	Variant Detected
Klebsiella pneumoniae group	*Klebsiella pneumoniae*	CTX-M-1
		CTX-M-3
		CTX-M-10
		CTX-M-11
		CTX-M-12
		CTX-M-15*
		CTX-M-22
		CTX-M-28
		CTX-M-32
		CTX-M-52
		CTX-M-54
		CTX-M-55
		CTX-M-57
		CTX-M-60
		CTX-M-62
		CTX-M-71
		CTX-M-72
		CTX-M-96
		CTX-M-155
		CTX-M-156
		CTX-M-157
		CTX-M-173
		CTX-M-176
		CTX-M-183
		CTX-M-197
		CTX-M-204
Morganella morganii	*Morganella morganii*	CTX-M-3
		CTX-M-15*
		CTX-M-55

Target	Associated Organism	Variant Detected
Proteus mirabilis	*Proteus mirabilis*	CTX-M-1
		CTX-M-3
		CTX-M-15
		CTX-M-32
		CTX-M-66
		CTX-M-116
		CTX-M-136
		CTX-M-164
		CTX-M-167
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	CTX-M-15
		CTX-M-28
		CTX-M-32
Salmonella	*Salmonella enterica*	CTX-M-15
		CTX-M-53
		CTX-M-55
		CTX-M-57
		CTX-M-61
		CTX-M-88
	Salmonella sp.	CTX-M-3
	Salmonella typhimurium	CTX-M-37
		CTX-M-61
Serratia	*Serratia liquefaciens*	CTX-M-22
Serratia marcescens/ Serratia	*Serratia marcescens*	CTX-M-3
		CTX-M-15
		CTX-M-55
Stenotrophomonas maltophilia	*Stenotrophomonas maltophilia*	CTX-M-15

Table 30: Predicted (*in silico*) reactivity for CTX-M-2 group

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	CTX-M-2
		CTX-M-5
		CTX-M-43
		CTX-M-115
Enterobacter cloacae complex	*Enterobacter cloacae*	CTX-M-5
Escherichia coli	*Escherichia coli*	CTX-M-2*
		CTX-M-44
		CTX-M-56
		CTX-M-92
		CTX-M-97
		CTX-M-124
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	CTX-M-2
		CTX-M-35
		CTX-M-59
		CTX-M-141
		CTX-M-165

Target	Associated Organism	Variant Detected
		CTX-M-200
Morganella morganii	*Morganella morganii*	CTX-M-2
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	CTX-M-2
		CTX-M-20
		CTX-M-171
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	CTX-1-2
		CTX-M-2
Salmonella	*Salmonella enterica*	CTX-M-2
		CTX-M-4
		CTX-M-5
		CTX-M-6
		CTX-M-7
	Salmonella typhimurium	CTX-M-2
		CTX-M-4
		CTX-M-5
		CTX-M-7

Table 31: Predicted (*in silico*) reactivity for CTX-M-8 group

Target	Associated Organism	Variant Detected
Citrobacter	*Citrobacter amalonaticus*	CTX-M-8
Enterobacter cloacae Complex	*Enterobacter cloacae*	CTX-M-8
Escherichia coli	*Escherichia coli*	CTX-M-8*
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	CTX-M-8
		CTX-M-63
Morganella morganii	*Morganella morganii*	CTX-M-63
Salmonella	*Salmonella enterica*	CTX-M-8

Table 32: Predicted (*in silico*) reactivity for CTX-M-9 group

Target	Associated Organism	Variant Detected
Citrobacter	*Citrobacter freundii*	CTX-M-14
		CTX-M-65
Enterobacter non-*cloacae* Complex	*Enterobacter aerogenes*	CTX-M-9
Enterobacter cloacae Complex	*Enterobacter cloacae*	CTX-M-9*
		CTX-M-13
		CTX-M-14
		CTX-M-64
		CTX-M-125
Escherichia coli	*Escherichia coli*	CTX-M-1/CTX-M-65
		CTX-M-9
		CTX-M-13
		CTX-M-14
		CTX-M-14/CTX-M-15
		CTX-M-15
		CTX-M-16
		CTX-M-19
		CTX-M-21
		CTX-M-24
		CTX-M-27
		CTX-M-38
		CTX-M-47
		CTX-M-51
		CTX-M-64
		CTX-M-65
		CTX-M-67
		CTX-M-73
		CTX-M-82
		CTX-M-87
		CTX-M-93
		CTX-M-97
		CTX-M-98
		CTX-M-102
		CTX-M-104
		CTX-M-105
		CTX-M-106
		CTX-M-121
		CTX-M-122
		CTX-M-126
		CTX-M-129
		CTX-M-130
		CTX-M-132
		CTX-M-134
		CTX-M-137
		CTX-M-148
		CTX-M-161
		CTX-M-168
		CTX-M-173
		CTX-M-174
		CTX-M-176
		CTX-M-177
		CTX-M-191
		CTX-M-195
		CTX-M-196
		CTX-M-198
		CTX-M-199

Target	Associated Organism	Variant Detected
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	CTX-M-9
		CTX-M-13
		CTX-M-14
		CTX-M-17
		CTX-M-18
		CTX-M-19
		CTX-M-24
		CTX-M-38
		CTX-M-46
		CTX-M-48
		CTX-M-49
		CTX-M-50
		CTX-M-65
		CTX-M-81
		CTX-M-99
		CTX-M-104
		CTX-M-147
		CTX-M-159
		CTX-M-201
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	CTX-M-13
		CTX-M-14
		CTX-M-24
		CTX-M-65
		CTX-M-90
Salmonella	*Salmonella enterica*	CTX-M-9
		CTX-M-14
		CTX-M-25
		CTX-M-27
		CTX-M-65
		CTX-M-83
		CTX-M-84
		CTX-M-85
		CTX-M-86
		CTX-M-143
Serratia	*Serratia liquefaciens*	CTX-M-14

Table 33: Predicted (*in silico*) reactivity for CTX-M-25 group

Target	Associated Organism	Variant Detected
Escherichia coli	*Escherichia coli*	CTX-M-25
		CTX-M-39
		CTX-M-94
		CTX-M-100
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*[A]*	CTX-M-26
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	CTX-M-41
		CTX-M-89
		CTX-M-91
		CTX-M-160
Salmonella	*Salmonella enterica*	CTX-M-25

Table 34: Predicted (*in silico*) reactivity for IMP

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	IMP-1
		IMP-2
		IMP-4
		IMP-5
		IMP-8
		IMP-10
		IMP-11
		IMP-14
		IMP-19
		IMP-55
		IMP-61
Citrobacter	*Citrobacter freundii*	IMP-1
		IMP-4
		IMP-8
		IMP-23
		IMP-38
Enterobacter non-*cloacae* Complex	*Enterobacter aerogenes*	IMP-4
Enterobacter cloacae Complex	*Enterobacter cloacae*	IMP-1
		IMP-4
		IMP-8
		IMP-11
		IMP-26
		IMP-34
		IMP-60
Enterobacter cloacae Complex	*Enterobacter hormaechei*	IMP-13
		IMP-14
Escherichia coli	*Escherichia coli*	IMP-1
		IMP-4
		IMP-6

Target	Associated Organism	Variant Detected
		IMP-8
		IMP-14
		IMP-30
		IMP-52
		IMP-59
		IMP-66
Klebsiella oxytoca	*Klebsiella oxytoca*	IMP-1
		IMP-4
		IMP-8
		IMP-28
		IMP-34
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	IMP-1
		IMP-4*
		IMP-6
		IMP-8
		IMP-10
		IMP-13
		IMP-19
		IMP-26
		IMP-32
		IMP-38
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	IMP-1
		IMP-27
		IMP-64
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	IMP-1*
		IMP-2
		IMP-4
		IMP-6
		IMP-7
		IMP-9
		IMP-10
		IMP-11
		IMP-13
		IMP-14*
		IMP-15
		IMP-16
		IMP-17
		IMP-18
		IMP-19
		IMP-20
		IMP-21
		IMP-22
		IMP-25
		IMP-26
		IMP-29
		IMP-30
		IMP-33
		IMP-34
		IMP-37
		IMP-40
		IMP-41
		IMP-43
		IMP-44
		IMP-45
		IMP-48
		IMP-49
		IMP-51
		IMP-53
		IMP-54
		IMP-56
		IMP-62
		IMP-63
Salmonella	*Salmonella enterica*	IMP-4
Serratia marcescens/*Serratia*	*Serratia marcescens*	IMP-1
		IMP-2
		IMP-6
		IMP-8
		IMP-24
Stenotrophomonas maltophilia	*Stenotrophomonas maltophilia*	IMP-25

Table 35: Predicted (*in silico*) reactivity for KPC

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	KPC-2
		KPC-3
		KPC-10
Citrobacter	*Citrobacter freundii*	KPC-2*
		KPC-3
		KPC-18
Enterobacter non-*cloacae* Complex	*Enterobacter aerogenes*	KPC-2
		KPC-3
Enterobacter cloacae Complex	*Enterobacter cloacae*	KPC-2*
		KPC-3
		KPC-4
		KPC-13
		KPC-18
Escherichia coli	*Escherichia coli*	KPC-2
		KPC-3
		KPC-4
		KPC-5
		KPC-9
		KPC-18
		KPC-21
		KPC-28
Klebsiella oxytoca	*Klebsiella oxytoca*	KPC-2

Target	Associated Organism	Variant Detected
		KPC-3*
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	KPC-1
		KPC-2
		KPC-3*
		KPC-4
		KPC-5
		KPC-6
		KPC-7
		KPC-8
		KPC-11
		KPC-12
		KPC-14
		KPC-15
		KPC-16
		KPC-17
		KPC-19
		KPC-22
		KPC-24
		KPC-25
		KPC-26
		KPC-27
		KPC-30
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	KPC-2
		KPC-6*
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	KPC-2
		KPC-5*
Salmonella	*Salmonella enterica*	KPC-2
Serratia marcescens/*Serratia*	*Serratia marcescens*	KPC-2
		KPC-3

Table 36: Predicted (*in silico*) reactivity for VIM

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	VIM-1
		VIM-2
		VIM-6
		VIM-11
Citrobacter	*Citrobacter freundii*	VIM-1
		VIM-2
		VIM-4
		VIM-23
Enterobacter non-*cloacae* Complex	*Enterobacter aerogenes*	VIM-1
Enterobacter cloacae Complex	*Enterobacter cloacae*	VIM-1
		VIM-2
		VIM-4
		VIM-23
		VIM-31
		VIM-40
	Enterobacter hormaechei	VIM-1
		VIM-4
		VIM-23
	Enterobacter xiangfangensis	VIM-1
Escherichia coli	*Escherichia coli*[A]	VIM-1
		VIM-2
		VIM-19
		VIM-29
Klebsiella oxytoca	*Klebsiella oxytoca*	VIM-1
		VIM-2
		VIM-4
		VIM-32
		VIM-35
		VIM-1*
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	VIM-2
		VIM-4
		VIM-12
		VIM-19
		VIM-24
		VIM-26
		VIM-27*
		VIM-33
		VIM-34
		VIM-39
		VIM-42
		VIM-51
		VIM-52
Morganella morganii	*Morganella morganii*	VIM-4
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	VIM-1
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	VIM-1
		VIM-2*
		VIM-3
		VIM-4*
		VIM-5
		VIM-6
		VIM-8
		VIM-9
		VIM-10
		VIM-11
		VIM-14
		VIM-15
		VIM-16
		VIM-17

Target	Associated Organism	Variant Detected
		VIM-18
		VIM-20
		VIM-28
		VIM-30
		VIM-36
		VIM-37
		VIM-41
		VIM-43
		VIM-44
		VIM-45

Target	Associated Organism	Variant Detected
		VIM-46
		VIM-48
		VIM-50
Salmonella	*Salmonella enterica*	VIM-1
Serratia marcescens/*Serratia*	*Serratia marcescens*	VIM-2
		VIM-4
		VIM-54
Stenotrophomonas maltophilia	*Stenotrophomonas maltophilia*	VIM-2

A. Unspecified VIM variant detected in Analytical Reactivity (Inclusivity) study.

Table 37: Predicted (*in silico*) reactivity for OXA-23

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	OXA-23*
		OXA-49
		OXA-23/OXA-104
		OXA-23/OXA-64
		OXA-23/OXA-66
		OXA-23/OXA-69
		OXA-27
		OXA-65/OXA-239
		OXA-68
		OXA-146
		OXA-165
		OXA-166
		OXA-167
		OXA-168
		OXA-169
		OXA-170
		OXA-171
		OXA-183

Target	Associated Organism	Variant Detected
		OXA-225
		OXA-366
		OXA-398
		OXA-422
		OXA-423
		OXA-435
		OXA-440
		OXA-469
		OXA-481
		OXA-482
		OXA-483
		OXA-565
Escherichia coli	*Escherichia coli*	OXA-23
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	OXA-73
Proteus mirabilis	*Proteus mirabilis*	OXA-23

Table 38: Predicted (*in silico*) reactivity for OXA-48

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	OXA-48
Citrobacter	*Citrobacter freundii*	OXA-48
		OXA-181
Enterobacter non-*cloacae Complex*	*Enterobacter aerogenes*	OXA-244
Enterobacter cloacae Complex	*Enterobacter cloacae*	OXA-48
		OXA-163
		OXA-181
	Enterobacter hormaechei	OXA-370

Target	Associated Organism	Variant Detected
	Enterobacter ludwigii	OXA-48
Escherichia coli	*Escherichia coli*[A]	OXA-48
		OXA-163
		OXA-181
		OXA-204
		OXA-232
		OXA-244
		OXA-438
		OXA-439
		OXA-566

Target	Associated Organism	Variant Detected
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	OXA-1/OXA-48
		OXA-48*
		OXA-10
		OXA-162
		OXA-181*
		OXA-204
		OXA-232*
		OXA-244
		OXA-245
		OXA-247
		OXA-484
		OXA-505
		OXA-517
		OXA-519
	Klebsiella variicola	OXA-181
Morganella morganii	*Morganella morganii*	OXA-181
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	OXA-48
		OXA-244
Serratia marcescens/*Serratia*	*Serratia marcescens*	OXA-48
		OXA-405

A. Unspecified OXA variant detected in the Analytical Reactivity (Inclusivity) study.

Table 39: Predicted (*in silico*) reactivity for NDM

Target	Associated Organism	Variant Detected
Acinetobacter baumannii	*Acinetobacter baumannii*	NDM-1*
		NDM-2
Citrobacter	*Citrobacter braakii*	NDM-4
Citrobacter	*Citrobacter freundii*	NDM-1[A]
		NDM-4
		NDM-6
		NDM-7
Enterobacter non-*cloacae* Complex	*Enterobacter aerogenes*	NDM-1
		NDM-4
		NDM-5
		NDM-7
Enterobacter cloacae Complex	*Enterobacter cloacae*[B]	NDM-1
		NDM-4
		NDM-7
	Enterobacter hormaechei	NDM-1
	Enterobacter ludwigii	NDM-1
Escherichia coli	*Escherichia coli*	NDM-1
		NDM-3
		NDM-4
		NDM-5*
		NDM-6*
		NDM-7*
		NDM-8
		NDM-11
		NDM-12
		NDM-13
		NDM-15
		NDM-16
		NDM-17
		NDM-18
		NDM-19
Klebsiella oxytoca	*Klebsiella oxytoca*	NDM-1
		NDM-4
Klebsiella pneumoniae Group	*Klebsiella pneumoniae*	NDM-1*
		NDM-4
		NDM-5
		NDM-6
		NDM-7
		NDM-10
		NDM-16
Morganella morganii	Morganella morganii	NDM-1[C]
Proteus mirabilis/*Proteus*	*Proteus mirabilis*	NDM-1
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*	NDM-1
		NDM-5
Salmonella	*Salmonella enterica*	NDM-1
		NDM-5
Serratia marcescens/Serratia	*Serratia marcescens*	NDM-1
Stenotrophomonas maltophilia	*Stenotrophomonas maltophilia*	NDM-1

A. Detected in a *Citrobacter* species in the **Analytical Reactivity (Inclusivity) study**.

B. Unspecified NDM variant detected in the **Analytical Reactivity (Inclusivity) study**.

C. NDM-1 was detected in Morganella morganii in the **Analytical Reactivity (Inclusivity) study** but no sequences were available for *in silico* analysis.

Table 40: Predicted (*in silico*) Reactivity (Inclusivity) Results for Variants Not Detected

Resistance Marker	Variant Not Detected	Associated Organism	No. of Sequences
CTX-M-1	CTX-M-80	*Klebsiella pneumoniae*	3
	CTX-M-15		
	Not Specified		
IMP	IMP-31	*Pseudomonas aeruginosa*	2
	IMP-35		2
	IMP-7		1
NDM	NDM-1	*Escherichia coli*	6
		Klebsiella variicola	3
		Salmonella enterica	1
	NDM-3	*Acinetobacter baumannii*	1
	NDM-4	*Escherichia coli*	1
	NDM-9	*Cronobacter sakazakii*	1
		Escherichia coli	1
		Klebsiella pneumoniae	2
	Not specified	*Escherichia coli*	2
		Klebsiella pneumoniae	1
		Klebsiella sp	1
		Pseudomonas aeruginosa	1
VIM	VIM-1	*Pseudomonas aeruginosa*	3
		Providencia vermicola	1
	VIM-2	*Klebsiella pneumoniae*	1
	VIM-5	*Enterobacter cloacae*	2
		Klebsiella pneumoniae	3
	VIM-7	*Pseudomonas aeruginosa*	4
	VIM-13		3
	VIM-25	*Acinetobacter baumannii*	1
		Proteus mirabilis	2
	VIM-38	*Pseudomonas aeruginosa*	2
	VIM-47		2
	VIM-49		2
	Not specified		1
OXA-48	OXA-232	*Escherichia coli*	1

g. Analytical Specificity (Cross-Reactivity/Exclusivity)

The analytical specificity of the BCID-GN Panel was assessed by examining the potential for cross-reactivity of on-panel and off-panel analytes. Cross-reactivity of on-panel analytes was evaluated using data from the analytical reactivity study (See Table 20). Bacterial and fungal analytes were diluted and tested in triplicate at bottle positive concentrations. Cross reactivity was not observed for any of the on-panel organism strain with the exception of one strain of *K. pneumoniae* for which NDM was unexpectedly detected in all three replicates. The root cause of the detected NDM was not confirmed with an alternate PCR assay; however, it is not likely that *K. pneumoniae* cross-reacts with the NDM target of the BCID-GN Panel based on results from 32 additional strains of *K. pneumoniae* that did not demonstrate cross-reactivity for NDM.

A total of 128 off-panel bacterial and fungal strains were tested in triplicate at ~$1x10^9$ CFU/mL and ~$1x10^7$ CFU/mL, respectively. If these target concentrations could not be obtained, the organism was diluted 2-fold from the stock for use (indicated with an asterisk in Tables 41-44 below); these concentrations were at least as high as concentrations present at bottle positivity (1 x 10^8 CFU/mL for bacteria and 1 x 10^6 CFU/mL for fungi).

The following off-panel species demonstrated cross reactivity with ePlex BCID-GN Panel targets:

- *Acinetobacter anitratus* (at a concentration of >$1x10^4$ CFU/mL) cross-reacts with the BCID-GN *Acinetobacter baumannii* assay
- *Enterobacter cowanii* (at a concentration of >$1x10^8$ CFU/mL) cross-reacts with the BCID-GN *Enterobacter cloacae* complex assay
- *Escherichia hermanii* cross-reacts with the BCID-GN *Enterobacter* (non-*cloacae* complex) assay (at a concentration of >$1x10^6$ CFU/mL) and with the BCID-GN *Serratia* assay (at a concentration of >$1x10^7$ CFU/mL)
- *Fusobacterium periodonticum* (at a concentration of $5x10^8$ CFU/mL) and *Fusobacterium simiae* (at a concentration of $2.9x10^8$ CFU/mL) cross-react with the BCID-GN *Fusobacterium nucleatum* assay
- *Shigella* (at a concentration of $1x10^9$ CFU/mL) cross-reacts with the BCID-GN *Escherichia coli* assay.

Off-panel organisms demonstrating cross-reactivity with ePlex BCID-GN Panel targets are **bolded** in Tables 41-44 below.

Table 41: Cross Reactivity, Off-Panel gram-negative organisms evaluated

Gram Negative Organism	Strain ID
Acinetobacter haemolyticus	ATCC 19002
Acinetobacter lwoffii	ATCC 15309
Acinetobacter junii	ATCC 17908
***Acinetobacter anitratus*[A]**	ATCC 49139
Aeromonas hydrophila	JMI 938982
Aeromonas salmonicida	ATCC 33658

Gram Negative Organism	Strain ID
Aeromonas sobria	ATCC 35993
Bacteroides distasonis (Parabacteroides)	ATCC 8503
Bacteroides merdae	ATCC 43184
Bacteroides thetaiotaomicron	ATCC 29741
*Bacteroides vulgatus**	ATCC 8482

Gram Negative Organism	Strain ID
Bacteroides caccae	ATCC 700189
Bacteroides eggerthii	ATCC 27754
*Bacteroides ovatus**	ATCC BAA-1296
*Bacteroides ureolyticus**	ATCC 33387
Bordetella pertussis	ATCC 9797
Burkholderia cepacia	ATCC 25416
Citrobacter amalonaticus	ATCC BAA-2563
Citrobacter gillenii	ATCC 51640
Citrobacter sedlakii	ATCC 51493
Citrobacter farmer	ATCC 51112
Citrobacter murliniae	ATCC 51642
Edwardsiella tarda	ATCC 15947
***Enterobacter cowanii*[B]**	DSM-18146
Escherichia albertii	DSM-17582
Escherichia fergusonii	ATCC 35469
***Escherichia hermanii*[C]**	ATCC 33650
*Ewingella americana**	ATCC 33853
Eikenella corrodens	ATCC BAA-1152
*Fusobacterium naviforme**	ATCC 25832
Fusobacterium gonidiaformans	ATCC 25563
*Fusobacterium necrogenes**	ATCC 25556
***Fusobacterium periodonticum**[D]**	ATCC 33693
***Fusobacterium simiae**[D]**	ATCC 33568
Fusobacterium varium	ATCC 27725
*Fusobacterium russii**	ATCC 25533
Fusobacterium ulcerans	ATCC 49186
Haemophilus haemolyticus	ATCC 33390
Haemophilus parahaemolyticus	ATCC 10014
Hafnia alvei	ATCC 51815
*Kingella kingae**	ATCC 23331
Kluyvera cochleae	ATCC 51609
Legionella pneumoniae	ATCC 33823
Leclercia adecarboxylata	ATCC 700325
*Methylobacterium mesophilicum**	ATCC 29983
Neisseria gonorrhoeae	ATCC 19424
Neisseria mucosa	ATCC 19695
Neisseria sicca	ATCC 29193
Neisseria flavecens	ATCC 13115
Neisseria lactamica	ATCC 23970
Neisseria perflava	ATCC 14799
Ochrobactrum anthropi	ATCC BAA-749
Pantoea agglomerans	ATCC 14537
Pantoea ananatis	NRRL B-41502
Pasteurella aerogenes	ATCC 27883
Pasteurella multicida subsp multicide	ATCC 12945
Prevotella intermedia	ATCC 15032
*Prevotella corporis**	ATCC 33547
*Prevotella oralis**	ATCC 33269
*Prevotella nigrescens**	ATCC 33563

Gram Negative Organism	Strain ID
Providencia rettgeri	ATCC 9250
Providencia stuartii	ATCC 33672
Providencia alcalifaciens	ATCC 9886
Pseudomonas fluorescens	ATCC 13525
Pseudomonas putida	ATCC 49128
Pseudomonas alcaligenes	ATCC 14909
Ralstonia insidiosa	ATCC 49129
Ralstonia pickettii	ATCC 27511
Raoultella planticola (Klebsiella planticola)	ATCC 31900
Raoultella ornithinolytica	CDC# 0134
Raoultella terrigena (Klebsiella terrigena)	ATCC 55553
***Shigella boydii*[E]**	ATCC 9207
***Shigella sonnei*[E]**	ATCC 25931
***Shigella flexneri*[E]**	ATCC 9199
Vibrio furnissii	NCTC11218
Vibrio alginolyticus	ATCC 17749
Vibrio parahaemolyticus	ATCC 17802
Yersinia enterocolitica subsp *enterocolitica*	ATCC 9610
Yersinia ruckeri	ATCC 29473
Yersinia kristensenii	ATCC 33639

A. Cross-reactivity seen with *Acinetobacter baumannii* at a concentration > $1x10^4$ CFU/mL.
B. Cross-reactivity seen with *Enterobacter cloacae* complex at a concentration > $1x10^8$ CFU/mL.
C. Cross-reactivity seen with *Enterobacter* (non-*cloacae* complex) at a concentration > $1x10^6$ CFU/mL and *Serratia* at a concentration of > $1x10^7$ CFU/mL.
D. Cross-reactivity seen with the *Fusobacterium nucleatum* assay.
E. Cross-reactivity seen with the *Escherichia coli* assay.

Table 42: Cross-reactivity study, Off-Panel gram-positive organisms evaluated

Gram Positive Organisms	Strain ID	Gram Positive Organisms	Strain ID
Actinomyces odontolyticus	ATCC 17929	*Lactobacillus paracasei**	ATCC 25598
Clostridium perfringens	ATCC 13124	*Lactobacillus acidophilus**	ATCC 314
Corynebacterium jeikeium	ATCC BAA-949	*Lactobacillus crispatus*	ATCC 33197
Corynebacterium renale	ATCC 19412	*Lactobacillus rhamnosus*	ATCC 39595
Corynebacterium ulcerans	ATCC 51799	*Lactococcus lactis*	ATCC 49032
*Corynebacterium xerosis**	ATCC 373	*Listeria innocua*	ATCC 33090
Corynebacterium durum	ATCC 33449	*Listeria monocytogenes*	ATCC 7644
*Corynebacterium diphtheriae**	ATCC 13812	*Micrococcus luteus*	ATCC 10240
Corynebacterium pseudodiphtheriticum	ATCC 10700	*Peptostreptococcus anaerobius*	ATCC 27337
*Corynebacterium striatum**	ATCC 43735	*Propionibacterium acnes*	ATCC 11827
Corynebacterium urealyticum	ATCC 43044	*Rothia mucilaginosa*	ATCC 25296
Lactobacillus casei	ATCC 39392		

Table 43: Cross-reactivity, Off-Panel fungal organisms evaluated

Fungal Pathogens	Strain ID	Concentration Tested
*Aspergillus fumigatus**	ATCC 204305	2.50 x 10^6 CFU/mL
Candida orthopsilosis	ATCC 96139	1 x 10^7 CFU/mL
Candida metapsilosis	ATCC 96144	1 x 10^7 CFU/mL
Candida tropicalis	ATCC 1369	1 x 10^7 CFU/mL
Cryptococcus grubii	ATCC 208821	1 x 10^7 CFU/mL
Cryptococcus gattii	ATCC 76108	1 x 10^7 CFU/mL
Cryptococcus neoformans	ATCC 14116	1 x 10^7 CFU/mL
Geotrichum capitatum	ATCC 10663	1 x 10^7 CFU/mL
Histoplasma capsulatum	*In silico*	N/A
Penicillium marneffei	ATCC 200050	1 x 10^7 CFU/mL
Rhodotorula glutinis	ATCC 32765	1 x 10^7 CFU/mL
Rhodotorula mucilaginosa	ATCC 9449	1 x 10^7 CFU/mL
Rhodotorula minuta	ATCC 36236	1 x 10^7 CFU/mL
*Saccharomyces cervisiae**	ATCC 18824	5.55 x 10^6 CFU/mL
Trichosporon dermatis	ATCC MYA-4294	1 x 10^7 CFU/mL
Trichosporon mucoides	ATCC 90046	1 x 10^7 CFU/mL

Table 44: Cross-reactivity, Off-Panel resistance genes evaluated

Antimicrobial Resistance Genes	Strain ID	Concentration Tested
FOX (Carried by *Klebsiella oxytoca*)*[A]	JMI 954306	8 x 10^8 CFU/mL
MOX (Carried by *Aeromonas hydrophila*)	JMI 938982	1 x 10^9 CFU/mL
SME (Carried by *Serratia marcescens*)[A]	CDC #0091	1 x 10^9 CFU/mL
SHV (Carried by *Klebsiella pneumoniae*)[A]	CDC #0087	1 x 10^9 CFU/mL
TEM (Carried by *Escherichia coli*)[A]	NCTC 13351	1 x 10^9 CFU/mL

A. The on-panel organism associated with the resistance gene was detected by the BCID-GN panel as expected

In silico analysis was performed to identify any additional off-panel organisms for potential cross-reactivity with ePlex BCID-GN Panel targets. Results for gram-negative and gram-positive organisms that are predicted to generate false positive results with BCID-GN Panel are presented in Tables 45 and 46.

Table 45: Cross-reactivity, Off-Panel gram-negative organisms evaluated by *in silico* analysis

Cross-reactive Organism	ePlex BCID-GN Target	No. of Sequences	Predicted Cross-Reactive Sequences* n (%)
Fusobacterium hwasookii	*F. nucleatum*	10	5 (50%)
Haemophilus aegyptius	*H. influenzae*	3	3 (100%)
Klebsiella michiganensis	*Klebsiella oxytoca*	40	40 (100%)
Pseudomonas denitrificans	*Pseudomonas aeruginosa*	17	16 (94.1%)

Table 46: Cross-reactivity, Off-Panel gram-positive organisms evaluated by *in silico* analysis

Organism	Number of Sequences	Predicted Cross-Reactive Sequences n (%)
Brevibacterium halotolerans	3	3 (100%)
Domibacillus indicus	1	1 (100%)
Domibacillus robiginosus	1	1 (100%)
Salinibacillus aidingensis	2	1 (50%)
Terribacillus aidingensis	1	1 (100%
Terribacillus halophilus	2	1 (50%)
Terribacillus saccharophilus	1	1 (100%)
Planomicrobium okeanokoites	1	1 (100%)
Lactococcus chungangensis	4	4 (100%)
Lactococcus laudensis	1	1 (100%)
Lactococcus piscium	18	18 (100%)
Lactococcus plantarum	6	5 (83.8%)
Lactococcus raffinolactis	49	46 (93.9%)
Okadaella gastrococcus	4	4 (100%)

h. Interfering Substances:

At total of 18 substances that could be present in blood culture specimens or introduced during sample handling were evaluated for potential interference with performance of the BCID-GN Panel. Each endogenous or exogenous substances was added to simulated positive blood culture samples which contained simulated blood culture matrix (human whole blood that had been incubated in a blood culture bottle) and one of three different organism mixes. Each mix contained organisms inoculated at concentrations equivalent to bottle positivity. Potentially interfering substances were spiked at medically relevant concentrations and were evaluated both without organism with all testing performed in triplicate. On each day of testing, each organism mix was evaluated without any interfering substance to serve as a positive control (no interference) to which the test specimens were compared.

None of the 18 substances commonly found in blood culture specimens or as medications commonly used to treat skin or blood infections were found to inhibit the ePlex BCID-GN Panel. Two substances required further dilution of the substance under evaluation to determine the final concentration for which interference is no longer observed. Caspofungin was diluted from 5 µg/mL to 4.5 µg/mL and γ-globulin was diluted from 1.7 g/dL to 0.85 g/dL. The final testing concentration of caspofungin and γ-globulin are above the clinically relevant concentration of those individuals suspected of having a bloodstream infection.

Table 47 includes the concentrations of the twelve organisms evaluated in the study. Table 48 includes the potentially interfering substances tested as well as their respective concentrations.

Table 47: Organism mixes and test concentrations

Mix	Organism	Resistance Marker	Test Concentration (CFU/mL)
1	*Escherichia coli*	OXA	2×10^8
	Haemophilus influenzae	-	6×10^8
	Enterobacter cloacae	CTX-M	2×10^8
	Candida albicans (Pan *Candida* target)	-	1×10^6
2	*Klebsiella oxytoca*	-	9×10^8
	Pseudomonas aeruginosa	IMP	1×10^8
	Acinetobacter baumannii	-	4×10^8
	Enterobacter aerogenes	-	2×10^8
3	*Neisseria meningitidis*	-	3×10^7
	Serratia marcescens	-	1×10^9
	Bacteroides fragilis	-	4×10^8
	Staphylococcus aureus (Pan Gram-Positive target)	-	2×10^7

Table 48. Interfering substances and test concentrations

Endogenous Substances	Test Concentration
Hemoglobin	0.6 g/L
Triglycerides	1000 mg/dL
Bilirubin	60 µg/mL
γ-globulin	0.85 g/dL
Human genomic DNA	6×10^5 copies/mL

Exogenous Substances	Test Concentration
Fluconazole	25 mg/L
Vancomycin	30 mg/L
Ciprofloxacin	3 mg/L
Gentamicin sulfate	3 µg/mL
Imipenem	83 µg/mL
Ceftriaxone	0.23 mg/mL
Tetracycline	5 mg/L
Amoxicillin/Clavulanate	3.5 µg/mL
Heparin	0.9 U/mL
Sodium Polyanetholsulfonate	0.25% w/v
Amphotericin B	2 µg/mL
Flucytosine	90 µg/mL
Caspofungin	4.5 µg/mL

i. Competitive Inhibition:

A competitive inhibition study was conducted to evaluate the possibility of high concentration organisms interfering with the detection of lower concentrations targets in the same blood culture specimen. Four organism mixes covering eight clinically relevant co-infection organisms including both on and off-panel analytes were selected for the study based on those identified in the peer-reviewed literature as common polymicrobial co-infections. The four mixes were used to formulate test case scenarios by pairing one mix with organisms at bottle positive concentrations with a second mix with organism at concentrations typically present at bottle positivity plus eight hours. Testing was performed in triplicate for each test condition. A summary of the organisms and test concentrations evaluated in the study is presented in Table 49. Testing generated 100% detected results for all targeted organisms. There were no false positive results for samples with the non-targeted organism, C*orynebacterium striatum*. The study results demonstrated the ability of the ePlex BCID-GN Panel to detect two clinically relevant co-infecting organisms in a specimen at both high and low concentrations. Detailed study results are presented in Table 50.

Table 49: Competitive Inhibition, Organisms and Concentrations Tested

Organism	High Concentration	Low Concentration
Klebsiella pneumoniae	1 x 10^9 CFU/mL	9 x 10^8 CFU/mL
Escherichia coli (CTX-M+)	1 x 10^9 CFU/mL	2 x 10^8 CFU/mL
Enterobacter cloacae (VIM+)	7 x 10^8 CFU/mL	2 x 10^8 CFU/mL
Klebsiella oxytoca (KPC+)	1 x 10^9 CFU/mL	9 x 10^8 CFU/mL
Pseudomonas aeruginosa (IMP+)	8 x 10^8 CFU/mL	1 x 10^8 CFU/mL
Serratia marcescens	2 x 10^9 CFU/mL	1 x 10^9 CFU/mL
Staphylococcus aureus	1 x 10^8 CFU/mL	2 x 10^7 CFU/mL
Corynebacterium striatum[A]	2 x 10^9 CFU/mL	4 x 10^6 CFU/mL

A. Off-panel organism

Table 50: Percent Detection of all Organisms in Each Mix

Mix Combination	Low Titer Organisms (BP)	Conc. (CFU/mL)	% Detection	High Titer Organisms (BP + 8hrs)	Conc. (CFU/mL)	% Detection
1 & 2	*K. pneumoniae*	9×10^8	100	*E. coli* (CTX-M)	1×10^9	100
	C. striatum	4×10^6	0	*E. cloacae* (VIM)	7×10^8	100
2 & 1	*E. coli* (CTX-M)	2×10^8	100	*K. pneumoniae*	1×10^9	100
	E. cloacae (VIM)	2×10^8	100	*C. striatum*	2×10^9	0
1 & 3	*K. pneumoniae*	9×10^8	100	*K. oxytoca* (KPC)	1×10^9	100
	C. striatum	4×10^6	0	*P. aeruginosa* (IMP)	8×10^8	100
3 & 1	*K. oxytoca* (KPC)	9×10^8	100	*K. pneumoniae*	1×10^9	100
	P. aeruginosa (IMP)	1×10^8	100	*C. striatum*	2×10^9	0
1 & 4	*K. pneumoniae*	9×10^8	100	*S. aureus*	1×10^8	100
	C. striatum	4×10^6	0	*S. marcescens*	2×10^9	100
4 & 1	*S. aureus*	2×10^7	100	*K. pneumoniae*	1×10^9	100
	S. marcescens	1×10^9	100	*C. striatum*	2×10^9	0
2 & 3	*E. coli* (CTX-M)	2×10^8	100	*K. oxytoca* (KPC)	1×10^9	100
	E. cloacae (VIM)	2×10^8	100	*P. aeruginosa* (IMP)	8×10^8	100
3 & 2	*K. oxytoca* (KPC)	9×10^8	100	*E. coli* (CTX-M)	1×10^9	100
	P. aeruginosa (IMP)	1×10^8	100	*E. cloacae* (VIM)	7×10^8	100
2 & 4	*E. coli* (CTX-M)	2×10^8	100	*S. aureus*	1×10^8	100
	E. cloacae (VIM)	2×10^8	100	*S. marcescens*	2×10^9	100
4 & 2	*S. aureus*	2×10^7	100	*E. coli* (CTX-M)	1×10^9	100
	S. marcescens	1×10^9	100	*E. cloacae* (VIM)	7×10^8	100
3 & 4	*K. oxytoca* (KPC)	9×10^8	100	*S. aureus*	1×10^8	100
	P. aeruginosa (IMP)	1×10^8	100	*S. marcescens*	2×10^9	100
4 & 3	*S. aureus*	2×10^7	100	*K. oxytoca* (KPC)	1×10^9	100
	S. marcescens	1×10^9	100	*P. aeruginosa* (IMP)	8×10^8	100

j. Specimen matrix equivalency study: Evaluation of blood culture bottle types

Thirteen different bottle types indicated for use with three different blood culture monitoring systems (BacT/Alert, BACTEC and VersaTREK) were evaluated analytically

with the ePlex BCID-GN panel. Results from this study will be used to support the analytical performance claims of the ePlex BCID-GN Panel as well as to assess for the potential presence of non-viable organisms or organism nucleic acids in blood culture media for each bottle type.

Testing was performed using three organism mixes with organism concentrations approximating bottle positivity. Each of the blood culture bottle types were tested with the manufacturer's recommended ratio of human whole blood to media. The organisms evaluated that were also used for the competitive inhibition study, represent each of the eight multiplex primer amplification pools on the BCID-GN Panel. Two lots of the 13 different blood culture bottles were evaluated using five replicates of the three organism mixes. Three negative control (negative blood matrix, no organism mix) replicates were also tested per lot of each bottle type. The thirteen bottle types evaluated are listed in Table 51 below.

Twelve of the bottle types evaluated showed no interference for any of the targets tested. One of three lots of the BACTEC Lytic Anaerobic bottles tested showed reduced sensitivity (i.e., false negative results) in initial testing for four analytes. Additional replicates tested with a new bottle lot generated the expected positive results. The following are the results observed for BACTEC Lytic Anaerobic bottles for the initially discordant analytes:

- Pan *Candida* was not detected in 2/15 sample replicates
- *Enterobacter cloacae* was not detected in 1/15 sample replicates
- *Escherichia coli* was not detected in 1/15 sample replicates
- CTX-M was not detected in 2/15 sample replicates

A limitation is included in the package insert stating that the ePlex BCID-GN Panel may have reduced sensitivity with BACTEC Lytic Anaerobic bottles.

Table 51: Sample Matrix Equivalency (Bottle Evaluation) Bottle Types

Manufacturer	Bottle Brand	Bottle Type	Study Outcome
BD	BACTEC	Plus Aerobic	No interference observed
BD	BACTEC	Plus Anaerobic	No interference observed
BD	BACTEC	Standard Aerobic	No interference observed
BD	BACTEC	Standard Anaerobic	No interference observed
BD	BACTEC	Peds Plus	No interference observed
BD	BACTEC	Lytic Anaerobic	False negative results were observed for Pan Candida, *Enterobacter cloacae*, *Escherichia coli*, CTX-M and OXA in one of three lots.*
bioMérieux	BACT/ALERT	SA Standard Aerobic	No interference observed
bioMérieux	BACT/ALERT	SN Standard Anaerobic	No interference observed
bioMérieux	BACT/ALERT	FA Plus	No interference observed
bioMérieux	BACT/ALERT	FN Plus	No interference observed
bioMérieux	BACT/ALERT	PF Plus	No interference observed
Thermo Scientific	VersaTREK	REDOX 1 EZ Draw Aerobic	No interference observed
Thermo Scientific	VersaTREK	REDOX 2 EZ Draw Anaerobic	No interference observed

* 2/15 replicates were false negative for Pan Candida; 1/15 replicates was false negative for Enterobacter cloacae; 1/15 replicates was false negative for Escherichia coli (OXA-48); 2/15 replicates were false negative for CTX-M

k. Carry-Over/Cross-contamination:

Potential run-to-run carryover or cross-contamination was evaluated by determining whether a high positive sample in one cartridge would cause false positive results in subsequently tested cartridges. It is noted that the configuration of the ePlex instrument mitigates the risk for carry-over/cross-contamination because each bay functions independently and is physically separated from other bays, therefore, not allowing cartridges to interact with one another. The carry-over/cross-contamination rate of the ePlex BCID-GN Panel and ePlex Instrument was tested in a checker board approach by running high positive and negative samples interspersed in all towers of a four-tower ePlex instrument over five consecutive runs on separate days. The high positive sample consisted of a multi-organism mix of four bacterial and one fungal species, covering eight assay targets, prepared at concentrations higher than typically found in positive blood cultures (i.e., 1×10^9 CFU/mL for bacteria and 1×10^7 CFU/mL for fungi). Negative blood culture matrix was used for negative samples.

Testing resulted in detection of all expected targets for samples containing organism. No carry-over/cross-contamination (i.e., no false positive results) was observed when testing consecutively within bays or in adjacent bays over 120 total runs.

l. Assay cutoffs:

Assay cutoff analysis was conducted to define signal boundaries that would discriminate between positive and negative detections for each ePlex BCID-GN Panel analyte and for the internal control. Clinical specimens, contrived positive blood culture specimens, and organisms spiked at or above the determined LoD were used to establish assay-cutoffs. Contrived samples were prepared by spiking organisms at low concentrations into simulated negative blood culture matrix and grown to bottle positivity on a blood culture continuous monitoring system. An appropriate boundary was established per target wherein specificity (limit the number of potential false positives) and sensitivity (limit the number of potential false negatives) were maximized and verified by ROC analysis. The boundary set points that were determined during the cutoff study are listed in Table 52.

Table 52: Summary of BCID-GN Panel cutoffs

Target	Boundary (nA)
Acinetobacter baumannii	10
Bacteroides fragilis	20
Citrobacter	20
Cronobacter sakazakii	20
Enterobacter aerogenes	20
Enterobacter amnigenus	10
Enterobacter asburiae	25
Enterobacter cloacae	20
Escherichia coli	30
Fusobacterium necrophorum	40
Fusobacterium nucleatum	20
Haemophilus influenzae	20
Klebsiella oxytoca	20
Klebsiella pneumoniae	40
Morganella morganii	20
Neisseria meningitidis	40
Proteus	20
Proteus mirabilis	60
Pseudomonas aeruginosa	5
Salmonella	5
Serratia	20
Serratia marcescens	20
Stenotrophomonas maltophilia	50
CTX-M	40
IMP	20
KPC	20
NDM	20
OXA-23	50
OXA-48	20
VIM	20
Pan Gram-Positive *Enterococcus faecalis*	60
Pan Gram-Positive *Bacillus*	50
Pan Gram-Positive *S. anginosus*	15
Pan Gram-Positive *Enterococcus*	15

Target	Boundary (nA)
Pan Gram-Positive *Strep*/*Staph*	50
Pan *Candida*	30
Schizosaccharomyces pombe	20
Synthetic control	20

2. Comparison studies:

   a. *Method comparison with predicate device:*

      N/A, Refer to the clinical studies section

   b. Matrix comparison:

      N/A

3. Clinical studies:

   The clinical performance of the ePlex BCID-GN Panel was evaluated during a multi-center study with testing comprised of prospective and retrospective positive blood culture specimens that contained gram-negative or gram-variable organisms by Gram stain. All blood culture bottles were flagged positive by a continuously monitoring blood culture system and removed from the system within eight hours of positivity and tested with the ePlex BCID-GN Panel.

   Bottle types evaluated in the clinical study included 13 different types from three different manufacturers as shown in Table 53:

   Table 53: Bottle types evaluated, prospective and retrospective testing

Bottle Types		
BacT/Alert SA	BACTEC - Peds Plus /F	VersaTREK - Redox 1 EZ Draw
BacT/Alert SN	BACTEC - Standard/10 Aerobic/F	VersaTREK - Redox 2
BACTEC Lytic/10 Anaerobic/	BACTEC - Standard/10 Anaerobic/F	VersaTREK - Redox 2 EZ Draw
BACTEC PLUS Aerobic/F	BACTEC Standard Anaerobic/F	
BACTEC PLUS Anaerobic/F	VersaTREK - Redox 1	

   Six clinical test sites, each having a 4-tower ePlex instrument (24 bays), participated in the prospective clinical evaluation of the ePlex BCID-GN Panel with testing performed by trained laboratory personnel. A total of seven ePlex BCID-GN Panel cartridge lots were used in the prospective clinical evaluation.

   Prospective specimens were collected at seven clinical sites in two phases. A total of 183

specimens were prospectively collected and frozen prior to testing and 171 specimens were prospectively collected and tested fresh (never frozen) for a total of 354 specimens tested across the 2 phases. One specimen was withdrawn due to organism identification from unacceptable methods. Of the 353 prospectively-collected specimens eligible for testing, 349 were evaluable (i.e., valid ePlex BCID-GN Panel results and a valid comparator result). Four specimens were not evaluable as they did not generate valid ePlex BCID-GN Panel results and were excluded from performance evaluations.

To supplement the prospective study for low prevalence organism targets, a total of 578 positive blood culture specimens (577 evaluable) were collected from 10 collection sites and retrospectively tested. One specimen was not evaluable because it did not have a final, valid ePlex BCID-GN Panel result and was excluded from performance evaluations.

Demographic information for each collection site is described in Tables 54 and 55 for the prospectively-collected and retrospectively collected specimen cohorts.

Table 54: Demographic Data by Collection Site (Prospective Specimens)

	All Sites N = 349 n (%)	**Site 1 N = 88 n (%)**	**Site 2 N = 23 n (%)**	**Site 3 N = 98 n (%)**	**Site 4 N = 58 n (%)**	**Site 5 N = 46 n (%)**	**Site 6 N = 28 n (%)**	**Site 7 N = 8 n (%)**
Sex								
Male	168 (48.1)	37 (42.0)	12 (52.2)	52 (53.1)	28 (48.3)	21 (45.7)	13 (46.4)	5 (62.5)
Female	181 (51.9)	51 (58.0)	11 (47.8)	46 (46.9)	30 (51.7)	25 (54.3)	15 (53.6)	3 (37.5)
Age								
<1 yr	7 (2.0)	2 (2.3)	0 (0.0)	4 (4.1)	0 (0.0)	1 (2.2)	0 (0.0)	0 (0.0)
1-17 yrs	10 (2.9)	4 (4.5)	1 (4.3)	3 (3.1)	1 (1.7)	1 (2.2)	0 (0.0)	0 (0.0)
18-44 yrs	50 (14.3)	10 (11.4)	3 (13.0)	20 (20.4)	3 (5.2)	8 (17.4)	6 (21.4)	0 (0.0)
45-64 yrs	124 (35.5)	35 (39.8)	9 (39.1)	28 (28.6)	21 (36.2)	14 (30.4)	13 (46.4)	4 (50.0)
65-84 yrs	125 (35.8)	29 (33.0)	8 (34.8)	35 (35.7)	25 (43.1)	17 (37.0)	7 (25.0)	4 (50.0)
85+ yrs	33 (9.5)	8 (9.1)	2 (8.7)	8 (8.2)	8 (13.8)	5 (10.9)	2 (7.1)	0 (0.0)

Table 55: Demographic Data by Collection Site (Retrospective Specimens)

	All Sites N = 577 n (%)	Site 1 N = 78 n (%)	Site 2 N = 73 n (%)	Site 3 N = 31 n (%)	Site 4 N = 93 n (%)	Site 5 N = 1 n (%)	Site 6 N = 80 n (%)	Site 7 N = 67 n (%)	Site 8 N = 48 n (%)	Site 9 N = 29 n (%)	Site 10 N = 77 n (%)
Sex											
Male	307 (53.2)	36 (46.2)	41 (56.2)	15 (48.4)	49 (52.7)	0 (0.0)	47 (58.8)	38 (56.7)	29 (60.4)	19 (65.5)	33 (42.9)
Female	270 (46.8)	42 (53.8)	32 (43.8)	16 (51.6)	44 (47.3)	1 (100)	33 (41.3)	29 (43.3)	19 (39.6)	10 (34.5)	44 (57.1)
Age											
<1 yr	9 (1.6)	1 (1.3)	0 (0.0)	0 (0.0)	3 (3.2)	0 (0.0)	2 (2.5)	0 (0.0)	1 (2.1)	0 (0.0)	2 (2.6)
1-17 yrs	20 (3.5)	1 (1.3)	0 (0.0)	1 (3.2)	8 (8.6)	0 (0.0)	6 (7.5)	0 (0.0)	0 (0.0)	1 (3.4)	3 (3.9)
18-44 yrs	78 (13.5)	13 (16.7)	7 (9.6)	2 (6.5)	10 (10.8)	1 (100)	15 (18.8)	8 (11.9)	8 (16.7)	6(20.7)	8 (10.4)
45-64 yrs	193 (33.4)	27 (34.6)	18 (24.7)	13 (41.9)	27 (29.0)	0 (0.0)	32 (40.0)	27 (40.3)	16 (33.3)	9 (31.0)	24 (31.2)
65-84 yrs	226 (39.2)	29 (37.2)	40 (54.8)	11 (35.5)	40 (43.0)	0 (0.0)	20 (25.0)	24 (35.8)	21 (43.8)	11 (37.9)	30 (39.0)
85+ yrs	49 (8.5)	7 (9.0)	8 (11.0)	4 (12.9)	5 (5.4)	0 (0.0)	5 (6.3)	6 (9.0)	2 (4.2)	2 (6.9)	10 (13.0)
Unknown	2 (0.3)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	2 (3.0)	0 (0.0)	0 (0.0)	0 (0.0)

The performance of the ePlex BCID-GN Panel for organism targets was evaluated in comparison to results from standard laboratory procedures for identification of blood culture isolates. Comparator testing included traditional and automated identification methods, MALDI-TOF, as well as microbiological and biochemical techniques. Identification for specimens with *Acinetobacter baumannii* (to distinguish between other *Acinetobacter* species) or *Candida parapsilosis* (to distinguish between *C. parapsilosis*, *C. metapsilosis* and *C. orthopsilosis*) identified by standard laboratory procedures was confirmed using analytically validated PCR assays followed by bi-directional sequencing. For antimicrobial resistance markers, results from the ePlex BCID-GN Panel were compared to analytically validated qPCR amplification assays followed by bi-directional sequencing in specimens with an applicable organism identified by culture.

The comparator methods used in the study for each ePlex BCID-GN Panel target are summarized in Table 56.

Table 56: Comparator method(s) for ePlex BCID-GN Panel targets

Target	Comparator Method
Acinetobacter baumannii	Standard laboratory procedures for organism ID. PCR/sequencing to confirm *Acinetobacter baumannii* and to differentiate between *Acinetobacter* species not targeted by the BCID-GN Panel
Bacteroides fragilis	Standard laboratory procedures for organism ID.
Citrobacter	
Cronobacter sakazakii	
Enterobacter cloacae complex	
Enterobacter (non-*cloacae* complex)	
Escherichia coli	
Fusobacterium necrophorum	
Fusobacterium nucleatum	
Haemophilus influenzae	
Klebsiella oxytoca	
Klebsiella pneumoniae group	
Morganella morganii	
Neisseria meningitidis	
Proteus	
Proteus mirabilis	
Pseudomonas aeruginosa	
Salmonella	
Serratia	
Serratia marcescens	
Stenotrophomonas maltophilia	
Pan Gram-Negative	
Pan *Candida*	Standard laboratory procedures for organism ID. PCR/sequencing to confirm *C. parapsilosis* or identify *C. metapsilosis, C. orthopsilosis.*
CTX-M, IMP, KPC, NDM, OXA, VIM	qPCR/sequencing in specimens with associated organism detected by comparator method. See Table 5 for organism associations.

Positive percent agreement (PPA) was calculated by dividing the number of true positive (TP) results by the sum of TP and false negative (FN) results, while negative percent agreement (NPA) was calculated by dividing the number of true negative (TN) results by the sum of TN and false positive (FP) results. A TP result being defined as a specimen where the detected ePlex BCID-GN Panel result matched the identified/detected comparator method result, while a TN result was one where a negative ePlex BCID-GN Panel result matched a negative comparator method result. The two-sided 95% confidence interval was also calculated.

Specimens for which FP and/or FN results (i.e., discordant results) were further

investigated using FDA-cleared devices, analytically validated PCR/sequencing assays or 16S sequencing to assess for the presence/absence of the target organism or resistance marker, as applicable. Results from discordant analyses are included as footnotes under the performance tables below.

A total of 777 contrived blood culture specimens were evaluated to supplement the prospective and retrospective specimen cohorts for low prevalence organism and resistance markers. Contrived specimens were prepared by spiking low concentrations of organism into blood culture bottle types appropriate for the organism being evaluated. Each bottle was also inoculated with human whole blood at volumes recommended by the bottle manufacturer. Bottles were incubated on a continuously monitoring blood culture system until flagged as positive. Contrived specimens were removed from the system within eight hours of positivity and stored frozen until the time of testing. Contrived specimens were batched together and tested along with other clinical specimens at five testing sites with all specimen identities blinded to test operators. A summary of the strains used to prepare contrived specimens as well as the numbers of specimens tested per strain are shown in Table 57.

Table 57: Contrived specimen study, organisms evaluated

Target	Organism	Strain	Independent Contrived Specimens Tested
Acinetobacter baumannii	*Acinetobacter baumannii*	ATCC19606	2
		ATCCBAA-2093	4
		ATCCBAA-747	5
		NCIMB12457	2
		NCTC13302	3
		NCTC13303	4
		NCTC13423	17
	Acinetobacter baumannii, NDM	CDC#0033	5
	Acinetobacter baumannii, OXA-23	ATCCBAA-1605	5
		NCTC13304	5
		NCTC13305	3
	***Acinetobacter baumannii* Total**		**55**
Bacteroides fragilis	*Bacteroides fragilis*	ATCC23745	8
		ATCC25285	8
		ATCC43860	8
		ATCC700786	8
		NCTC9343	8
	***Bacteroides fragilis* Total**		**40**
Citrobacter	*Citrobacter braakii*	ATCC43162	4
		ATCC51113	4
	Citrobacter freundii	ATCC43864	4
		ATCC8090	4
		NCTC8581	4

Target	Organism	Strain	Independent Contrived Specimens Tested
		NCTC9750	5
	Citrobacter freundii, CTX-M	JMI2047	6
	Citrobacter freundii, KPC	CDC#0116	4
	Citrobacter koseri	ATCC27156	4
	Citrobacter youngae	ATCC29935	4
	***Citrobacter* Total**		**43**
Cronobacter sakazakii	*Cronobacter sakazakii*	ATCC12868	2
		ATCC29004	2
		ATCC29544	4
		ATCCBAA-894	3
		FSLF6-0023	4
		FSLF6-0028	4
		FSLF6-0029	4
		FSLF6-0034	3
		FSLF6-0035	3
		FSLF6-0043	4
		FSLF6-0049	3
		FSLF6-0050	4
		FSLF6-0051	5
	***Cronobacter sakazakii* Total**		**45**
Enterobacter cloacae complex	*Enterobacter asburiae*	ATCC35953	2
		ATCC35955	1
		ATCC35956	4
		ATCC35957	1
	Enterobacter cloacae, CTX-M	CDC#0038	4
		NCTC13464	3
	Enterobacter cloacae, CTX-M, KPC	CDC#0163	2
	Enterobacter cloacae, CTX-M, NDM	CDC#0038	1
		JMI53571	12
	Enterobacter cloacae, VIM	CDC#0154	6
	Enterobacter hormaechei, KPC	ATCCBAA-2082	1
	***Enterobacter cloacae* complex Total**		**37**
*Enterobacter (*non-*cloacae* complex)	*Enterobacter aerogenes*	ATCC13048	3
		ATCC29010	3
		ATCC51697	3
	Enterobacter aerogenes, IMP	CDC#0161	5
	Enterobacter aerogenes, OXA-48	CDC#0074	12
	Enterobacter amnigenus	ATCC33072	3
		ATCC33731	3
		ATCC51816	4
	***Enterobacter (*non-*cloacae* complex) Total**		**36**
Escherichia coli	*Escherichia coli*, CTX-M	CDC#0086	3
		NCTC13452	3

Target	Organism	Strain	Independent Contrived Specimens Tested
		NCTC13461	6
		NCTC13463	2
	Escherichia coli, CTX-M, NDM	CDC#0137	6
		CDC#0151	2
	Escherichia coli, IMP	NCTC13476	7
	Escherichia coli, KPC	ATCCBAA-2340	2
		CDC#0114	4
	Escherichia coli, NDM	CDC#0118	6
		CDC#0150	5
		CDC#0151	4
	Escherichia coli, VIM	JMI32465	2
	***Escherichia coli* Total**		**52**
Fusobacterium necrophorum	*Fusobacterium necrophorum* subsp. *funduliforme*	ATCC51357	24
	Fusobacterium necrophorum subsp. *necrophorum*	ATCC27852	24
	***Fusobacterium necrophorum* Total**		**48**
Fusobacterium nucleatum	*Fusobacterium nucleatum*	ATCC23726	8
		ATCC25586	20
		ATCC31647	19
	***Fusobacterium nucleatum* Total**		**47**
Haemophilus influenzae	*Haemophilus influenzae*	ATCC10211	9
		ATCC43065	6
		ATCC49144	9
		NCTC12699	9
		NCTC8468	8
	***Haemophilus influenzae* Total**		**41**
Klebsiella oxytoca	*Klebsiella oxytoca*	ATCC43086	4
		ATCC43165	4
		ATCC43863	2
		ATCC49131	4
		ATCC51817	3
		ATCC700324	3
	***Klebsiella oxytoca* Total**		**20**
Klebsiella pneumoniae group	*Klebsiella pneumoniae*, CTX-M	NCTC13465	5
	Klebsiella pneumoniae, CTX-M, NDM	ATCCBAA-2146	3
	Klebsiella pneumoniae, CTX-M, OXA	CDC#0140	12
	Klebsiella pneumoniae, IMP	CDC#0034	8
		CDC#0080	6
	Klebsiella pneumoniae, KPC	CDC#0112	1
		CDC#0113	1
		CDC#0115	4
		CDC#0117	4

Target	Organism	Strain	Independent Contrived Specimens Tested
		CDC#0120	4
		CDC#0125	4
		CDC#0129	4
	Klebsiella pneumoniae, VIM	LMC_DR00015	6
		NCTC13439	5
		NCTC13440	5
	***Klebsiella pneumoniae* group Total**		**72**
Morganella morganii	*Morganella morganii*	148-200	8
		148-204	8
		148-205	8
		148-206	9
		148-209	7
	Morganella morganii, CTX-M1, NDM	CDC#0057	5
	Morganella morganii, KPC	CDC#0133	4
	***Morganella morganii* Total**		**49**
Neisseria meningitidis	*Neisseria meningitidis*	ATCC13077	9
		ATCC13090	8
		ATCC13102	8
		ATCC13113	3
		ATCC35561	10
		NCTC10026	6
	***Neisseria meningitidis* Total**		**44**
Proteus mirabilis	*Proteus mirabilis*, KPC	CDC#0155	4
	Proteus mirabilis, NDM	CDC#0159	5
	***Proteus mirabilis* Total**		**9**
Pseudomonas aeruginosa	*Pseudomonas aeruginosa*, IMP	CDC#0092	5
		CDC#0103	8
	Pseudomonas aeruginosa, KPC	CDC#0090	1
	Pseudomonas aeruginosa, VIM	CDC#0054	5
		CDC#0100	4
		CDC#0108	4
		NCTC13437	5
	***Pseudomonas aeruginosa* Total**		**32**
Salmonella	*Salmonella* 4,5,12:I:-	FSLS5-0580	2
	Salmonella Heidelberg	ATCC8326	2
	Salmonella Infantis	ATCCBAA-1675	2
	Salmonella Javiana	ATCC10721	1
	Salmonella Montevideo	ATCC8387	8
	Salmonella Muenchen	ATCC8388	1
	Salmonella Newport	ATCC6962	6
	Salmonella Typhimurium	ATCC13311	7
	Salmonella enterica subspecies *enterica Enteritidis* (Group D1)	ATCCBAA-708	6
	***Salmonella* Total**		**35**

Target	Organism	Strain	Independent Contrived Specimens Tested
Serratia	*Serratia ficaria*	ATCC33105	4
	Serratia grimesii	ATCC14460	3
	Serratia plymuthica	ATCC53858	3
	Serratia rubidaea	ATCC27593	4
		ATCC29025	3
	***Serratia* Total**		**17**
Serratia marcescens	*Serratia marcescens*	ATCC13880	2
		ATCC14041	3
		ATCC14756	3
		ATCC43861	4
		ATCC43862	3
		ATCC8100	3
	Serratia marcescens, IMP	LMC-DR23105	1
	***Serratia marcescens* Total**		**19**
Stenotrophomonas maltophilia	*Stenotrophomonas maltophilia*	148-201	7
		148-222	6
		148-223	7
		148-224	8
		148-225	8
	***Stenotrophomonas maltophilia* Total**		**36**

Clinical performance, organism targets:

Tables 58-78 include a summary of the ePlex BCID-GN Panel performance for each organism target for prospective, retrospective and contrived specimens. Performance is based on comparison of ePlex BCID-GN Panel results to results from comparator methods for prospective and retrospective specimens and based on comparison of ePlex BCID-GN Panel results to the expected result for contrived specimens.

Table 58: Clinical performance, *Acinetobacter baumannii*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Acinetobacter baumannii	Prospective (Fresh)	0/0	---	167/167	100 (97.8-100)
	Prospective (Frozen)	4/4	100 (51.0-100)	178/178	100 (97.9-100)
	Prospective (All)	**4/4**	**100 (51.0-100)**	**345/345**	**100 (98.9-100)**
	Retrospective	15/15	100 (79.6-100)	560/561	99.8 (99.0-100)
	Prospective / Retrospective	**19/19**	**100 (83.2-100)**	**905/906[A]**	**99.9 (99.4-100)**
	Contrived	55/55	100 (93.5-100)	722/722	100 (99.5-100)
	Overall	**74/74**	**100 (95.1-100)**	**1627/1628**	**99.9 (99.7-100)**

CI= Confidence Interval

A. *Acinetobacter baumanii* was detected in the 1/1 false positive specimen using PCR/sequencing.

Table 59: Clinical performance, *Bacteroides fragilis*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Bacteroides fragilis	Prospective (Fresh)	6/6	100 (61.0-100)	161/161	100 (97.7-100)
	Prospective (Frozen)	5/5	100 (56.6-100)	177/177	100 (97.9-100)
	Prospective (All)	**11/11**	**100 (74.1-100)**	**338/338**	**100 (98.9-100)**
	Retrospective	14/17	82.4 (59.0-93.8)	558/560	99.6 (98.7-99.9)
	Prospective / Retrospective	**25/28[A]**	**89.3 (72.8-96.3)**	**896/898[B]**	**99.8 (99.2-99.9)**
	Contrived	40/40	100 (91.2-100)	737/737	100 (99.5-100)
	Overall	**65/68**	**95.6 (87.8-98.5)**	**1633/1635**	**99.9 (99.6-100)**

A. *B. fragilis* was not detected in 2 false negative specimens, but PCR/sequencing instead detected *B. caccae* and *B. thetaiotaomicron*, which were not identified by standard laboratory procedures.
B. *B. fragilis* was detected in 2/2 false positive specimens using PCR/sequencing.

Table 60: Clinical performance, *Citrobacter*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Citrobacter	Prospective (Fresh)	3/3	100 (43.9-100)	163/164	99.4 (96.6-99.9)
	Prospective (Frozen)	2/2	100 (34.2-100)	178/180	98.9 (96.0-99.7)
	Prospective (All)	**5/5**	**100 (56.6-100)**	**341/344**	**99.1 (97.5-99.7)**
	Retrospective	20/21	95.2 (77.3-99.2)	555/556	99.8 (99.0-100)
	Prospective / Retrospective	**25/26**	**96.2 (81.1-99.3)**	**896/900[A]**	**99.6 (98.9-99.8)**
	Contrived	43/43	100 (91.8-100)	734/734	100 (99.5-100)
	Overall	**68/69**	**98.6 (92.2-99.7)**	**1630/1634**	**99.8 (99.4-99.9)**

A. *Citrobacter braakii* (2) and *Citrobacter freundii* (2) were detected in 4/4 false positive specimens using PCR/sequencing.

Table 61: Clinical performance, *Cronobacter sakazakii*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Cronobacter sakazakii	Prospective (Fresh)	0/0	---	167/167	100 (97.8-100)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**0/0**	**---**	**349/349**	**100 (98.9-100)**
	Retrospective	1/1	100 (20.7-100)	576/576	100 (99.3-100)
	Prospective / Retrospective	**1/1**	**100 (20.7-100)**	**925/925**	**100 (99.6-100)**
	Contrived	45/45	100 (92.1-100)	732/732	100 (99.5-100)
	Overall	**46/46**	**100 (92.3-100)**	**1657/1657**	**100 (99.8-100)**

Table 62: Clinical performance, *Enterobacter cloacae* complex

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Enterobacter cloacae complex	Prospective (Fresh)	12/12	100 (75.8-100)	153/155	98.7 (95.4-99.6)
	Prospective (Frozen)	7/7	100 (64.6-100)	173/175	98.9 (95.9-99.7)
	Prospective (All)	**19/19**	**100 (83.2-100)**	**326/330**	**98.8 (96.9-99.5)**
	Retrospective	47/50	94.0 (83.8-97.9)	526/527	99.8 (98.9-100)
	Prospective / Retrospective	**66/69[A]**	**95.7 (88.0-98.5)**	**852/857[B]**	**99.4 (98.6-99.8)**
	Contrived	35/37[C]	94.6 (82.3-98.5)	739/740	99.9 (99.2-100)
	Overall	**101/106**	**95.3 (89.4-98.0)**	**1591/1597**	**99.6 (99.2-99.8)**

A. A species of the *Enterobacter cloacae* complex was not detected in 1 false negative specimen, but PCR/sequencing and MALDI-TOF instead detected *E. coli.* Standard laboratory procedures identified *E. cloacae* only.
B. *E. cloacae* was detected in 2/5 false positive specimens using PCR/sequencing.
C. *E. cloacae* complex was not detected in 2 specimens containing *Enterobacter asburiae.*

Table 63: Clinical performance, *Enterobacter* (non-*cloacae* complex)

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Enterobacter -non-*cloacae* complex	Prospective (Fresh)	3/3	100 (43.9-100)	163/164	99.4 (96.6-99.9)
	Prospective (Frozen)	5/7	71.4 (35.9-91.8)	175/175	100 (97.9-100)
	Prospective (All)	**8/10**	**80.0 (49.0-94.3)**	**338/339**	**99.7 (98.3-99.9)**
	Retrospective	12/12	100 (75.8-100)	565/565	100 (99.3-100)
	Prospective / Retrospective	**20/22[A]**	**90.9 (72.2-97.5)**	**903/904[B]**	**99.9 (99.4-100)**
	Contrived	36/36	100 (90.4-100)	741/741	100 (99.5-100)
	Overall	**56/58**	**96.6 (88.3-99.0)**	**1644/1645**	**99.9 (99.7-100)**

A. A species of the *Enterobacter* non-*cloacae* complex was not detected in 2 false negative specimens. Standard laboratory procedures identified *E. aerogenes* and PCR/sequencing detected *E. cloacae*.,.
B. A species of the *Enterobacter* non-*cloacae* complex was not detected in the false positive specimen using PCR/sequencing.

Table 64: Clinical performance, *Escherichia coli*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Escherichia coli	Prospective (Fresh)	59/60	98.3 (91.1-99.7)	106/107	99.1 (94.9-99.8)
	Prospective (Frozen)	72/73	98.6 (92.6-99.8)	109/109	100 (96.6-100)
	Prospective (All)	**131/133**	**98.5 (94.7-99.6)**	**215/216**	**99.5 (97.4-99.9)**
	Retrospective	132/140	94.3 (89.1-97.1)	435/437	99.5 (98.3-99.9)
	Prospective / Retrospective	**263/273**	**96.3 (93.4-98.0)**	**650/653[A]**	**99.5 (98.7-99.8)**
	Contrived	52/52	100 (93.1-100)	725/725	100 (99.5-100)
	Overall	**315/325**	**96.9 (94.4-98.3)**	**1375/1378**	**99.8 (99.4-99.9)**

A. *E. coli* was detected in 3/3 false positive specimens using PCR/sequencing.

Table 65: Clinical performance, *Fusobacterium necrophorum*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Fusobacterium necrophorum	Prospective (Fresh)	0/0	---	167/167	100 (97.8-100)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**0/0**	**---**	**349/349**	**100 (98.9-100)**
	Retrospective	1/1	100 (20.7-100)	576/576	100 (99.3-100)
	Prospective / Retrospective	**1/1**	**100 (20.7-100)**	**925/925**	**100 (99.6-100)**
	Contrived	47/48	97.9 (89.1-99.6)	729/729	100 (99.5-100)
	Overall	**48/49**	**98.0 (89.3-99.6)**	**1654/1654**	**100 (99.8-100)**

Table 66: Clinical performance, *Fusobacterium nucleatum*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Fusobacterium nucleatum	Prospective (Fresh)	0/0	---	167/167	100 (97.8-100)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**0/0**	**---**	**349/349**	**100 (98.9-100)**
	Retrospective	5/5	100 (56.6-100)	571/572	99.8 (99.0-100)
	Prospective / Retrospective	**5/5**	**100 (56.6-100)**	**920/921[A]**	**99.9 (99.4-100)**
	Contrived	47/47	100 (92.4-100)	730/730	100 (99.5-100)
	Overall	**52/52**	**100 (93.1-100)**	**1650/1651**	**99.9 (99.7-100)**

A. *F. nucleatum* was detected in the 1/1 false positive specimen using PCR/sequencing.

Table 67: Clinical performance, *Haemophilus influenzae*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Haemophilus influenzae	Prospective (Fresh)	3/3	100 (43.9-100)	164/164	100 (97.7-100)
	Prospective (Frozen)	4/4	100 (51.0-100)	178/178	100 (97.9-100)
	Prospective (All)	**7/7**	**100 (64.6-100)**	**342/342**	**100 (98.9-100)**
	Retrospective	7/7	100 (64.6-100)	570/570	100 (99.3-100)
	Prospective / Retrospective	**14/14**	**100 (78.5-100)**	**912/912**	**100 (99.6-100)**
	Contrived	41/41	100 (91.4-100)	736/736	100 (99.5-100)
	Overall	**55/55**	**100 (93.5-100)**	**1648/1648**	**100 (99.8-100)**

Table 68: Clinical performance, *Klebsiella oxytoca*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Klebsiella oxytoca	Prospective (Fresh)	4/6	66.7 (30.0-90.3)	160/161	99.4 (96.6-99.9)
	Prospective (Frozen)	7/7	100 (64.6-100)	175/175	100 (97.9-100)
	Prospective (All)	**11/13**	**84.6 (57.8-95.7)**	**335/336**	**99.7 (98.3-99.9)**
	Retrospective	29/34	85.3 (69.9-93.6)	541/543	99.6 (98.7-99.9)
	Prospective / Retrospective	**40/47[A]**	**85.1 (72.3-92.6)**	**876/879[B]**	**99.7 (99.0-99.9)**
	Contrived	20/20	100 (83.9-100)	757/757	100 (99.5-100)
	Overall	**60/67**	**89.6 (80.0-94.8)**	**1633/1636**	**99.8 (99.5-99.9)**

A. *K.* oxytoca was not detected in 2 false negative specimens using PCR/sequencing, but 16S sequencing instead detected *Raoultella ornithinolytica* and *Raoultella planticola*, which were not identified by standard laboratory procedures.
B. *Klebsiella oxytoca* was detected in 3/3 false positive specimens using PCR/sequencing.

Table 69: Clinical performance, *Klebsiella pneumoniae* group

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Klebsiella pneumoniae group	Prospective (Fresh)	29/30	96.7 (83.3-99.4)	136/137	99.3 (96.0-99.9)
	Prospective (Frozen)	29/31	93.5 (79.3-98.2)	151/151	100 (97.5-100)
	Prospective (All)	**58/61**	**95.1 (86.5-98.3)**	**287/288**	**99.7 (98.1-99.9)**
	Retrospective	106/108	98.1 (93.5-99.5)	466/469	99.4 (98.1-99.8)
	Prospective / Retrospective	**164/169[A]**	**97.0 (93.3-98.7)**	**753/757[B]**	**99.5 (98.6-99.8)**
	Contrived	72/72	100 (94.9-100)	705/705	100 (99.5-100)
	Overall	**236/241**	**97.9 (95.2-99.1)**	**1458/1462**	**99.7 (99.3-99.9)**

A. *K. pneumoniae* was not detected in 1 false negative specimen, but PCR/sequencing and MALDI-TOF instead detected *K. oxytoca*, which was not identified by standard laboratory procedures.
B. *K. pneumoniae* was detected in 4/4 false positive specimens using PCR/sequencing.

Table 70: Clinical performance, *Morganella morganii*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Morganella morganii	Prospective (Fresh)	3/3	100 (43.9-100)	164/164	100 (97.7-100)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**3/3**	**100 (43.9-100)**	**346/346**	**100 (98.9-100)**
	Retrospective	10/10	100 (72.2-100)	566/567	99.8 (99.0-100)
	Prospective / Retrospective	**13/13**	**100 (77.2-100)**	**912/913[A]**	**99.9 (99.4-100)**
	Contrived	49/49	100 (92.7-100)	728/728	100 (99.5-100)
	Overall	**62/62**	**100 (94.2-100)**	**1640/1641**	**99.9 (99.7-100)**

A. *M. morganii* was detected in 1/1 false positive clinical specimens using PCR/sequencing.

Table 71: Clinical performance, *Neisseria meningitidis*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Neisseria meningitidis	Prospective (Fresh)	0/0	---	167/167	100 (97.8-100)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**0/0**	**---**	**349/349**	**100 (98.9-100)**
	Retrospective	0/0	---	576/577	99.8 (99.0-100)
	Prospective / Retrospective	**0/0**	**---**	**925/926[A]**	**99.9 (99.4-100)**
	Contrived	44/44	100 (92.0-100)	733/733	100 (99.5-100)
	Overall	**44/44**	**100 (92.0-100)**	**1658/1659**	**99.9 (99.7-100)**

A. *N. meningitidis* was not detected in the false positive specimen using PCR/sequencing.

Table 72: Clinical performance, *Proteus*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Proteus	Prospective (Fresh)	7/8	87.5 (52.9-97.8)	159/159	100 (97.6-100)
	Prospective (Frozen)	15/15	100 (79.6-100)	167/167	100 (97.8-100)
	Prospective (All)	**22/23**	**95.7 (79.0-99.2)**	**326/326**	**100 (98.8-100)**
	Retrospective	54/55	98.2 (90.4-99.7)	522/522	100 (99.3-100)
	Prospective / Retrospective	**76/78**	**97.4 (91.1-99.3)**	**848/848**	**100 (99.5-100)**
	Contrived	9/9	100 (70.1-100)	768/768	100 (99.5-100)
	Overall	**85/87**	**97.7 (92.0-99.4)**	**1616/1616**	**100 (99.8-100)**

Table 73: Clinical performance, *Proteus mirabilis*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Proteus mirabilis	Prospective (Fresh)	7/8	87.5 (52.9-97.8)	159/159	100 (97.6-100)
	Prospective (Frozen)	15/15	100 (79.6-100)	167/167	100 (97.8-100)
	Prospective (All)	**22/23**	**95.7 (79.0-99.2)**	**326/326**	**100 (98.8-100)**
	Retrospective	50/51	98.0 (89.7-99.7)	526/526	100 (99.3-100)
	Prospective / Retrospective	**72/74**	**97.3 (90.7-99.3)**	**852/852**	**100 (99.6-100)**
	Contrived	9/9	100 (70.1-100)	768/768	100 (99.5-100)
	Overall	**81/83**	**97.6 (91.6-99.3)**	**1620/1620**	**100 (99.8-100)**

Table 74: Clinical performance, *Pseudomonas aeruginosa*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Pseudomonas aeruginosa	Prospective (Fresh)	10/10	100 (72.2-100)	157/157	100 (97.6-100)
	Prospective (Frozen)	17/18	94.4 (74.2-99.0)	163/164	99.4 (96.6-99.9)
	Prospective (All)	**27/28**	**96.4 (82.3-99.4)**	**320/321**	**99.7 (98.3-99.9)**
	Retrospective	56/60	93.3 (84.1-97.4)	514/517	99.4 (98.3-99.8)
	Prospective / Retrospective	**83/88**	**94.3 (87.4-97.5)**	**834/838[A]**	**99.5 (98.8-99.8)**
	Contrived	32/32	100 (89.3-100)	745/745	100 (99.5-100)
	Overall	**115/120**	**95.8 (90.6-98.2)**	**1579/1583**	**99.7 (99.4-99.9)**

A. *P. aeruginosa* was detected in 2/4 false positive specimens using PCR/sequencing.

Table 75: Clinical performance, *Salmonella*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Salmonella	Prospective (Fresh)	2/2	100 (34.2-100)	165/165	100 (97.7-100)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**2/2**	**100 (34.2-100)**	**347/347**	**100 (98.9-100)**
	Retrospective	18/19	94.7 (75.4-99.1)	558/558	100 (99.3-100)
	Prospective / Retrospective	**20/21[A]**	**95.2 (77.3-99.2)**	**905/905**	**100 (99.6-100)**
	Contrived	34/35	97.1 (85.5-99.5)	742/742	100 (99.5-100)
	Overall	**54/56**	**96.4 (87.9-99.0)**	**1647/1647**	**100 (99.8-100)**

A. *Salmonella* was not detected in 1 false negative specimen, but PCR/sequencing instead detected *E. coli*, which was not identified by standard laboratory procedures.

Table 76: Clinical performance, *Serratia*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Serratia	Prospective (Fresh)	6/6	100 (61.0-100)	161/161	100 (97.7-100)
	Prospective (Frozen)	4/4	100 (51.0-100)	178/178	100 (97.9-100)
	Prospective (All)	**10/10**	**100 (72.2-100)**	**339/339**	**100 (98.9-100)**
	Retrospective	34/34	100 (89.8-100)	542/543	99.8 (99.0-100)
	Prospective / Retrospective	**44/44**	**100 (92.0-100)**	**881/882[A]**	**99.9 (99.4-100)**
	Contrived	36/36	100 (90.4-100)	741/741	100 (99.5-100)
	Overall	**80/80**	**100 (95.4-100)**	**1622/1623**	**99.9 (99.7-100)**

A. *S. marcescens* was detected in the 1/1 false positive specimen using PCR/sequencing.

Table 77: Clinical performance, *Serratia marcescens*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Serratia marcescens	Prospective (Fresh)	5/5	100 (56.6-100)	162/162	100 (97.7-100)
	Prospective (Frozen)	4/4	100 (51.0-100)	178/178	100 (97.9-100)
	Prospective (All)	**9/9**	**100 (70.1-100)**	**340/340**	**100 (98.9-100)**
	Retrospective	34/34	100 (89.8-100)	542/543	99.8 (99.0-100)
	Prospective / Retrospective	**43/43**	**100 (91.8-100)**	**882/883[A]**	**99.9 (99.4-100)**
	Contrived	19/19	100 (83.2-100)	758/758	100 (99.5-100)
	Overall	**62/62**	**100 (94.2-100)**	**1640/1641**	**99.9 (99.7-100)**

A. *S. marcescens* was detected in the 1/1 false positive specimen using PCR/sequencing.

Table 78: Clinical performance, *Stenotrophomonas maltophilia*

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Stenotrophomonas maltophilia	Prospective (Fresh)	2/3	66.7 (20.8-93.9)	164/164	100 (97.7-100)
	Prospective (Frozen)	1/1	100 (20.7-100)	181/181	100 (97.9-100)
	Prospective (All)	**3/4**	**75.0 (30.1-95.4)**	**345/345**	**100 (98.9-100)**
	Retrospective	8/10	80.0 (49.0-94.3)	566/567	99.8 (99.0-100)
	Prospective / Retrospective	**11/14**	**78.6 (52.4-92.4)**	**911/912[A]**	**99.9 (99.4-100)**
	Contrived	36/36	100 (90.4-100)	741/741	100 (99.5-100)
	Overall	**47/50**	**94.0 (83.8-97.9)**	**1652/1653**	**99.9 (99.7-100)**

A. *S. maltophilia* was detected in the 1/1 false positive specimen using PCR/sequencing.

Performance of the ePlex BCID-GN Panel for the Pan *Candida* and Pan Gram-Positive assay targets is summarized in Table 79 below. In addition to the evaluable prospective and retrospective specimens that contained gram-negative organisms by Gram stain (Intended use population), the clinical performance of the Pan *Candida* and Pan Gram-Positive targets was evaluated with an additional 741 non-intended use retrospective specimens that contained gram-positive or fungal organisms by Gram stain; these specimens are denoted as retrospective (non-intended use) specimens and were used to supplement the prospective intended use specimens.

Pan *Candida* target performance: For the intended use population (specimens with gram-negative organisms observed by Gram stain), there were eight specimens with targeted *Candida* species identified by the comparator method. All eight specimens were mixed cultures, including three specimens with false negative results for the ePlex BCID-GN Pan *Candida* assay.

It is noted that the Pan *Candida* assay is designed to detect only four *Candida* species (*Candida albicans, C. glabrata, C. parapsilosis,* and *C. krusei*). In the clinical study there were six specimens for which non-targeted *Candida* species were identified either by

comparator testing or PCR/sequencing performed during discordant analysis. The non-targeted *Candida* species identified in these six specimens included *C. dubliniensis (1), C. lusitaniae (3),* and *C. tropicalis (2).* The ePlex BCID-GN Panel generated the expected result for each of these specimens.

Pan Gram-Positive performance: For the intended use population (specimens with gram-negative bacteria observed by Gram stain), there were a total of 17 false negative specimens for the Pan Gram-positive assay target (i.e., PPA for prospective specimens was 73.9% (17/23); PPA for retrospective specimens was 80% (44/55)). For the six false negative prospective specimens, two specimens were determined to have *Paenibacillus* species based on additional testing of discrepant specimens. Three of the other four false negative prospective specimens and all 11 of the false negative retrospective specimens were mixed cultures containing gram-negative organisms that were correctly detected by the ePlex BCID-GN Panel.

The assay labeling includes the following two limitations regarding mixed cultures, with the second limitation specifying the lower sensitivity of the ePlex BCID Panel for the Pan *Candida* and Pan Gram-Positive assay targets.

- In mixed cultures, the ePlex BCID-GN Panel may not identify all organisms in the specimen, depending upon the concentration of each target present.

- The BCID-GN Pan *Candida* and Pan Gram-Positive assays are designed to detect *Candida* and gram-positive microorganisms in mixed blood cultures where these microorganisms might be missed by Gram stain. Lower than expected clinical sensitivity was observed for these analytes, which was likely due to the presence of *Candida* spp. or gram-positive microorganisms in mixed cultures at concentrations below the respective limits of detection for the Pan *Candida* and Pan GP targets.

Table 79: Clinical performance, Pan *Candida* and Pan Gram-Positive targets

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	**% (95% CI)**	**TN/TN+FP**	**% (95% CI)**
Pan *Candida*	Prospective (Fresh)	1/1	100 (20.7-100)	165/166	99.4 (96.7-99.9)
	Prospective (Frozen)	0/0	---	182/182	100 (97.9-100)
	Prospective (All)	**1/1**	**100 (20.7-100)**	**347/348[A]**	**99.7 (98.4-99.9)**
	Retrospective	4/7[B]	57.1 (25.0-84.2)	569/570[C]	99.8 (99.0-100)
	Retrospective (Non-Intended Use)	99/102[D]	97.1 (91.7-99.0)	638/639[E]	99.8 (99.1-100)
	Contrived	0/0	---	777/777	100 (99.5-100)
Pan Gram-Positive	Prospective (Fresh)	7/8	87.5 (52.9-97.8)	155/159	97.5 (93.7-99.0)
	Prospective (Frozen)	10/15	66.7 (41.7-84.8)	164/167	98.2 (94.9-99.4)
	Prospective (All)	**17/23[F]**	**73.9 (53.5-87.5)**	**319/326[G]**	**97.9 (95.6-99.0)**
	Retrospective	44/55[H]	80.0 (67.6-88.4)	512/522[I]	98.1 (96.5-99.0)
	Retrospective (Non-Intended Use)	567/571	99.3 (98.2-99.7)	165/170[J]	97.1 (93.3-98.7)
	Contrived	0/0	---	776/777	99.9 (99.3-100)

A. *Candida glabrata* was detected in the 1/1 false positive specimen using PCR/sequencing.
B. 3 of 3 (100%) false negative results occurred in specimens in mixed infections with bacterial organisms where the BCID-GN Panel results were correct for the other infections in these specimens.
C. *Candida albicans* was detected in the 1/1 false positive specimen using PCR/sequencing.
D. 2 of 3 (67%) false negative results occurred in specimens in mixed infections with bacterial organisms where the BCID-GN Panel results were correct for the other infections in these specimens.
E. *Candida glabrata* was detected in the 1/1 false positive specimen using PCR/sequencing.
F. *Bacillus* (the gram-positive organism identified by standard laboratory procedures) was not detected in 2 false negative specimens using PCR/sequencing, but 16S sequencing instead detected *Paenibacillus lautus* and *Paenibacillus urinalis*, which were not identified by standard laboratory procedures. 3 of the remaining 4 (75%) false negative results occurred in specimens in mixed infections with bacterial organisms where the BCID-GN Panel correctly detected the gram-negative organisms present.
G. *Enterococcus* (1), *Staphylococcus* (3), or *Streptococcus* (2) were detected in 6/7 false positive specimens using PCR/sequencing (testing was not performed for the remaining 1/7 false positive specimens).
H. 11 of 11 (100%) false negative results occurred in specimens in mixed infections with bacterial organisms where the BCID-GN Panel correctly detected the gram-negative organisms present.
I. *Enterococcus* (2), *Staphylococcus* (1), or *Streptococcus* (5) were detected in 8/10 false positive specimens using PCR/sequencing (testing was not performed for the remaining 2/10 false positive specimens).
J. *Bacillus* (1) or *Streptococcus* (1) were detected in 2/5 false positive specimens using PCR/sequencing (testing was not performed for the remaining 3/5 false positive specimens).

Performance for ePlex BCID-GN Panel for genus and group assays, stratification by species

The ePlex BCID-GN Panel reports genus or group level results for *Citrobacter*, *Enterobacter cloacae* complex, *Enterobacter* non-*cloacae* complex, *Proteus*, *Salmonella*, *Serratia*, Pan *Candida*, and Pan Gram-Positive assay targets. Clinical performance is described in Table 80 for intended use specimens (i.e., bottles containing gram-negative organisms by Gram stain). Performance is stratified by species identified by comparator methods.

Table 81 includes performance for the Pan-*Candida* and Pan-Gram Positive assays for non-intended use specimens (i.e., bottles with gram-positive or fungal organisms by Gram stain). This data was used to supplement the intended use specimens due to the lower prevalence of *Candida* spp. and gram-positive organisms observed in intended use specimens.

Table 80: Performance, genus and group assays stratified by species

Target Species Detected by Comparator Method	Prospective Specimens		Retrospective Specimens		Contrived Specimens		Combined Specimens	
	Sensitivity/PPA		Sensitivity/PPA		Sensitivity/PPA		Sensitivity/PPA	
	TP/TP+FN	% (95% CI)	TP/TP+FN	% (95% CI)	TP/TP+FN	% (95% CI)	TP/TP+FN	% (95% CI)
Citrobacter	5/5	100 (56.6-100)	20/21	95.2 (77.3-99.2)	43/43	100 (91.8-100)	68/69	98.6 (92.2-99.7)
Citrobacter braakii			2/3	66.7 (20.8-93.9)	8/8	100 (67.6-100)	10/11	90.9 (62.3-98.4)
Citrobacter freundii	4/4	100 (51.0-100)	13/13	100 (77.2-100)	27/27	100 (87.5-100)	44/44	100 (92.0-100)
Citrobacter koseri	1/1	100 (20.7-100)	4/4	100 (51.0-100)	4/4	100 (51.0-100)	9/9	100 (70.1-100)
Citrobacter youngae			1/1	100 (20.7-100)	4/4	100 (51.0-100)	5/5	100 (56.6-100)
Enterobacter (non-*cloacae* complex)	8/10	80.0 (49.0-94.3)	12/12	100 (75.8-100)	36/36	100 (90.4-100)	56/58	96.6 (88.3-99.0)
Enterobacter aerogenes	7/9	77.8 (45.3-93.7)	12/12	100 (75.8-100)	26/26	100 (87.1-100)	45/47	95.7 (85.8-98.8)
Enterobacter amnigenus					10/10	100 (72.2-100)	10/10	100 (72.2-100)
Enterobacter gergoviae	1/1	100 (20.7-100)					1/1	100 (20.7-100)
Enterobacter cloacae complex	19/19	100 (83.2-100)	47/50	94.0 (83.8-97.9)	35/37	94.6 (82.3-98.5)	101/106	95.3 (89.4-98.0)
Enterobacter asburiae					6/8	75.0 (40.9-92.9)	6/8	75.0 (40.9-92.9)
Enterobacter cloacae	19/19	100 (83.2-100)	46/49	93.9 (83.5-97.9)	28/28	100 (87.9-100)	93/96	96.9 (91.2-98.9)
Enterobacter hormaechei			1/1	100 (20.7-100)	1/1	100 (20.7-100)	2/2	100 (34.2-100)
Proteus	22/23	95.7 (79.0-99.2)	54/55	98.2 (90.4-99.7)	9/9	100 (70.1-100)	85/87	97.7 (92.0-99.4)
Proteus mirabilis	22/23	95.7 (79.0-99.2)	50/51	98.0 (89.7-99.7)	9/9	100 (70.1-100)	81/83	97.6 (91.6-99.3)
Proteus vulgaris			5/5	100 (56.6-100)			5/5	100 (56.6-100)
Salmonella	2/2	100 (34.2-100)	18/19	94.7 (75.4-99.1)	34/35	97.1 (85.5-99.5)	54/56	96.4 (87.9-99.0)
Salmonella	2/2	100 (34.2-100)	15/15	100 (79.6-100)			17/17	100 (81.6-100)
Salmonella 4,5,12:i:-					2/2	100 (34.2-100)	2/2	100 (34.2-100)
Salmonella Heidelberg					2/2	100 (34.2-100)	2/2	100 (34.2-100)
Salmonella Infantis					2/2	100 (34.2-100)	2/2	100 (34.2-100)
Salmonella Javiana					1/1	100 (20.7-100)	1/1	100 (20.7-100)
Salmonella Montevideo					7/8	87.5 (52.9-97.8)	7/8	87.5 (52.9-97.8)
Salmonella Muenchen					1/1	100 (20.7-100)	1/1	100 (20.7-100)
Salmonella Newport					6/6	100 (61.0-100)	6/6	100 (61.0-100)
Salmonella Typhimurium					7/7	100 (64.6-100)	7/7	100 (64.6-100)
Salmonella choleraesuis subsp. Arizonae			0/1	0.0 (0.0-79.3)			0/1	0.0 (0.0-79.3)
Salmonella enterica subsp. enterica Enteritidis (Group D1)					6/6	100 (61.0-100)	6/6	100 (61.0-100)
Salmonella enterica subsp. enterica serovar Typhimurium			1/1	100 (20.7-100)			1/1	100 (20.7-100)
Salmonella Typhi			2/2	100 (34.2-100)			2/2	100 (34.2-100)
Serratia	10/10	100 (72.2-100)	34/34	100 (89.8-100)	36/36	100 (90.4-100)	80/80	100 (95.4-100)
Serratia ficaria					4/4	100 (51.0-100)	4/4	100 (51.0-100)
Serratia grimesii					3/3	100 (43.9-100)	3/3	100 (43.9-100)

Target Species Detected by Comparator Method	Prospective Specimens		Retrospective Specimens		Contrived Specimens		Combined Specimens	
	Sensitivity/PPA		Sensitivity/PPA		Sensitivity/PPA		Sensitivity/PPA	
	TP/TP+FN	% (95% CI)	TP/TP+FN	% (95% CI)	TP/TP+FN	% (95% CI)	TP/TP+FN	% (95% CI)
Serratia liquefaciens	1/1	100 (20.7-100)					1/1	100 (20.7-100)
Serratia marcescens	9/9	100 (70.1-100)	34/34	100 (89.8-100)	19/19	100 (83.2-100)	62/62	100 (94.2-100)
Serratia plymuthica					3/3	100 (43.9-100)	3/3	100 (43.9-100)
Serratia rubidaea					7/7	100 (64.6-100)	7/7	100 (64.6-100)
Pan *Candida*	1/1	100 (20.7-100)	4/7	57.1 (25.0-84.2)	N/A	N/A	5/8	62.5 (30.6-86.3)
Candida albicans	1/1	100 (20.7-100)	2/4	50.0 (15.0-85.0)			3/5	60.0 (23.1-88.2)
Candida glabrata			1/2	50.0 (9.5-90.5)			1/2	50.0 (9.5-90.5)
Candida krusei			1/1	100 (20.7-100)			1/1	100 (20.7-100)
Candida parapsilosis								
Pan Gram-*Positive*	17/23	73.9 (53.5-87.5)	44/55	80.0 (67.6-88.4)	N/A	N/A	61/78	78.2 (67.8-85.9)
Bacillus (unspeciated)	1/4	25.0 (4.6-69.9)					1/4	25.0 (4.6-69.9)
Enterococcus (unspeciated)			0/1	0.0 (0.0-79.3)			0/1	0.0 (0.0-79.3)
Enterococcus casseliflavus			0/1	0.0 (0.0-79.3)			0/1	0.0 (0.0-79.3)
Enterococcus faecalis	5/7	71.4 (35.9-91.8)	18/20	90.0 (69.9-97.2)			23/27	85.2 (67.5-94.1)
Enterococcus faecium	1/1	100 (20.7-100)	8/9	88.9 (56.5-98.0)			9/10	90.0 (59.6-98.2)
Staphylococcus (unspeciated)	3/3	100 (43.9-100)	4/6	66.7 (30.0-90.3)			7/9	77.8 (45.3-93.7)
Staphylococcus aureus	2/2	100 (34.2-100)	5/6	83.3 (43.6-97.0)			7/8	87.5 (52.9-97.8)
Staphylococcus cohnii	1/1	100 (20.7-100)					1/1	100 (20.7-100)
Staphylococcus epidermidis	1/1	100 (20.7-100)	2/3	66.7 (20.8-93.9)			3/4	75.0 (30.1-95.4)
Staphylococcus haemolyticus	2/2	100 (34.2-100)					2/2	100 (34.2-100)
Staphylococcus hominis	1/1	100 (20.7-100)	3/3	100 (43.9-100)			4/4	100 (51.0-100)
Streptococcus	0/1	0.0 (0.0-79.3)					0/1	0.0 (0.0-79.3)
Streptococcus - viridans group	1/1	100 (20.7-100)	0/1	0.0 (0.0-79.3)			1/2	50.0 (9.5-90.5)
Streptococcus anginosus group	1/1	100 (20.7-100)	4/5	80.0 (37.6-96.4)			5/6	83.3 (43.6-97.0)
Streptococcus infantarius			1/1	100 (20.7-100)			1/1	100 (20.7-100)
Streptococcus mitis group			1/1	100 (20.7-100)			1/1	100 (20.7-100)
Streptococcus oralis			1/1	100 (20.7-100)			1/1	100 (20.7-100)
Streptococcus pneumoniae			1/1	100 (20.7-100)			1/1	100 (20.7-100)
Streptococcus salivarius			1/1	100 (20.7-100)			1/1	100 (20.7-100)

Table 81: Pan Candida and Pan-Gram positive assay targets, performance stratified by species for non-intended use specimens (Specimens with gram-positive or fungal organisms by Gram stain).

Target Species Detected by Comparator Method	Retrospective (Non-Intended Use) Specimens	
	Sensitivity/PPA	
	TP/TP+FN	% (95% CI)
Pan Candida	99/102	97.1 (91.7-99.0)
Candida albicans	47/48	97.9 (89.1-99.6)
Candida glabrata	37/38	97.4 (86.5-99.5)
Candida krusei	3/3	100 (43.9-100)
Candida parapsilosis	15/16	93.8 (71.7-98.9)
Pan Gram-Positive	567/571	99.3 (98.2-99.7)
Bacillus (unspeciated)	4/4	100 (51.0-100)
Bacillus cereus	4/5	80.0 (37.6-96.4)
Bacillus cereus group - not anthracis	1/1	100 (20.7-100)
Enterococcus	1/1	100 (20.7-100)
Enterococcus faecalis	34/34	100 (89.8-100)
Enterococcus faecium	14/14	100 (78.5-100)
Coagulase-negative *Staphylococci* (CoNS)	14/14	100 (78.5-100)
Staphylococcus (unspeciated)	86/86	100 (95.7-100)
Staphylococcus aureus	173/174	99.4 (96.8-99.9)
Staphylococcus auricularis	3/3	100 (43.9-100)
Staphylococcus capitis	7/7	100 (64.6-100)
Staphylococcus cohnii	1/1	100 (20.7-100)
Staphylococcus epidermidis	86/87	98.9 (93.8-99.8)
Staphylococcus haemolyticus	6/6	100 (61.0-100)
Staphylococcus hominis	19/19	100 (83.2-100)
Staphylococcus hominis ssp *hominis*	21/21	100 (84.5-100)
Staphylococcus intermedius	1/1	100 (20.7-100)
Staphylococcus lugdunensis	1/1	100 (20.7-100)
Staphylococcus saccharolyticus	1/1	100 (20.7-100)
Staphylococcus saprophyticus	1/1	100 (20.7-100)
Staphylococcus sciuri	1/1	100 (20.7-100)
Staphylococcus simulans	2/2	100 (34.2-100)
Staphylococcus warneri	4/4	100 (51.0-100)
Alpha Hemolytic *Streptococcus*	1/1	100 (20.7-100)
Beta Hemolytic *Streptococci*, Group G	1/1	100 (20.7-100)
Gamma Hemolytic *Streptococcus*	1/1	100 (20.7-100)
Streptococcus (unspeciated)	9/9	100 (70.1-100)
Streptococcus - viridans group	17/17	100 (81.6-100)
Streptococcus agalactiae	21/21	100 (84.5-100)
Streptococcus anginosus	2/2	100 (34.2-100)

Target Species Detected by Comparator Method	Retrospective (Non-Intended Use) Specimens	
	Sensitivity/PPA	
	TP/TP+FN	% (95% CI)
Streptococcus bovis	2/2	100 (34.2-100)
Streptococcus constellatus ssp *constellatus*	1/1	100 (20.7-100)
Streptococcus dysgalactiae (Group G)	4/4	100 (51.0-100)
Streptococcus gordonii	1/1	100 (20.7-100)
Streptococcus intermedius	1/1	100 (20.7-100)
Streptococcus mitis	11/12	91.7 (64.6-98.5)
Streptococcus mitis group	2/2	100 (34.2-100)
Streptococcus pneumoniae	22/22	100 (85.1-100)
Streptococcus pyogenes	9/9	100 (70.1-100)
Streptococcus salivarius	2/2	100 (34.2-100)

Clinical performance, resistance marker targets:

Results for ePlex BCID Panel resistance marker targets are only reported when an associated organism is detected in the same specimen. The assay report will report 'N/A' for a resistance marker if an applicable organism is not detected in the specimen (See Table 5 above for associated organisms specifically reported with the six targeted resistance markers on the ePlex BCID-GN Panel).

Clinical performance for the ePlex BCID-GN Panel assays for CTX-M, KPC, IMP, NDM, OXA and VIM targets are described in Tables 82-99 below. Note that there were no prospective or retrospective specimens that harbored IMP, NDM, or VIM beta-lactamase genes; therefore, performance was established for these analytes using contrived specimens.

Clinical Performance, CTX-M:

Table 82 includes the clinical performance for detection of CTX-M for prospective, retrospective and contrived specimens that contain an applicable gram-negative organism based on the comparator method (for clinical specimens) or the inoculated organism (for contrived specimens).

Table 83 includes the clinical performance summary for detection of CTX-M and any applicable organism by the BCID-GN Panel in comparison to detection/identification by comparator methods. Results include all prospective and retrospective specimens combined.

Table 84 includes the clinical performance for detection of CTX-M for prospective, retrospective and contrived specimens stratified by organism identified by the comparator method (for clinical specimens) and based on the organism used for inoculation (for contrived specimens).

Table 82: Clinical performance, CTX-M

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
CTX-M	Prospective (Fresh)	10/13	76.9 (49.7-91.8)	127/127	100 (97.1-100)
	Prospective (Frozen)	12/16	75.0 (50.5-89.8)	144/144	100 (97.4-100)
	Prospective (All)	**22/29**	**75.9 (57.9-87.8)**	**271/271**	**100 (98.6-100)**
	Retrospective	52/56	92.9 (83.0-97.2)	483/483	100 (99.2-100)
	Prospective / Retrospective	**74/85[A]**	**87.1 (78.3-92.6)**	**754/754**	**100 (99.5-100)**
	Contrived	75/75	100 (95.1-100)	437/437	100 (99.1-100)
	Overall	**149/160**	**93.1 (88.1-96.1)**	**1191/1191**	**100 (99.7-100)**

A. In 3/11 false negative specimens, CTX-M signal was above the threshold for detection; however, an associated organism was not detected by the BCID-GN Panel and the CTX-M target was reported as 'N/A'. Further testing of the 8/11 remaining false negative specimens indicated that 7 of the 8 specimens may have been contaminated during the original comparator extraction process and misidentified as having CTX-M present. Specifically, results for 7 of the 8 specimens were negative for CTX-M from the following additional testing: 1) qPCR testing of 2 repeat extractions from the original specimen, 2) qPCR testing of an extraction from the isolate, and 3) testing of the original specimen with an FDA-cleared multiplex assay. For these 7 specimens, the originally extracted specimen was re-tested with qPCR and CTX-M was again detected, suggesting contamination during the original extraction process. The remaining 8th specimen was positive for CTX-M from the repeat extractions, negative for CTX-M from the isolate, and negative for CTX-M when tested with an FDA-cleared multiplex assay. These inconsistent detection results suggest the 8th specimen may be a true low-copy CTX-M positive specimen.

Table 83: Distribution of CTX-M in clinical specimens

	Comparator Method			
BCID-GN	**Org+/ARG+**	**Org+/ARG-**	**Org-**	**Total**
Org+/ARG+	74	0	2	76
Org+/ARG-	8[A]	741	2	751
Org-	3[B]	13	83	99
Total	**85**	**754**	**87**	**926**

% Agreement (95% CI) for Org+/ARG+: 74/85=87.1% (78.3-92.6)
% Agreement (95% CI) for Org+/ARG-: 741/754=98.3% (97.1-99.0)
% Agreement (95% CI) for Org-: 83/87=95.4% (88.8-98.2)

A. Further testing of these 8 false negative specimens indicated that 7 of the 8 specimens may have been contaminated during the original extraction process and misidentified as having CTX-M present. Specifically, results for 7 of the 8 specimens were negative for CTX-M from the following additional testing: 1) qPCR testing of 2 repeat extractions from the original specimen, 2) qPCR testing of an extraction from the isolate, and 3) testing of the original specimen with an FDA-cleared multiplex assay. For these 7 specimens, the originally extracted specimen was re-tested with qPCR and CTX-M was again detected, suggesting contamination during the original extraction process. The remaining 8th specimen was positive for CTX-M from the repeat extractions, negative for CTX-M from the isolate, and negative for CTX-M when tested with an FDA-cleared multiplex assay. These inconsistent detection results suggest the 8th specimen may be a true low-copy CTX-M positive specimen.
B. For these 3 specimens, CTX-M signal was above the threshold for detection; however, an associated organism was not detected by the ePlex BCID-GN Panel and the CTX-M target was reported as 'N/A'.

Table 84: Clinical performance of CTX-M target in specimens with associated organisms detected by comparator methods

Species Detected by Comparator Method		Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Acinetobacter baumannii	Prospective	0/0	---	4/4	100 (51.0-100)
	Retrospective	0/0	---	15/15	100 (79.6-100)
	Contrived	0/0	---	55/55	100 (93.5-100)
	Combined	0/0	---	74/74	100 (95.1-100)
Citrobacter	Prospective	0/0	---	5/5	100 (56.6-100)
	Retrospective	1/1	100 (20.7-100)	20/20	100 (83.9-100)
	Contrived	6/6	100 (61.0-100)	37/37	100 (90.6-100)
	Combined	7/7	100 (64.6-100)	62/62	100 (94.2-100)
Enterobacter (non-*cloacae* complex)	Prospective	0/0	---	10/10	100 (72.2-100)
	Retrospective	0/0	---	12/12	100 (75.8-100)
	Contrived	0/0	---	36/36	100 (90.4-100)
	Combined	0/0	---	58/58	100 (93.8-100)
Enterobacter cloacae complex	Prospective	0/0	---	19/19	100 (83.2-100)
	Retrospective	0/0	---	50/50	100 (92.9-100)
	Contrived	22/22	100 (85.1-100)	15/15	100 (79.6-100)
	Combined	22/22	100 (85.1-100)	84/84	100 (95.6-100)
Escherichia coli	Prospective	16/18	88.9 (67.2-96.9)	115/115	100 (96.8-100)
	Retrospective	35/37	94.6 (82.3-98.5)	103/103	100 (96.4-100)
	Contrived	22/22	100 (85.1-100)	30/30	100 (88.6-100)
	Combined	73/77	94.8 (87.4-98.0)	248/248	100 (98.5-100)
Klebsiella oxytoca	Prospective	0/1	0.0 (0.0-79.3)	12/12	100 (75.8-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	20/20	100 (83.9-100)
	Combined	0/1	0.0 (0.0-79.3)	66/66	100 (94.5-100)
Klebsiella pneumoniae group	Prospective	5/5	100 (56.6-100)	56/56	100 (93.6-100)
	Retrospective	14/15	93.3 (70.2-98.8)	93/93	100 (96.0-100)
	Contrived	20/20	100 (83.9-100)	52/52	100 (93.1-100)
	Combined	39/40	97.5 (87.1-99.6)	201/201	100 (98.1-100)
Morganella morganii	Prospective	0/0	---	3/3	100 (43.9-100)
	Retrospective	0/0	---	10/10	100 (72.2-100)
	Contrived	5/5	100 (56.6-100)	44/44	100 (92.0-100)
	Combined	5/5	100 (56.6-100)	57/57	100 (93.7-100)
Proteus	Prospective	2/5	40.0 (11.8-76.9)	18/18	100 (82.4-100)
	Retrospective	2/3	66.7 (20.8-93.9)	52/52	100 (93.1-100)
	Contrived	0/0	---	9/9	100 (70.1-100)
	Combined	4/8	50.0 (21.5-78.5)	79/79	100 (95.4-100)
Proteus mirabilis	Prospective	2/5	40.0 (11.8-76.9)	18/18	100 (82.4-100)
	Retrospective	2/3	66.7 (20.8-93.9)	48/48	100 (92.6-100)
	Contrived	0/0	---	9/9	100 (70.1-100)
	Combined	4/8	50.0 (21.5-78.5)	75/75	100 (95.1-100)
Pseudomonas aeruginosa	Prospective	0/1	0.0 (0.0-79.3)	27/27	100 (87.5-100)
	Retrospective	0/0	---	60/60	100 (94.0-100)
	Contrived	0/0	---	32/32	100 (89.3-100)
	Combined	0/1	0.0 (0.0-79.3)	119/119	100 (96.9-100)
Salmonella	Prospective	0/0	---	2/2	100 (34.2-100)
	Retrospective	1/1	100 (20.7-100)	18/18	100 (82.4-100)
	Contrived	0/0	---	35/35	100 (90.1-100)
	Combined	1/1	100 (20.7-100)	55/55	100 (93.5-100)
Serratia	Prospective	0/0	---	10/10	100 (72.2-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	36/36	100 (90.4-100)
	Combined	0/0	---	80/80	100 (95.4-100)
Serratia marcescens	Prospective	0/0	---	9/9	100 (70.1-100)

Species Detected by Comparator Method		Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	19/19	100 (83.2-100)
	Combined	0/0	---	62/62	100 (94.2-100)
Stenotrophomonas maltophilia	Prospective	0/0	---	4/4	100 (51.0-100)
	Retrospective	0/0	---	10/10	100 (72.2-100)
	Contrived	0/0	---	36/36	100 (90.4-100)
	Combined	0/0	---	50/50	100 (92.9-100)

Clinical performance, IMP:

Table 85 includes the clinical performance for detection of IMP for prospective, retrospective and contrived specimens that contain an applicable gram-negative organism based on the comparator method (for clinical specimens) or the inoculated organism (for contrived specimens). No prospective or retrospective specimens were found to contain IMP by the ePlex BCID-GN Panel or by the comparator method.

Table 86 includes a performance summary for detection of IMP and any applicable organism by the ePlex BCID-GN Panel in comparison to detection/identification by comparator methods. Results include all prospective and retrospective specimens combined.

Table 87 includes the performance for detection of IMP for contrived specimens stratified by organism.

Table 85: Clinical performance, IMP

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
IMP	Prospective (Fresh)	0/0	---	138/138	100 (97.3-100)
	Prospective (Frozen)	0/0	---	159/159	100 (97.6-100)
	Prospective (All)	0/0	---	297/297	100 (98.7-100)
	Retrospective	0/0	---	532/532	100 (99.3-100)
	Prospective / Retrospective	0/0	---	829/829	100 (99.5-100)
	Contrived	40/40	100 (91.2-100)	436/436	100 (99.1-100)
	Overall	40/40	100 (91.2-100)	1265/1265	100 (99.7-100)

Table 86: Distribution of IMP in clinical specimens

	Comparator Method			
BCID-GN	**Org+/ARG+**	**Org+/ARG-**	**Org-**	**Total**
Org+/ARG+	0	0	0	0
Org+/ARG-	0	812	4	816
Org-	0	17	93	110
Total	**0**	**829**	**97**	**926**

% Agreement (95% CI) for Org+/ARG+: 0/0= N/A
% Agreement (95% CI) for Org+/ARG-: 812/829=97.9% (96.7-98.7)
% Agreement (95% CI) for Org-: 93/97=95.9% (89.9-98.4)

Table 87: Clinical performance for IMP in contrived specimens

Species Detected by Comparator Method	**Sensitivity/PPA**		**Specificity/NPA**	
	TP/TP+FN	**% (95% CI)**	**TN/TN+FP**	**% (95% CI)**
Acinetobacter baumannii	0/0	---	74/74	100 (95.1-100)
Citrobacter	0/0	---	69/69	100 (94.7-100)
Enterobacter (non-*cloacae* complex)	5/5	100 (56.6-100)	53/53	100 (93.2-100)
Enterobacter cloacae complex	0/0	---	106/106	100 (96.5-100)
Escherichia coli	7/7	100 (64.6-100)	318/318	100 (98.8-100)
Klebsiella oxytoca	0/0	---	67/67	100 (94.6-100)
Klebsiella pneumoniae group	14/14	100 (78.5-100)	227/227	100 (98.3-100)
Morganella morganii	0/0	---	62/62	100 (94.2-100)
Proteus	0/0	---	87/87	100 (95.8-100)
Proteus mirabilis	0/0	---	83/83	100 (95.6-100)
Pseudomonas aeruginosa	13/13	100 (77.2-100)	107/107	100 (96.5-100)
Salmonella	0/0	---	56/56	100 (93.6-100)
Serratia	1/1	100 (20.7-100)	79/79	100 (95.4-100)
Serratia marcescens	1/1	100 (20.7-100)	61/61	100 (94.1-100)

Clinical performance, KPC:

Table 88 includes the clinical performance for KPC for prospective, retrospective and contrived specimens that contain an applicable gram-negative organism based on the comparator method (for clinical specimens) or the inoculated organism (for contrived specimens).

Table 89 includes the clinical performance summary for detection of KPC and any applicable organism by the BCID-GN Panel in comparison to detection/identification by comparator methods. Results include all prospective and retrospective specimens combined.

Table 90 includes the clinical performance for detection of KPC for prospective, retrospective and contrived specimens stratified by organism identified by the comparator method (for clinical specimens) and based on the organism used for inoculation (for contrived specimens).

Table 88: Clinical performance, KPC

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
KPC	Prospective (Fresh)	2/2	100 (34.2-100)	136/136	100 (97.3-100)
	Prospective (Frozen)	1/1	100 (20.7-100)	158/158	100 (97.6-100)
	Prospective (All)	3/3	100 (43.9-100)	294/294	100 (98.7-100)
	Retrospective	4/5	80.0 (37.6-96.4)	527/528	99.8 (98.9-100)
	Prospective / Retrospective	7/8	87.5 (52.9-97.8)	821/822	99.9 (99.3-100)
	Contrived	44/44	100 (92.0-100)	477/477	100 (99.2-100)
	Overall	51/52	98.1 (89.9-99.7)	1298/1299	99.9 (99.6-100)

Table 89: Distribution of KPC in clinical specimens

	Comparator Method			
BCID-GN	**Org+/ARG+**	**Org+/ARG-**	**Org-**	**Total**
Org+/ARG+	7	1	0	8
Org+/ARG-	1	804	4	809
Org-	0	17	92	109
Total	**8**	**822**	**96**	**926**

% Agreement (95% CI) for Org+/ARG+: 7/8=87.5% (52.9-97.8)
% Agreement (95% CI) for Org+/ARG-: 804/822=97.8% (96.6-98.6)
% Agreement (95% CI) for Org-: 92/96=95.8% (89.8-98.4)

Table 90: Clinical performance of KPC target in specimens with associated organisms detected by comparator methods

Species Detected by Comparator Method		Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Acinetobacter baumannii	Prospective	0/0	---	4/4	100 (51.0-100)
	Retrospective	0/0	---	15/15	100 (79.6-100)
	Contrived	0/0	---	55/55	100 (93.5-100)
	Combined	0/0	---	74/74	100 (95.1-100)
Citrobacter	Prospective	0/0	---	5/5	100 (56.6-100)
	Retrospective	0/0	---	21/21	100 (84.5-100)
	Contrived	4/4	100 (51.0-100)	39/39	100 (91.0-100)
	Combined	4/4	100 (51.0-100)	65/65	100 (94.4-100)
Cronobacter sakazakii	Prospective	---	---	---	---
	Retrospective	0/0	---	1/1	100 (20.7-100)
	Contrived	0/0	---	45/45	100 (92.1-100)
	Combined	0/0	---	46/46	100 (92.3-100)
Enterobacter (non-*cloacae* complex)	Prospective	0/0	---	10/10	100 (72.2-100)
	Retrospective	0/0	---	12/12	100 (75.8-100)
	Contrived	0/0	---	36/36	100 (90.4-100)
	Combined	0/0	---	58/58	100 (93.8-100)
Enterobacter cloacae complex	Prospective	0/0	---	19/19	100 (83.2-100)
	Retrospective	0/1	0.0 (0.0-79.3)	49/49	100 (92.7-100)
	Contrived	3/3	100 (43.9-100)	34/34	100 (89.8-100)
	Combined	3/4	75.0 (30.1-95.4)	102/102	100 (96.4-100)
Escherichia coli	Prospective	1/1	100 (20.7-100)	132/132	100 (97.2-100)
	Retrospective	0/0	---	140/140	100 (97.3-100)
	Contrived	6/6	100 (61.0-100)	46/46	100 (92.3-100)
	Combined	7/7	100 (64.6-100)	318/318	100 (98.8-100)

Species Detected by Comparator Method		Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Klebsiella oxytoca	Prospective	0/0	---	13/13	100 (77.2-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	20/20	100 (83.9-100)
	Combined	0/0	---	67/67	100 (94.6-100)
Klebsiella pneumoniae group	Prospective	2/2	100 (34.2-100)	59/59	100 (93.9-100)
	Retrospective	4/4	100 (51.0-100)	103/104	99.0 (94.8-99.8)
	Contrived	22/22	100 (85.1-100)	50/50	100 (92.9-100)
	Combined	28/28	100 (87.9-100)	212/213	99.5 (97.4-99.9)
Morganella morganii	Prospective	0/0	---	3/3	100 (43.9-100)
	Retrospective	0/0	---	10/10	100 (72.2-100)
	Contrived	4/4	100 (51.0-100)	45/45	100 (92.1-100)
	Combined	4/4	100 (51.0-100)	58/58	100 (93.8-100)
Proteus	Prospective	0/0	---	23/23	100 (85.7-100)
	Retrospective	0/0	---	55/55	100 (93.5-100)
	Contrived	4/4	100 (51.0-100)	5/5	100 (56.6-100)
	Combined	4/4	100 (51.0-100)	83/83	100 (95.6-100)
Proteus mirabilis	Prospective	0/0	---	23/23	100 (85.7-100)
	Retrospective	0/0	---	51/51	100 (93.0-100)
	Contrived	4/4	100 (51.0-100)	5/5	100 (56.6-100)
	Combined	4/4	100 (51.0-100)	79/79	100 (95.4-100)
Pseudomonas aeruginosa	Prospective	0/0	---	28/28	100 (87.9-100)
	Retrospective	0/0	---	60/60	100 (94.0-100)
	Contrived	1/1	100 (20.7-100)	31/31	100 (89.0-100)
	Combined	1/1	100 (20.7-100)	119/119	100 (96.9-100)
Salmonella	Prospective	0/0	---	2/2	100 (34.2-100)
	Retrospective	0/0	---	19/19	100 (83.2-100)
	Contrived	0/0	---	35/35	100 (90.1-100)
	Combined	0/0	---	56/56	100 (93.6-100)
Serratia	Prospective	0/0	---	10/10	100 (72.2-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	36/36	100 (90.4-100)
	Combined	0/0	---	80/80	100 (95.4-100)
Serratia marcescens	Prospective	0/0	---	9/9	100 (70.1-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	19/19	100 (83.2-100)
	Combined	0/0	---	62/62	100 (94.2-100)

Clinical performance, NDM:

Table 91 includes the clinical performance for detection of NDM for prospective, retrospective and contrived specimens that contain an applicable gram-negative organism based on the comparator method (for clinical specimens) or the inoculated organism (for contrived specimens). No prospective or retrospective specimens were found to contain IMP by the ePlex BCID-GN Panel or by the comparator method.

Table 92 includes a performance summary for detection of NDM and any applicable organism by the ePlex BCID-GN Panel in comparison to detection/identification by comparator methods. Results include all prospective and retrospective specimens combined.

Table 93 includes the performance for detection of NDM for contrived specimens stratified by organism.

Table 91: Clinical performance, NDM

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
NDM	Prospective (Fresh)	0/0	---	138/138	100 (97.3-100)
	Prospective (Frozen)	0/0	---	159/159	100 (97.6-100)
	Prospective (All)	0/0	---	297/297	100 (98.7-100)
	Retrospective	0/0	---	532/532	100 (99.3-100)
	Prospective / Retrospective	0/0	---	829/829	100 (99.5-100)
	Contrived	54/54	100 (93.4-100)	422/422	100 (99.1-100)
	Overall	54/54	100 (93.4-100)	1251/1251	100 (99.7-100)

Table 92: Distribution of NDM in clinical specimens

	Comparator Method			
BCID-GN	**Org+/ARG+**	**Org+/ARG-**	**Org-**	**Total**
Org+/ARG+	0	0	0	0
Org+/ARG-	0	812	4	816
Org-	0	17	93	110
Total	**0**	**829**	**97**	**926**

% Agreement (95% CI) for Org+/ARG+: 0/0= N/A
% Agreement (95% CI) for Org+/ARG-: 812/829=97.9% (96.7-98.7)
% Agreement (95% CI) for Org-: 93/97=95.9% (89.9-98.4)

.

Table 93: Clinical performance of NDM Target in contrived specimens by organism

Species Detected by Comparator Method	Sensitivity/PPA		Specificity/NPA	
	TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Acinetobacter baumannii	5/5	100 (56.6-100)	69/69	100 (94.7-100)
Citrobacter	0/0	---	69/69	100 (94.7-100)
Enterobacter (non-*cloacae* complex)	0/0	---	58/58	100 (93.8-100)
Enterobacter cloacae complex	13/13	100 (77.2-100)	93/93	100 (96.0-100)
Escherichia coli	23/23	100 (85.7-100)	302/302	100 (98.7-100)
Klebsiella oxytoca	0/0	---	67/67	100 (94.6-100)

Species Detected by Comparator Method	Sensitivity/PPA		Specificity/NPA	
	TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Klebsiella pneumoniae group	3/3	100 (43.9-100)	238/238	100 (98.4-100)
Morganella morganii	5/5	100 (56.6-100)	57/57	100 (93.7-100)
Proteus	5/5	100 (56.6-100)	82/82	100 (95.5-100)
Proteus mirabilis	5/5	100 (56.6-100)	78/78	100 (95.3-100)
Pseudomonas aeruginosa	0/0	---	120/120	100 (96.9-100)
Salmonella	0/0	---	56/56	100 (93.6-100)
Serratia	0/0	---	80/80	100 (95.4-100)
Serratia marcescens	0/0	---	62/62	100 (94.2-100)

Clinical Performance, OXA: Note that the OXA assay on the BCID-GN Panel is designed to detect OXA-23 and OXA-48 only.

Table 94 includes the clinical performance for OXA for prospective, retrospective and contrived specimens that contain an applicable gram-negative organism based on the comparator method (for clinical specimens) or the inoculated organism (for contrived specimens).

Table 95 includes the clinical performance summary for detection of OXA and any applicable organism by the BCID-GN Panel in comparison to detection/identification by comparator methods. Results include all prospective and retrospective specimens combined.

Table 96 includes the clinical performance for detection of OXA for prospective, retrospective and contrived specimens stratified by organism identified by the comparator method (for clinical specimens) and based on the organism used for inoculation (for contrived specimens).

Table 94: Clinical performance, OXA

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
OXA	Prospective (Fresh)	0/1	0.0 (0.0-79.3)	137/137	100 (97.3-100)
	Prospective (Frozen)	1/1	100 (20.7-100)	158/158	100 (97.6-100)
	Prospective (All)	1/2	50.0 (9.5-90.5)	295/295	100 (98.7-100)
	Retrospective	9/11	81.8 (52.3-94.9)	519/521	99.6 (98.6-99.9)
	Prospective / Retrospective	10/13[A]	76.9 (49.7-91.8)	814/816	99.8 (99.1-99.9)
	Contrived	37/37	100 (90.6-100)	439/439	100 (99.1-100)
	Overall	47/50	94.0 (83.8-97.9)	1253/1255	99.8 (99.4-100)

A. In 1/3 false negative specimens, OXA signal was above the threshold for detection; however, an associated organism was not detected by the BCID-GN Panel and the OXA target was reported as 'N/A'. One additional false negative specimen was tested with an FDA-cleared multiplex assay and OXA was not detected. The isolate from the remaining false negative specimen tested negative for OXA-23 and OXA-48 by qPCR.

Table 95: Distribution of OXA in clinical specimens

	Comparator Method			
BCID-GN	**Org+/ARG+**	**Org+/ARG-**	**Org-**	**Total**
Org+/ARG+	10	2	0	12
Org+/ARG-	2[A]	798	4	804
Org-	1[B]	16	93	110
Total	13	816	97	926

% Agreement (95% CI) for Org+/ARG+: 10/13=76.9% (49.7-91.8)
% Agreement (95% CI) for Org+/ARG-: 798/816=97.8% (96.5-98.6)
% Agreement (95% CI) for Org-: 93/97=95.9% (89.9-98.4)

A. One false negative specimen was tested with an FDA-cleared multiplex assay and OXA was not detected. The isolate from the remaining false negative specimen tested negative for OXA-23 and OXA-48 by qPCR.
B. For this specimen, OXA signal was above the threshold for detection; however, an associated organism was not detected by the ePlex BCID-GN Panel and the OXA target was reported as 'N/A'.

Table 96: Clinical performance of OXA target in specimens with associated organisms detected by comparator methods

Species Detected by Comparator Method		**Sensitivity/PPA**		**Specificity/NPA**	
		TP/TP+FN	**% (95% CI)**	**TN/TN+FP**	**% (95% CI)**
Acinetobacter baumannii	Prospective	1/1	100 (20.7-100)	3/3	100 (43.9-100)
	Retrospective	7/7	100 (64.6-100)	8/8	100 (67.6-100)
	Contrived	13/13	100 (77.2-100)	42/42	100 (91.6-100)
	Combined	21/21	100 (84.5-100)	53/53	100 (93.2-100)
Citrobacter	Prospective	0/0	---	5/5	100 (56.6-100)
	Retrospective	0/0	---	21/21	100 (84.5-100)
	Contrived	0/0	---	43/43	100 (91.8-100)
	Combined	0/0	---	69/69	100 (94.7-100)
Enterobacter (non-*cloacae* complex)	Prospective	0/0	---	10/10	100 (72.2-100)
	Retrospective	0/0	---	12/12	100 (75.8-100)
	Contrived	12/12	100 (75.8-100)	24/24	100 (86.2-100)
	Combined	12/12	100 (75.8-100)	46/46	100 (92.3-100)
Enterobacter cloacae complex	Prospective	0/0	---	19/19	100 (83.2-100)
	Retrospective	0/0	---	50/50	100 (92.9-100)
	Contrived	0/0	---	37/37	100 (90.6-100)
	Combined	0/0	---	106/106	100 (96.5-100)
Escherichia coli	Prospective	0/1	0.0 (0.0-79.3)	132/132	100 (97.2-100)
	Retrospective	1/2	50.0 (9.5-90.5)	138/138	100 (97.3-100)
	Contrived	0/0	---	52/52	100 (93.1-100)
	Combined	1/3	33.3 (6.1-79.2)	322/322	100 (98.8-100)
Klebsiella oxytoca	Prospective	0/0	---	13/13	100 (77.2-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	20/20	100 (83.9-100)
	Combined	0/0	---	67/67	100 (94.6-100)
Klebsiella pneumoniae group	Prospective	0/0	---	61/61	100 (94.1-100)
	Retrospective	0/1	0.0 (0.0-79.3)	107/107	100 (96.5-100)
	Contrived	12/12	100 (75.8-100)	60/60	100 (94.0-100)
	Combined	12/13	92.3 (66.7-98.6)	228/228	100 (98.3-100)
Morganella morganii	Prospective	0/0	---	3/3	100 (43.9-100)
	Retrospective	0/0	---	10/10	100 (72.2-100)

Species Detected by Comparator Method		Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
	Contrived	0/0	---	49/49	100 (92.7-100)
	Combined	0/0	---	62/62	100 (94.2-100)
Proteus	Prospective	0/0	---	23/23	100 (85.7-100)
	Retrospective	1/1	100 (20.7-100)	53/54	98.1 (90.2-99.7)
	Contrived	0/0	---	9/9	100 (70.1-100)
	Combined	1/1	100 (20.7-100)	85/86	98.8 (93.7-99.8)
Proteus mirabilis	Prospective	0/0	---	23/23	100 (85.7-100)
	Retrospective	0/0	---	50/51	98.0 (89.7-99.7)
	Contrived	0/0	---	9/9	100 (70.1-100)
	Combined	0/0	---	82/83	98.8 (93.5-99.8)
Pseudomonas aeruginosa	Prospective	0/0	---	28/28	100 (87.9-100)
	Retrospective	0/0	---	59/60	98.3 (91.1-99.7)
	Contrived	0/0	---	32/32	100 (89.3-100)
	Combined	0/0	---	119/120	99.2 (95.4-99.9)
Salmonella	Prospective	0/0	---	2/2	100 (34.2-100)
	Retrospective	0/0	---	19/19	100 (83.2-100)
	Contrived	0/0	---	35/35	100 (90.1-100)
	Combined	0/0	---	56/56	100 (93.6-100)
Serratia	Prospective	0/0	---	10/10	100 (72.2-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	36/36	100 (90.4-100)
	Combined	0/0	---	80/80	100 (95.4-100)
Serratia marcescens	Prospective	0/0	---	9/9	100 (70.1-100)
	Retrospective	0/0	---	34/34	100 (89.8-100)
	Contrived	0/0	---	19/19	100 (83.2-100)
	Combined	0/0	---	62/62	100 (94.2-100)

Clinical performance, VIM:

Table 97 includes the clinical performance for detection of VIM for prospective, retrospective and contrived specimens that contain an applicable gram-negative organism based on the comparator method (for clinical specimens) or the inoculated organism (for contrived specimens). No prospective or retrospective specimens were found to contain VIM by the ePlex BCID-GN Panel or by the comparator method.

Table 98 includes a performance summary for detection of VIM and any applicable organism by the ePlex BCID-GN Panel in comparison to detection/identification by comparator methods. Results include all prospective and retrospective specimens combined.

Table 99 includes the performance for detection of VIM for contrived specimens stratified by organism.

Table 97: Clinical performance, VIM

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
VIM	Prospective (Fresh)	0/0	---	138/138	100 (97.3-100)

Target	Specimen Type	Sensitivity/PPA		Specificity/NPA	
		TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
	Prospective (Frozen)	0/0	---	159/159	100 (97.6-100)
	Prospective (All)	0/0	---	297/297	100 (98.7-100)
	Retrospective	0/0	---	532/532	100 (99.3-100)
	Prospective / Retrospective	0/0	---	829/829	100 (99.5-100)
	Contrived	42/42	100 (91.6-100)	434/434	100 (99.1-100)
	Overall	42/42	100 (91.6-100)	1263/1263	100 (99.7-100)

Table 98: Distribution of VIM in clinical specimens

	Comparator Method			
BCID-GN	Org+/ARG+	Org+/ARG-	Org-	Total
Org+/ARG+	0	0	0	0
Org+/ARG-	0	812	4	816
Org-	0	17	93	110
Total	0	829	97	926

% Agreement (95% CI) for Org+/ARG+: 0/0= N/A
% Agreement (95% CI) for Org+/ARG-: 812/829=97.9% (96.7-98.7)
% Agreement (95% CI) for Org-: 93/97=95.9% (89.9-98.4)

.

Table 99: Clinical performance of VIM target in contrived specimens by organism

Species Detected by Comparator Method	Sensitivity/PPA		Specificity/NPA	
	TP/TP+FN	% (95% CI)	TN/TN+FP	% (95% CI)
Acinetobacter baumannii	0/0	---	74/74	100 (95.1-100)
Citrobacter	0/0	---	69/69	100 (94.7-100)
Enterobacter (non-*cloacae* complex)	0/0	---	58/58	100 (93.8-100)
Enterobacter cloacae complex	6/6	100 (61.0-100)	100/100	100 (96.3-100)
Escherichia coli	2/2	100 (34.2-100)	323/323	100 (98.8-100)
Klebsiella oxytoca	0/0	---	67/67	100 (94.6-100)
Klebsiella pneumoniae group	16/16	100 (80.6-100)	225/225	100 (98.3-100)
Morganella morganii	0/0	---	62/62	100 (94.2-100)
Proteus	0/0	---	87/87	100 (95.8-100)
Proteus mirabilis	0/0	---	83/83	100 (95.6-100)
Pseudomonas aeruginosa	18/18	100 (82.4-100)	102/102	100 (96.4-100)
Salmonella	0/0	---	56/56	100 (93.6-100)
Serratia	0/0	---	80/80	100 (95.4-100)
Serratia marcescens	0/0	---	62/62	100 (94.2-100)

Resistance markers, comparison to antimicrobial susceptibility testing

A supplemental comparison of the CTX-M BCID-GN Panel target versus phenotypic antimicrobial susceptibility testing (AST) results for extended spectrum beta-lactamase (ESBL) activity, Ceftazidime, Ceftriaxone, and Aztreonam, and a combination of the four results (ESBL/Combo) is provided in Table 100 for clinical isolates with available AST results. In total, 162 isolates had ESBL confirmatory testing and 770 isolates had ESBL confirmatory testing and/or AST results for one or more of Ceftazidime, Ceftriaxone, or

Aztreonam. A true positive (TP) result was defined where CTX-M was detected by the BCID-GN Panel and the isolate was positive for the ESBL confirmatory test or resistant (R) or intermediate (I) to the specific antimicrobial. A false negative (FN) result was defined similarly when CTX-M was not detected by the BCID-GN Panel. PPA was calculated as 100 x (TP/(TP+FN)). A true negative (TN) result was defined where CTX-M was not detected by the BCID-GN Panel and the isolate was negative for the ESBL confirmatory test or susceptible (S) to the specific antimicrobial. A false positive (FP) result was defined similarly when CTX-M was detected by the BCID-GN Panel and the isolate was susceptible to the specific antimicrobial. NPA was calculated as 100 x (TN/(TN+FP)).

For the ESBL/Combo analysis, the phenotypic AST result was positive or negative based on the ESBL confirmatory test, if available. If the ESBL confirmatory test was not available, then the phenotypic AST result was considered positive if any of the three antimicrobials were resistant or intermediate, otherwise, if AST results were susceptible, the result was considered negative. Note: ESBL resistance can be due to mechanisms other than acquisition of the CTX-M resistance gene.

Table 100: BCID-GN Panel target CTX-M target, comparison to phenotypic AST results for ESBL confirmatory testing, Ceftazidime, Ceftriaxone, and Aztreonam

Associated Organism	ESBL confirmatory		Ceftazidime		Ceftriaxone		Aztreonam		ESBL/Combo	
	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)
A. baumannii	---	---	0/10 (0.0%)	3/3 (100%)	0/11 (0.0%)	1/1 (100%)	0/3 (0.0%)	0/0	0/15 (0.0%)	2/2 (100%)
Citrobacter	---	---	0/2 (0.0%)	14/15 (93.3%)	0/3 (0.0%)	21/22 (95.5%)	0/1 (0.0%)	16/17 (94.1%)	0/3 (0.0%)	21/22 (95.5%)
Enterobacter	---	---	0/2 (0.0%)	12/12 (100%)	0/2 (0.0%)	18/18 (100%)	0/1 (0.0%)	9/9 (100%)	0/3 (0.0%)	17/17 (100%)
E. cloacae complex	---	---	0/10 (0.0%)	30/30 (100%)	0/17 (0.0%)	39/39 (100%)	0/5 (0.0%)	25/25 (100%)	0/18 (0.0%)	39/39 (100%)
E. coli	28/30 (93.3%)	82/82 (100%)	29/38 (76.3%)	159/167 (95.2%)	47/59 (79.7%)	180/181 (99.4%)	23/27 (85.2%)	118/120 (98.3%)	47/53 (88.7%)	196/197 (99.5%)
K. oxytoca	0/0	10/10 (100%)	0/1 (0.0%)	29/29 (100%)	0/2 (0.0%)	40/40 (100%)	0/2 (0.0%)	20/20 (100%)	0/2 (0.0%)	41/41 (100%)
K. pneumoniae group	7/11 (63.6%)	25/25 (100%)	13/23 (56.5%)	83/83 (100%)	17/27 (63.0%)	115/115 (100%)	10/12 (83.3%)	72/72 (100%)	17/26 (65.4%)	119/119 (100%)
M. morganii			0/1 (0.0%)	6/6 (100%)	0/2 (0.0%)	9/9 (100%)	0/0	4/4 (100%)	0/2 (0.0%)	9/9 (100%)
Proteus	3/3 (100%)	0/0	1/2 (50.0%)	46/48 (95.8%)	4/7 (57.1%)	59/59 (100%)	2/3 (66.7%)	35/36 (97.2%)	4/8 (50.0%)	59/59 (100%)
P. mirabilis	3/3 (100%)	0/0	1/2 (50.0%)	43/45 (95.6%)	4/5 (80.0%)	56/56 (100%)	2/3 (66.7%)	32/33 (97.0%)	4/6 (66.7%)	56/56 (100%)
P. aeruginosa			0/7 (0.0%)	46/46 (100%)	0/20 (0.0%)	0/0	0/9 (0.0%)	25/25 (100%)	0/32 (0.0%)	41/41 (100%)
Salmonella	1/1 (100%)	0/0	0/0	6/6 (100%)	1/2 (50.0%)	12/12 (100%)	1/1 (100%)	8/8 (100%)	1/2 (50.0%)	15/15 (100%)
Serratia			0/1 (0.0%)	25/25 (100%)	0/4 (0.0%)	32/32 (100%)	0/1 (0.0%)	20/20 (100%)	0/4 (0.0%)	34/34 (100%)

Associated Organism	ESBL confirmatory		Ceftazidime		Ceftriaxone		Aztreonam		ESBL/Combo	
	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)
S. marcescens			0/1 (0.0%)	24/24 (100%)	0/4 (0.0%)	31/31 (100%)	0/1 (0.0%)	19/19 (100%)	0/4 (0.0%)	33/33 (100%)
S. maltophilia			0/2 (0.0%)	2/2 (100%)	0/3 (0.0%)	0/0	0/5 (0.0%)	0/0	0/5 (0.0%)	2/2 (100%)
Any Organism CI	42/48 (87.5%) (75.3-94.1)	117/117 (100%) (96.8-100)	44/102 (43.1%) (33.9-52.8)	528/541 (97.6%) (95.9-98.6)	73/168 (43.5%) (36.2-51.0)	613/615 (99.7%) (98.8-99.9)	38/74 (51.4%) (40.2-62.4)	403/408 (98.8%) (97.2-99.5)	73/183 (39.9%) (33.1-47.1)	684/686 (99.7%) (98.9-99.9)

CI= confidence interval

A supplemental comparison of the five BCID-GN Panel carbapenemase resistance gene targets (OXA, KPC, IMP, NDM, VIM) versus phenotypic AST testing for Ertapenem, Imipenem, and Meropenem is provided in Table 101 for clinical isolates with available AST results. In total, 731 isolates had AST results for 1 or more of Ertapenem, Imipenem, or Meropenem. A true positive (TP) result was defined where OXA, KPC, IMP, NDM and/or VIM was detected by the BCID-GN Panel and the isolate was resistant (R) or intermediate (I) to Ertapenem, Imipenem, or Meropenem. A false negative (FN) result was defined similarly when OXA, KPC, IMP, NDM and/or VIM was not detected by the BCID-GN Panel and the isolate was resistant or intermediate to any of the three antimicrobials. PPA was calculated as 100 x (TP/(TP+FN)). A true negative (TN) result was defined where OXA, KPC, IMP, NDM and/or VIM were not detected by the BCID-GN Panel and the isolate was susceptible for all carbapenems evaluated with AST. A false positive (FP) result was defined similarly when OXA, KPC, IMP, NDM and/or VIM was detected by the BCID-GN Panel and the isolate was susceptible for all carbapenems evaluated by AST. NPA was calculated as 100 x (TN/(TN+FP)). Note: Resistance to carbapenems can be due to mechanisms other than acquisition of the OXA, KPC, IMP, NDM and/or VIM resistance genes.

Table 101: Clinical performance of the BCID-GN Panel resistance gene targets compared to phenotypic AST results for Ertapenem, Imipenem, and Meropenem

Associated Organism	OXA		KPC		IMP		NDM		VIM		Any Resistance Marker	
	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)
A. baumannii	8/9 (88.9%)	9/9 (100%)	0/9 (0.0%)	9/9 (100%)	0/9 (0.0%)	9/9 (100%)	0/9 (0.0%)	9/9 (100%)	0/9 (0.0%)	9/9 (100%)	8/9 (88.9%)	9/9 (100%)
Citrobacter	0/1 (0.0%)	22/22 (100%)	0/1 (0.0%)	22/22 (100%)	0/1 (0.0%)	22/22 (100%)	0/1 (0.0%)	22/22 (100%)	0/1 (0.0%)	22/22 (100%)	0/1 (0.0%)	22/22 (100%)
C. sakazakii	---	---	0/0	1/1 (100%)	---	---	---	---	---	---	---	---
Enterobacter	0/0	15/15 (100%)	0/0	15/15 (100%)	0/0	15/15 (100%)	0/0	15/15 (100%)	0/0	15/15 (100%)	0/0	15/15 (100%)
E. cloacae complex	0/1 (0.0%)	51/51 (100%)	0/1 (0.0%)	51/51 (100%)	0/1 (0.0%)	51/51 (100%)	0/1 (0.0%)	51/51 (100%)	0/1 (0.0%)	51/51 (100%)	0/1 (0.0%)	51/51 (100%)

Associated Organism	OXA		KPC		IMP		NDM		VIM		Any Resistance Marker	
	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)	PPA TP/TP+FN (%)	NPA TN/TN+FP (%)
E. coli	0/2 (0.0%)	247/248 (99.6%)	1/2 (50.0%)	248/248 (100%)	0/2 (0.0%)	248/248 (100%)	0/2 (0.0%)	248/248 (100%)	0/2 (0.0%)	248/248 (100%)	1/2 (50.0%)	247/248 (99.6%)
K. oxytoca	0/0	40/40 (100%)	0/0	40/40 (100%)	0/0	40/40 (100%)	0/0	40/40 (100%)	0/0	40/40 (100%)	0/0	40/40 (100%)
K. pneumoniae group	0/8 (0.0%)	136/136 (100%)	5/8 (62.5%)	135/136 (99.3%)	0/8 (0.0%)	136/136 (100%)	0/8 (0.0%)	136/136 (100%)	0/8 (0.0%)	136/136 (100%)	5/8 (62.5%)	135/136 (99.3%)
M. morganii	0/0	10/10 (100%)	0/0	10/10 (100%)	0/0	10/10 (100%)	0/0	10/10 (100%)	0/0	10/10 (100%)	0/0	10/10 (100%)
Proteus	0/0	61/62 (98.4%)	0/0	62/62 (100%)	0/0	62/62 (100%)	0/0	62/62 (100%)	0/0	62/62 (100%)	0/0	61/62 (98.4%)
P. mirabilis	0/0	57/57 (100%)	0/0	57/57 (100%)	0/0	57/57 (100%)	0/0	57/57 (100%)	0/0	57/57 (100%)	0/0	57/57 (100%)
P. aeruginosa	0/5 (0.0%)	65/66 (98.5%)	0/5 (0.0%)	66/66 (100%)	0/5 (0.0%)	66/66 (100%)	0/5 (0.0%)	66/66 (100%)	0/5 (0.0%)	66/66 (100%)	0/5 (0.0%)	65/66 (98.5%)
Salmonella	0/0	8/8 (100%)	0/0	8/8 (100%)	0/0	8/8 (100%)	0/0	8/8 (100%)	0/0	8/8 (100%)	0/0	8/8 (100%)
Serratia	0/0	38/38 (100%)	0/0	38/38 (100%)	0/0	38/38 (100%)	0/0	38/38 (100%)	0/0	38/38 (100%)	0/0	38/38 (100%)
S. marcescens	0/0	37/37 (100%)	0/0	37/37 (100%)	0/0	37/37 (100%)	0/0	37/37 (100%)	0/0	37/37 (100%)	0/0	37/37 (100%)
Any Organism CI	8/26 (30.8%) (16.5-50.0)	796/799 (99.6%) (98.9-99.9)	6/26 (23.1%) (11.0-42.1)	799/800 (99.9%) (99.3-100)	0/26 (0.0%) (0.0-12.9)	799/799 (100%) (99.5-100)	0/26 (0.0%) (0.0-12.9)	799/799 (100%) (99.5-100)	0/26 (0.0%) (0.0-12.9)	799/799 (100%) (99.5-100)	14/26 (53.8%) (35.5-71.2)	796/800 (99.5%) (98.7-99.8)

CI= confidence interval

Clinical Study, performance of BCID-GN Panel for Specimens with multiple analytes

The ePlex BCID-GN Panel identified multiple bacterial analytes in 103 prospective and retrospective specimens. Of the 349 prospective specimens tested in the clinical study, 320 (91.7%) had single detections by the ePlex BCID-GN Panel. A total of 29 (8.2%) specimens had multiple organisms detected; 22 (6.2%) specimens had double detections, and seven specimens (2.0%) had triple detections.

Of the 29 specimens with multi-detections by the BCID-GN Panel, 13 (44.8%) had one or more organism targets detected by BCID-GN that were not identified by the comparator method (i.e., false positive ePlex BCID-GN Panel result).

All multi-detection combinations as detected by the ePlex BCID-GN Panel during the prospective clinical evaluation are presented in Table 102. False positive BCID-GN Panel results are described as discrepant organisms in the far right column of the table.

Table 102: Multi-detections identified by the ePlex BCID-GN Panel (Prospective specimens)

Distinct Co-Detection Combinations Detected by the ePlex BCID-GN Panel in Prospective Clinical Specimens				Number of Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A,B]
Target 1	**Target 2**	**Target 3**	**Resistance Marker**		
A. baumannii	Pan Gram-Positive			2 (0)	
Citrobacter	*E. cloacae* complex	*K. oxytoca*		2 (2)	*Citrobacter* (2), *E. cloacae* complex (2)
Citrobacter	*K. oxytoca*	*K. pneumoniae* group		1 (1)	*Citrobacter* (1)
Citrobacter	*P. mirabilis*	Pan Gram-Positive		1 (1)	Pan Gram-Positive (1)
E. cloacae complex	*E. coli*	*K. pneumoniae* group		1 (0)	
E. cloacae complex	Pan *Candida*	Pan Gram-Positive		1 (0)	
E. cloacae complex	Pan Gram-Positive			2 (0)	
E. coli	*K. oxytoca*			2 (1)	*K. oxytoca* (1)
E. coli	*K. pneumoniae* group		CTX-M	1 (1)	*E. coli* (1)
E. coli	Pan Gram-Positive			2 (1)	Pan Gram-Positive (1)
Enterobacter	*K. pneumoniae* group			1 (1)	*Enterobacter* (1)
K. oxytoca	Pan Gram-Positive			1 (0)	
K. oxytoca	*S. marcescens*			1 (0)	
K. pneumoniae group	*P. mirabilis*			1 (0)	
K. pneumoniae group	Pan Gram-Positive			2 (0)	
K. pneumoniae group	Pan Gram-Positive		CTX-M, KPC	1 (1)	Pan Gram-Positive (1)
M. morganii	*P. mirabilis*			1 (0)	
P. aeruginosa	*P. mirabilis*	Pan Gram-Positive		1 (0)	
P. aeruginosa	Pan Gram-Positive			1 (0)	
P. mirabilis	Pan Gram-Positive			3 (2)	Pan Gram-Positive (2)
P. mirabilis	Pan Gram-Positive		CTX-M	1 (0)	

A. A discrepant organism or resistance marker is defined as one that was detected by the BCID-GN Panel but not by the comparator method(s).

B. 12/13 false positive organisms were investigated using PCR/sequencing; the discrepant organism was detected in 11/13 and not detected in 1. One false positive Pan Gram-Positive specimen was not tested.

i. In 3/3 false positive *Citrobacter* specimens, *Citrobacter* was detected.
ii. In 2/2 false positive *E. cloacae* complex specimens, *E. cloacae* complex was detected.
iii. An *Enterobacter* species was not detected in the 1 false positive *Enterobacter* (non-*cloacae* complex) specimen.
iv. In 1/1 false positive *E. coli* specimen, *E. coli* was detected.
v. In 1/1 false positive *K. oxytoca* specimen, *K. oxytoca* was detected.
vi. In 4/4 false positive Pan Gram-Positive specimens, a Pan Gram-Positive organism was detected.

Of the 349 prospective specimens evaluated in the clinical study, 34 specimens (9.7%) had multiple organisms identified by culture. Of these 34 specimens, 10 specimens (29.4%) had one or more organisms that were not identified by the ePlex BCID-GN Panel (i.e., false negative ePlex BCID-GN Panel result).

Due to the significant percentage of false negative results for the ePlex BCID-GN Panel for blood culture specimens with multiple organisms, a limitation is included in the package insert stating the following:

- In mixed cultures, the ePlex BCID-GN Panel may not identify all organisms in the specimen, depending upon the concentration of each target present.

Table 103 includes results from the prospective clinical study for specimens with multiple organisms identified by comparator methods. False negative BCID-GN Panel results are described as discrepant organisms in the far right column of the table.

Table 103: Clinical performance, multi-detections identified by the comparator method(s) (Prospective Specimens)

Distinct Co-Detection Combinations Detected by the Comparator Method(s) in Prospective Clinical Specimens					Number Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A]
Target 1	**Target 2**	**Target 3**	**Target 4**	**Resistance Marker**		
A. baumannii	*E. faecium*	*Staphylococcus*			1 (0)	
A. baumannii	*Staphylococcus*				1 (0)	
*Achromobacter xylosoxidans**	*E. cloacae*				1 (0)	
*Acinetobacter lwoffii**	*Staphylococcus hominis*				1 (0)	
*Acinetobacter pittii**	*S. aureus*				1 (0)	
*Aerococcus viridans**	*K. oxytoca*	*S. epidermidis*	*Staphylococcus cohnii*		1 (0)	
*Aerococcus viridans**	*Staphylococcus hominis*				1 (0)	
B. fragilis	*Clostridium species**				1 (0)	
Bacillus	*E. cloacae*				1 (0)	
*C. acnes**	*E. coli*				1 (0)	
C. albicans	*E. cloacae*	*E. faecalis*			1 (0)	
C. freundii	*P. mirabilis*	*Providencia stuartii**			1 (0)	
*Candida lusitaniae**	*S. liquefaciens*				1 (0)	
*Citrobacter amalonaticus**	*E. coli*				1 (0)	

Distinct Co-Detection Combinations Detected by the Comparator Method(s) in Prospective Clinical Specimens					Number Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A]
Target 1	**Target 2**	**Target 3**	**Target 4**	**Resistance Marker**		
E. aerogenes	*K. oxytoca*	*Leclercia adecarboxylata**			2 (2)	*E. aerogenes* (2)
E. aerogenes	*P. aeruginosa*				1 (1)	*P. aeruginosa* (1)
E. cloacae	*E. coli*	*K. pneumoniae*			1 (0)	
E. cloacae	*S. aureus*				1 (0)	
E. coli	*E. faecalis*				1 (0)	
E. coli	*K. pneumoniae*				1 (1)	*E. coli* (1)
E. coli	*P. mirabilis*	*Providencia stuartii**	*S. anginosus* group	CTX-M	1 (1)	*E. coli* (1)
E. faecalis	*K. pneumoniae*				3 (1)	*E. faecalis* (1)
E. faecalis	*M. morganii*	*P. mirabilis*			1 (1)	*E. faecalis* (1)
E. faecalis	*Providencia stuartii**				1 (0)	
*Enterobacteriaceae**	*K. pneumoniae*				1 (0)	
K. pneumoniae	*Staphylococcus haemolyticus*	non-fermenting gram-negative bacilli*			1 (1)	*K. pneumoniae* (1)
*Lactococcus lactis**	*P. mirabilis*				1 (0)	
*Micrococcus luteus**	*Sphingomonas paucimobilis**				1 (0)	
P. aeruginosa	*P. mirabilis*	*Streptococcus - viridans* group			1 (0)	
P. aeruginosa	*S. maltophilia*				1 (1)	*S. maltophilia* (1)
P. aeruginosa	*Staphylococcus haemolyticus*				1 (0)	
P. mirabilis	*Staphylococcus*				1 (0)	
S. maltophilia	*Streptococcus*				1 (1)	*Streptococcus* (1)

* Indicates an off-panel organism not detected by the BCID-GN Panel.

A. A discrepant organism or resistance marker is defined as one that was detected by the comparator method(s) but not by the BCID-GN Panel. Excludes organisms that are not targeted by the BCID-GN Panel.

Of the 577 retrospective specimens tested in the clinical study, 503 specimens (87.2%) had single detections by the BCID-GN Panel. A total of 74 specimens (12.8%) had multiple organisms detected; 62 (10.7%) specimens had two organisms detected, and 12 (2.1%) specimens had three organisms detected.

Of the 74 specimens with multi-detections by the BCID-GN Panel, 23 (31.1%) specimens

had one or more organism targets detected by BCID-GN that were not identified by the comparator method (i.e., false positive for the BCID-GN Panel).

All multi-detection combinations as detected by the ePlex BCID-GN Panel during the prospective clinical evaluation are presented in Table 104. False positive BCID-GN Panel results are described as discrepant organisms in the far right column of the table.

Table 104: Co-Detections Identified by the ePlex BCID-GN Panel (Retrospective Specimens)

Distinct Co-Detection Combinations Detected by the ePlex BCID-GN Panel in Retrospective Clinical Specimens				Number Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A,B]
Target 1	**Target 2**	**Target 3**	**Resistance Marker**		
A. baumannii	*K. pneumoniae* group	Pan Gram-Positive	CTX-M, OXA	1 (1)	*A. baumannii* (1), *K. pneumoniae* group (1), Pan Gram-Positive (1)
A. baumannii	Pan Gram-Positive			2 (0)	
A. baumannii	Pan Gram-Positive		OXA	4 (1)	Pan Gram-Positive (1)
B. fragilis	*E. cloacae* complex	Pan Gram-Positive		1 (1)	*B. fragilis* (1)
B. fragilis	*E. coli*			2 (1)	*B. fragilis* (1)
B. fragilis	Pan Gram-Positive			1 (0)	
Citrobacter	*E. cloacae* complex			1 (1)	*E. cloacae* complex (1)
Citrobacter	*E. coli*			1 (0)	
Citrobacter	*K. oxytoca*			1 (1)	*Citrobacter* (1)
Citrobacter	*K. oxytoca*	*K. pneumoniae* group		1 (1)	*K. oxytoca* (1)
Citrobacter	*K. pneumoniae* group			1 (0)	
Citrobacter	*K. pneumoniae* group	Pan Gram-Positive	CTX-M	1 (0)	
Citrobacter	*M. morganii*	Pan Gram-Positive		1 (1)	*M. morganii* (1)
Citrobacter	Pan Gram-Positive			3 (2)	Pan Gram-Positive (2)
E. cloacae complex	*K. pneumoniae* group			1 (0)	
E. cloacae complex	*P. aeruginosa*	Pan Gram-Positive		1 (1)	*P. aeruginosa* (1)
E. cloacae complex	Pan *Candida*			1 (1)	Pan *Candida* (1)
E. cloacae complex	Pan Gram-Positive			2 (1)	Pan Gram-Positive (1)
E. coli	*K. oxytoca*			1 (0)	
E. coli	*K. oxytoca*	Pan Gram-Positive		1 (0)	
E. coli	*K. pneumoniae* group			2 (0)	

Distinct Co-Detection Combinations Detected by the ePlex BCID-GN Panel in Retrospective Clinical Specimens				Number Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A,B]
Target 1	**Target 2**	**Target 3**	**Resistance Marker**		
E. coli	*M. morganii*			1 (0)	
E. coli	*P. mirabilis*			3 (0)	
E. coli	*P. mirabilis*	Pan Gram-Positive		1 (0)	
E. coli	Pan Gram-Positive			8 (2)	Pan Gram-Positive (2)
E. coli	Pan Gram-Positive		CTX-M	1 (0)	
Enterobacter	Pan *Candida*			1 (0)	
Enterobacter	Pan Gram-Positive			1 (0)	
H. influenzae	*N. meningitidis*	*P. aeruginosa*		1 (1)	*N. meningitidis* (1), *P. aeruginosa* (1)
K. oxytoca	*K. pneumoniae* group			2 (1)	*K. pneumoniae* group (1)
K. oxytoca	Pan Gram-Positive			3 (2)	Pan Gram-Positive (2)
K. oxytoca	*S. marcescens*			1 (1)	*S. marcescens* (1)
K. pneumoniae group	Pan Gram-Positive			4 (1)	Pan Gram-Positive (1)
K. pneumoniae group	Pan Gram-Positive	*S. marcescens*		1 (1)	*K. pneumoniae* group (1)
K. pneumoniae group	*S. maltophilia*			1 (0)	
M. morganii	*P. aeruginosa*	Pan Gram-Positive		1 (1)	*P. aeruginosa* (1)
M. morganii	*P. mirabilis*			1 (0)	
M. morganii	Pan Gram-Positive	*Proteus*		1 (0)	
P. aeruginosa	Pan Gram-Positive			1 (0)	
P. mirabilis	Pan Gram-Positive			5 (0)	
Pan *Candida*	Pan Gram-Positive			2 (0)	
Pan Gram-Positive	*S. maltophilia*			1 (0)	
Pan Gram-Positive	*S. marcescens*			3 (0)	

A. A discrepant organism or resistance marker is defined as one that was detected by the BCID-GN Panel but not by the comparator method(s).

B. 24/26 false positive organisms were investigated using PCR/sequencing; the discrepant organism was detected in 21/24, not detected in 2, and was indeterminate for one organism.

i. In 1/1 false positive *A.baumannii* specimen, *A. baumannii* was detected.
ii. In 2/2 false positive *B. fragilis* specimens, *B. fragilis* was detected.
iii. In 1/1 false positive *Citrobacter* specimen, *Citrobacter* was detected.
iv. In the one false positive *E. cloacae* complex specimen, PCR/sequencing was indeterminate.
v. In 1/1 false positive K. oxytoca specimen, K. oxytoca was detected.
vi. In 3/3 false positive *K. pneumoniae* group specimens, *K. pneumoniae* group was detected.
vii. In 1/1 false positive specimen, *M. morganii* was detected.
viii. *N. meningitidis* was not detected in the 1 *N. meningitidis* false positive specimen.
ix. In 2/3 false positive *P. aeruginosa* specimens, *P. aeruginosa* was detected. *P. aeruginosa* was not detected in the remaining specimen.
x. In 1/1 false positive pan *Candida* specimen, Pan *Candida* was detected.

xi. In 8/8 false positive Pan Gram-Positive specimens, a Pan Gram-Positive organism was detected.
xii. In 1/1 false positive *S. marcescens* specimen, *S. marcescens* was detected.

Of the 577 retrospective specimens evaluated in the clinical study, 77 specimens (9.7%) had multiple organisms identified by the culture comparator. Of these 77 specimens, 28 specimens (36.4%) had one or more organisms that were not identified by the ePlex BCID-GN Panel (i.e., false negative).

Table 105 includes results from the retrospective clinical study for specimens with multiple organisms identified by comparator methods. False negative BCID-GN Panel results are described as discrepant organisms in the far right column of the table.

Table 105: Co-Detections identified by the comparator method(s) (Retrospective Specimens)

Distinct Co-Detection Combinations Detected by the Comparator Method(s) in Retrospective Clinical Specimens					**Number Specimens (Number Discrepant)**	**Discrepant Organism(s) / Resistance Marker(s)[A]**
Target 1	**Target 2**	**Target 3**	**Target 4**	**Resistance Marker**		
A. baumannii	*E. faecalis*			OXA	2 (0)	
A. baumannii	*E. faecalis*	*S. aureus*			1 (0)	
A. baumannii	*E. faecium*			OXA	1 (0)	
A. baumannii	*Staphylococcus*				1 (0)	
*Acinetobacter radioresistens**	*P. vulgaris*			OXA	1 (0)	
*Aeromonas caviae**	*E. coli*	*Enterococcus casseliflavus*	*K. oxytoca*		1 (1)	*E. casseliflavus* (1)
*Aeromonas veronii**	*E. cloacae*				1 (1)	*E. cloacae* (1)
B. fragilis	*S. anginosus* group				1 (0)	
C. albicans	*E. faecalis*				1 (0)	
C. albicans	*E. faecium*	*Staphylococcus hominis*			1 (1)	*C. albicans* (1)
C. albicans	*P. aeruginosa*				1 (1)	*C. albicans* (1)
C. albicans	*S. epidermidis*				1 (0)	
C. braakii	*E. cloacae*	*K. oxytoca*			1 (1)	*C. braakii* (1), *K. oxytoca* (1)
C. braakii	*E. coli*				1 (0)	
C. braakii	*Streptococcus oralis*				1 (0)	
C. freundii	*Enterococcus*				1 (1)	*Enterococcus* (1)
C. freundii	*K. pneumoniae*				2 (0)	
C. freundii	*K. pneumoniae*	*Staphylococcus hominis*		CTX-M	1 (0)	

Distinct Co-Detection Combinations Detected by the Comparator Method(s) in Retrospective Clinical Specimens					Number Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A]
Target 1	**Target 2**	**Target 3**	**Target 4**	**Resistance Marker**		
C. glabrata	*E. aerogenes*	*Staphylococcus*			1 (1)	*Staphylococcus* (1)
C. glabrata	*P. mirabilis*				1 (1)	*C. glabrata* (1)
C. koseri	*E. faecalis*				1 (0)	
C. krusei	*S. epidermidis*				1 (1)	*S. epidermidis* (1)
C. youngae	*K. oxytoca*				1 (1)	*K. oxytoca* (1)
*Clostridium perfringens**	*E. coli*				1 (0)	
E. aerogenes	*S. anginosus* group				1 (0)	
E. cloacae	*E. coli*				1 (1)	*E. coli* (1)
E. cloacae	*E. faecalis*				1 (1)	*E. faecalis* (1)
E. cloacae	*E. faecium*				1 (0)	
E. cloacae	*E. faecium*	*Staphylococcus hominis*			1 (0)	
E. cloacae	*K. pneumoniae*				1 (0)	
E. cloacae	*M. morganii*				1 (1)	*E. cloacae* (1)
E. cloacae	*S. anginosus* group				1 (0)	
E. cloacae	*S. maltophilia*				1 (1)	*S. maltophilia* (1)
E. coli	*E. faecalis*				2 (0)	
E. coli	*E. faecalis*			CTX-M	1 (0)	
E. coli	*E. faecalis*	*K. pneumoniae*			1 (1)	*E. coli* (1)
E. coli	*E. faecalis*	*P. mirabilis*			1 (0)	
E. coli	*E. faecium*				1 (0)	
E. coli	*E. faecium*			CTX-M	1 (1)	*E. faecium* (1)
E. coli	*K. oxytoca*	*Streptococcus infantarius*			1 (0)	
E. coli	*P. aeruginosa*				1 (1)	*P. aeruginosa* (1)
E. coli	*P. mirabilis*				1 (1)	*E. coli* (1)
E. coli	*P. mirabilis*	*P. vulgaris*	*Streptococcus - viridans* group		1 (1)	*S. viridans* group (1)
E. coli	*Propionibacteria**				1 (0)	
E. coli	*S. anginosus gp*				1 (1)	*S. anginosus* group (1)
E. coli	*S. aureus*				1 (0)	
E. coli	*S. pneumoniae*				1 (0)	

Distinct Co-Detection Combinations Detected by the Comparator Method(s) in Retrospective Clinical Specimens					Number Specimens (Number Discrepant)	Discrepant Organism(s) / Resistance Marker(s)[A]
Target 1	**Target 2**	**Target 3**	**Target 4**	**Resistance Marker**		
E. coli	*Staphylococcus*				1 (0)	
E. faecalis	*K. pneumoniae*				1 (1)	*K. pneumoniae* (1)
E. faecalis	*M. morganii*				1 (0)	
E. faecalis	*M. morganii*	*P. vulgaris*			1 (0)	
E. faecalis	*P. aeruginosa*	*S. aureus*			1 (1)	*P. aeruginosa* (1)
E. faecalis	*P. mirabilis*				3 (0)	
E. faecalis	*S. maltophilia*				1 (1)	*E. faecalis* (1)
E. faecalis	*S. marcescens*				1 (0)	
E. faecium	*K. pneumoniae*				1 (0)	
E. faecium	*P. aeruginosa*				1 (0)	
E. faecium	*P. mirabilis*				1 (0)	
K. oxytoca	*S. anginosus* group				1 (0)	
K. pneumoniae	*P. aeruginosa*				1 (1)	*P. aeruginosa* (1)
K. pneumoniae	*S. aureus*				2 (1)	*S. aureus* (1)
K. pneumoniae	*Staphylococcus*				1 (1)	*Staphylococcus* (1)
P. aeruginosa	*S. maltophilia*				1 (1)	S. maltophilia (1)
P. mirabilis	*Peptostreptococcus anaerobius**				1 (0)	
P. mirabilis	*Providencia stuartii**				2 (1)	*P. mirabilis* (1)
P. mirabilis	*Staphylococcus*				1 (0)	
*Pseudomonas putida**	*S. epidermidis*	*S. maltophilia*			1 (0)	
S. aureus	*S. marcescens*				1 (0)	
S. marcescens	*Staphylococcus*				1 (0)	
S. marcescens	*Streptococcus mitis* group	*Streptococcus salivarius*			1 (0)	

* Indicates an off-panel organism not detected by the BCID-GN Panel.

A. A discrepant organism or resistance marker is defined as one that was detected by the comparator method(s) but not by the BCID-GN Panel. Excludes organisms not targeted by the BCID-GN Panel.

Instrument Errors/Invalid Rates:

A total of 2460 prospective, retrospective, and contrived specimens were initially tested in the clinical study. Of these, 23/2460 (0.9%) did not complete the run and the specimen was retested according to assay instructions.

A total of 2334/2460 (94.9%, 95% CI: 93.9%-95.7%) specimens generated valid results for an initial invalid rate of 5.1% (95% CI: 4.3%-6.1%)

Upon repeat testing of the 126 specimens with initially invalid results, 1/126 (0.8%) did not complete the run and the specimen was retested. After repeat testing, all 126 specimens completed testing with 114/126 (90.5%) specimens generating valid results for a repeat invalid rate of 9.5% (95% CI: 5.5%-15.9%)

Overall, after final testing, 12/2460 (0.5%, 95% CI: 0.3%-0.9%) specimens had final, invalid results, resulting in a final validity rate of 2448/2460 (99.5%, 95% CI: 99.1%-99.7%).

4. Clinical cut-off:

   Not applicable

5. Expected values/Reference range:

   A prospective, multicenter clinical study was conducted to evaluate the clinical performance of the ePlex BCID-GN Panel in positive blood culture specimens. A total of 349 specimens were prospectively collected at seven clinical sites in two phases from patients of all ages and genders. In the first phase from June 2014 through July 2016, 182 specimens were prospectively collected and frozen; from June through July 2018, 167 specimens were prospectively collected and tested fresh (never frozen). The expected values of individual analytes based on the ePlex BCID-GN Panel results in prospective specimens are summarized by age group and by site in Tables 106 and 107.

Table 106: Expected values by age group (Prospective specimens)

Target	All Ages (N=349) n (%)	Age <1 (N=7) n (%)	Age 1-17 (N=10) n (%)	Age 18-44 (N=50) n (%)	Age 45-64 (N=124) n (%)	Age 65-84 (N=125) n (%)	Age 85+ (N=33) n (%)
Acinetobacter baumannii	4 (1.1)	0 (0.0)	0 (0.0)	1 (2.0)	2 (1.6)	1 (0.8)	0 (0.0)
Bacteroides fragilis	11 (3.2)	0 (0.0)	0 (0.0)	2 (4.0)	4 (3.2)	2 (1.6)	3 (9.1)
Citrobacter	8 (2.3)	0 (0.0)	0 (0.0)	2 (4.0)	1 (0.8)	2 (1.6)	3 (9.1)
Cronobacter sakazakii	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Enterobacter (non-*cloacae* complex)	9 (2.6)	0 (0.0)	1 (10.0)	2 (4.0)	5 (4.0)	1 (0.8)	0 (0.0)
Enterobacter cloacae complex	23 (6.6)	3 (42.9)	1 (10.0)	6 (12.0)	5 (4.0)	8 (6.4)	0 (0.0)
Escherichia coli	132 (37.8)	2 (28.6)	2 (20.0)	16 (32.0)	41 (33.1)	55 (44.0)	16 (48.5)
Fusobacterium necrophorum	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Fusobacterium nucleatum	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Haemophilus influenzae	7 (2.0)	0 (0.0)	0 (0.0)	3 (6.0)	1 (0.8)	1 (0.8)	2 (6.1)
Klebsiella oxytoca	12 (3.4)	0 (0.0)	0 (0.0)	3 (6.0)	8 (6.5)	1 (0.8)	0 (0.0)
Klebsiella pneumoniae group	59 (16.9)	1 (14.3)	1 (10.0)	10 (20.0)	26 (21.0)	17 (13.6)	4 (12.1)
Morganella morganii	3 (0.9)	0 (0.0)	0 (0.0)	0 (0.0)	2 (1.6)	1 (0.8)	0 (0.0)
Neisseria meningitidis	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Proteus	22 (6.3)	0 (0.0)	0 (0.0)	0 (0.0)	6 (4.8)	13 (10.4)	3 (9.1)
Proteus mirabilis	22 (6.3)	0 (0.0)	0 (0.0)	0 (0.0)	6 (4.8)	13 (10.4)	3 (9.1)

Target	All Ages (N=349) n (%)	Age <1 (N=7) n (%)	Age 1-17 (N=10) n (%)	Age 18-44 (N=50) n (%)	Age 45-64 (N=124) n (%)	Age 65-84 (N=125) n (%)	Age 85+ (N=33) n (%)
Pseudomonas aeruginosa	28 (8.0)	0 (0.0)	2 (20.0)	3 (6.0)	12 (9.7)	10 (8.0)	1 (3.0)
Salmonella	2 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	1 (0.8)	0 (0.0)
Serratia	10 (2.9)	0 (0.0)	1 (10.0)	0 (0.0)	5 (4.0)	4 (3.2)	0 (0.0)
Serratia marcescens	9 (2.6)	0 (0.0)	1 (10.0)	0 (0.0)	4 (3.2)	4 (3.2)	0 (0.0)
Stenotrophomonas maltophilia	3 (0.9)	0 (0.0)	0 (0.0)	3 (6.0)	0 (0.0)	0 (0.0)	0 (0.0)
Pan *Candida*	2 (0.6)	1 (14.3)	0 (0.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
Pan Gram-Positive	24 (6.9)	1 (14.3)	2 (20.0)	5 (10.0)	7 (5.6)	7 (5.6)	2 (6.1)
CTX-M	24 (6.9)	0 (0.0)	0 (0.0)	2 (4.0)	7 (5.6)	12 (9.6)	3 (9.1)
IMP	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
KPC	3 (0.9)	0 (0.0)	0 (0.0)	1 (2.0)	1 (0.8)	1 (0.8)	0 (0.0)
NDM	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
OXA	1 (0.3)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
VIM	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)

Table 107: Expected values by collection site (Prospective specimens)

Target	All Sites (N=349) n (%)	Site 1 (N=88) n (%)	Site 2 (N=23) n (%)	Site 3 (N=98) n (%)	Site 4 (N=58) n (%)	Site 5 (N=46) n (%)	Site 6 (N=28) n (%)	Site 7 (N=8) n (%)
Acinetobacter baumannii	4 (1.1)	3 (3.4)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Bacteroides fragilis	11 (3.2)	2 (2.3)	3 (13.0)	3 (3.1)	2 (3.4)	1 (2.2)	0 (0.0)	0 (0.0)
Citrobacter	8 (2.3)	2 (2.3)	0 (0.0)	3 (3.1)	1 (1.7)	1 (2.2)	1 (3.6)	0 (0.0)
Cronobacter sakazakii	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Enterobacter (non-*cloacae* complex)	9 (2.6)	2 (2.3)	0 (0.0)	4 (4.1)	0 (0.0)	1 (2.2)	2 (7.1)	0 (0.0)
Enterobacter cloacae complex	23 (6.6)	3 (3.4)	1 (4.3)	10 (10.2)	1 (1.7)	6 (13.0)	2 (7.1)	0 (0.0)
Escherichia coli	132 (37.8)	30 (34.1)	8 (34.8)	37 (37.8)	25 (43.1)	17 (37.0)	12 (42.9)	3 (37.5)
Fusobacterium necrophorum	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Fusobacterium nucleatum	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Haemophilus influenzae	7 (2.0)	1 (1.1)	0 (0.0)	2 (2.0)	2 (3.4)	1 (2.2)	1 (3.6)	0 (0.0)
Klebsiella oxytoca	12 (3.4)	5 (5.7)	0 (0.0)	3 (3.1)	1 (1.7)	2 (4.3)	1 (3.6)	0 (0.0)
Klebsiella pneumoniae group	59 (16.9)	17 (19.3)	3 (13.0)	20 (20.4)	5 (8.6)	7 (15.2)	4 (14.3)	3 (37.5)
Morganella morganii	3 (0.9)	0 (0.0)	1 (4.3)	2 (2.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Neisseria meningitidis	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Proteus	22 (6.3)	9 (10.2)	0 (0.0)	5 (5.1)	5 (8.6)	2 (4.3)	1 (3.6)	0 (0.0)
Proteus mirabilis	22 (6.3)	9 (10.2)	0 (0.0)	5 (5.1)	5 (8.6)	2 (4.3)	1 (3.6)	0 (0.0)
Pseudomonas aeruginosa	28 (8.0)	5 (5.7)	2 (8.7)	10 (10.2)	8 (13.8)	2 (4.3)	1 (3.6)	0 (0.0)
Salmonella	2 (0.6)	1 (1.1)	0 (0.0)	0 (0.0)	0 (0.0)	1 (2.2)	0 (0.0)	0 (0.0)
Serratia	10 (2.9)	1 (1.1)	2 (8.7)	1 (1.0)	3 (5.2)	3 (6.5)	0 (0.0)	0 (0.0)
Serratia marcescens	9 (2.6)	1 (1.1)	2 (8.7)	1 (1.0)	3 (5.2)	2 (4.3)	0 (0.0)	0 (0.0)
Stenotrophomonas maltophilia	3 (0.9)	1 (1.1)	0 (0.0)	0 (0.0)	1 (1.7)	0 (0.0)	1 (3.6)	0 (0.0)
Pan *Candida*	2 (0.6)	0 (0.0)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)	1 (3.6)	0 (0.0)

Target	All Sites (N=349) n (%)	Site 1 (N=88) n (%)	Site 2 (N=23) n (%)	Site 3 (N=98) n (%)	Site 4 (N=58) n (%)	Site 5 (N=46) n (%)	Site 6 (N=28) n (%)	Site 7 (N=8) n (%)
Pan Gram-Positive	24 (6.9)	15 (17.0)	1 (4.3)	5 (5.1)	1 (1.7)	0 (0.0)	2 (7.1)	0 (0.0)
CTX-M	24 (6.9)	9 (10.2)	1 (4.3)	3 (3.1)	4 (6.9)	5 (10.9)	2 (7.1)	0 (0.0)
IMP	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
KPC	3 (0.9)	1 (1.1)	0 (0.0)	1 (1.0)	1 (1.7)	0 (0.0)	0 (0.0)	0 (0.0)
NDM	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
OXA	1 (0.3)	1 (1.1)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
VIM	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)

N. Instrument Name:

ePlex Instrument

O. System Descriptions:

1. Modes of Operation:
   Does the applicant's device contain the ability to transmit data to a computer, webserver, or mobile device?
   Yes ____X____ or No _____

   After specimens have been loaded in the ePlex BCID-GN Panel cartridge, the remaining processing steps are executed under control of the ePlex instrument.

   Does the applicant's device transmit data to a computer, webserver, or mobile device using wireless transmission?
   Yes ___X_____ or No _____

2. Software:
   FDA has reviewed applicant's Hazard Analysis and software development processes for this line of product types:
   Yes ___X_____ or No _____

3. Specimen Identification:

   Specimen identification can be entered manually or via barcode.

4. Specimen Sampling and Handling:

   The operator opens an ePlex BCID-GN Panel cartridge pouch and writes the accession ID or places a barcode label with accession ID on the ePlex BCID-GN Panel cartridge. After mixing the blood culture bottle by inversion, the operator transfers sufficient volume of blood culture specimen from the bottle to a sterile tube. The operator then pipettes 50 µl of the positive blood culture specimen into the sample loading port of the ePlex BCID-GN Panel cartridge. The barcode scanner will read both the accession ID

barcode and the cartridge barcode. The loaded cartridge is then inserted into the bay and the BCID-GN Panel test is initiated.

5. Calibration:

   N/A

6. Quality Control:

   See M.1.c above.

P. Other Supportive Instrument Performance Characteristics Data Not Covered In The "Performance Characteristics" Section above:

N/A

Q. Proposed Labeling:

The labeling supports the finding of substantial equivalence for this device.

R. Conclusion:

The submitted information in this premarket notification is complete and supports a substantial equivalence decision.